

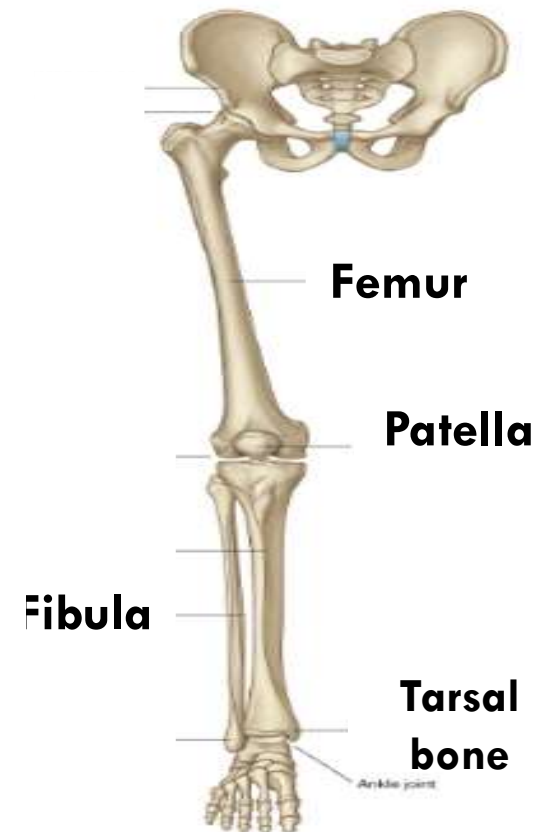
Preface UL vs LL

UPPER LIMB	LOWER LIMB
Prehension	Locomotion, Wt bearing
Smaller and weaker bones	Larger and strong
Joints are smaller and less stable	Larger and more stable
Muscles are smaller and attaches to smaller bony areas	Antigravity muscles more developed
Pectoral girdle is made of two bones	Pelvic girdle is made of single bone
Preaxial border faces laterally	Faces medially

Preface

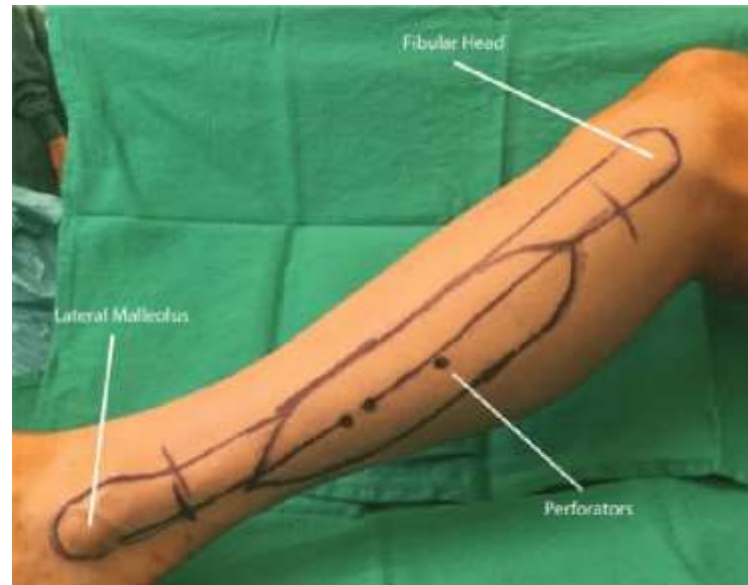
2

- **Hip region**
 - Hip/innominate/ bone
- **Thigh region**
 - Femur
- **Knee region**
 - Patella/knee cap/
- **Leg region**
 - Tibia
 - Fibula
- **Foot region/foot bone**
 - 7 Tarsal bones
 - 5 Metatarsal bones
 - 14 Phalanges



THE LOWER LIMB

- ◉ The longest region
- ◉ The risky region
- ◉ The donor site



Function of Lower Limb

- **Locomotion**

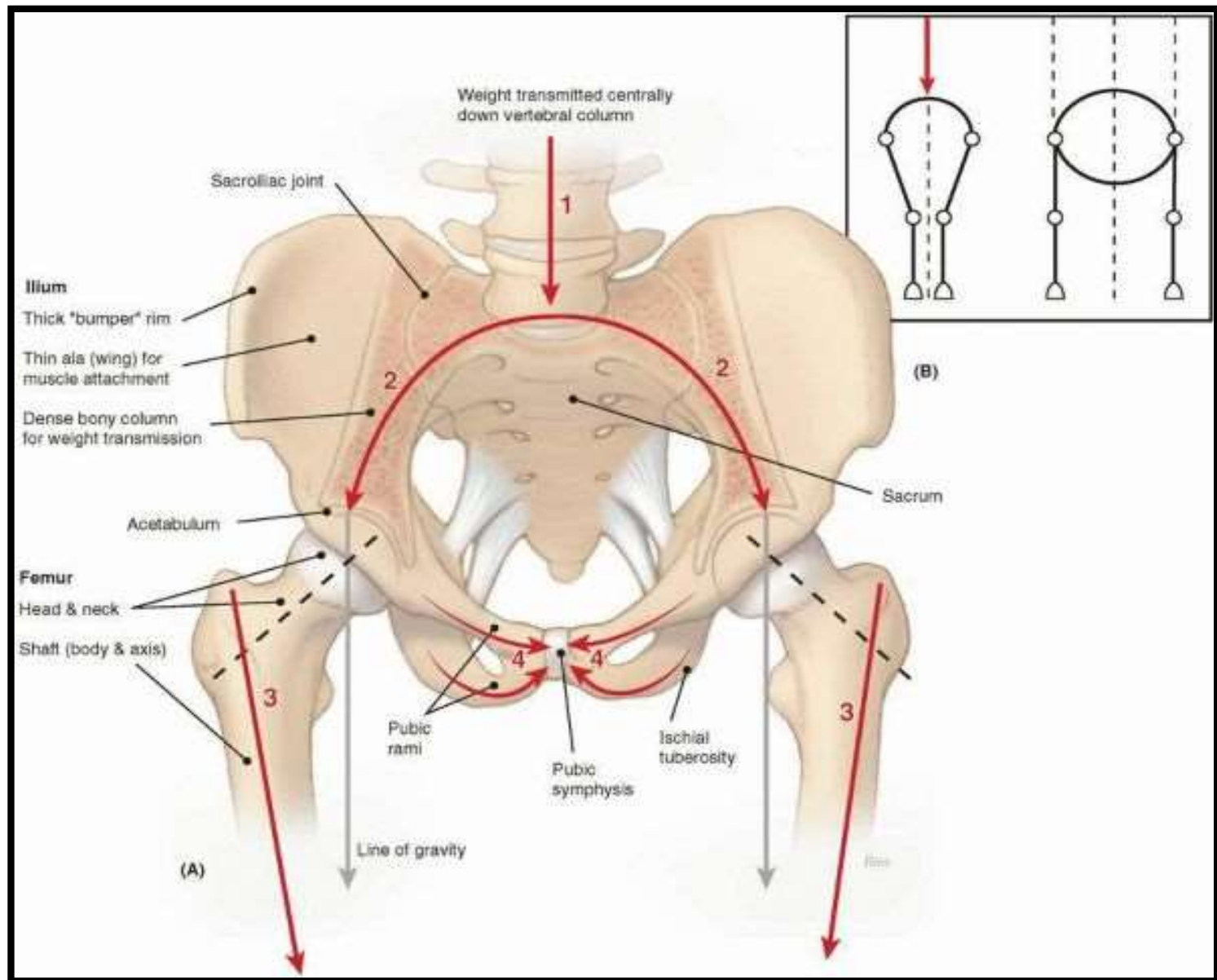
- **Support body wei**

- **Maintain balance**

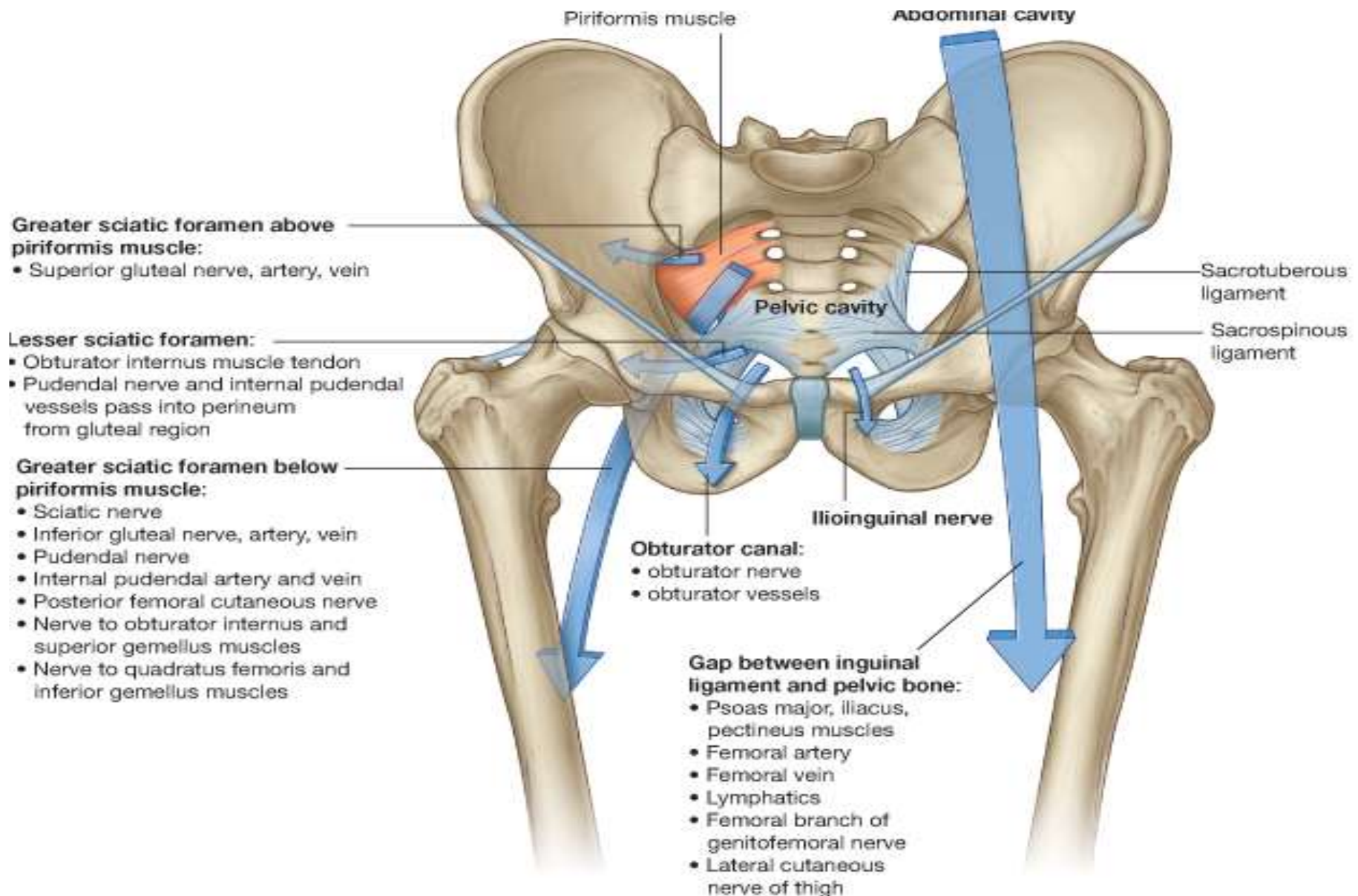
- **Connected to trunk by pelvic girdle**



Weight transfer from upper body to the lower ones



TRANSITION FROM ABDOMEN AND PELVIS TO LOWER LIMB

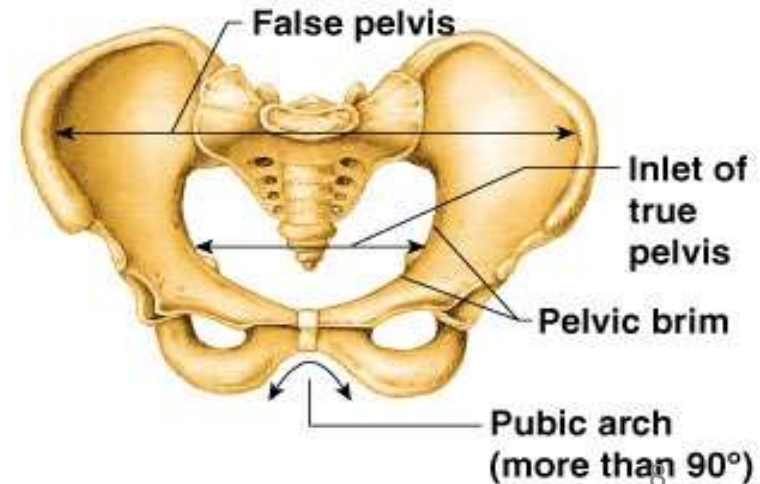
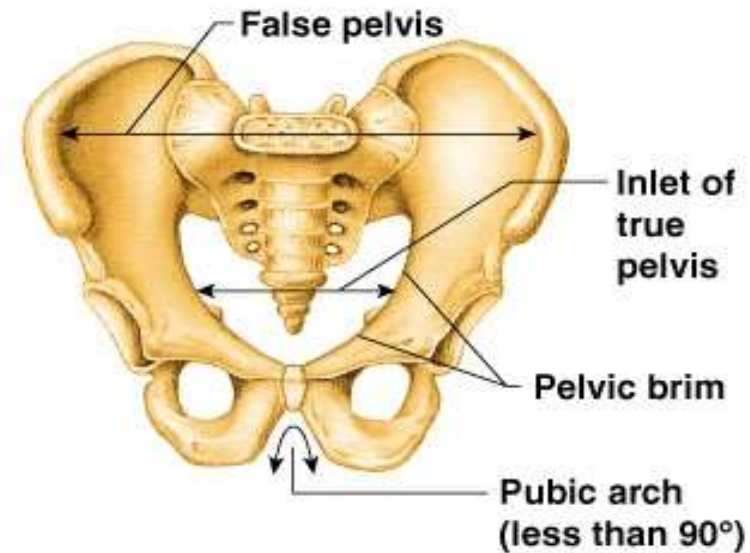
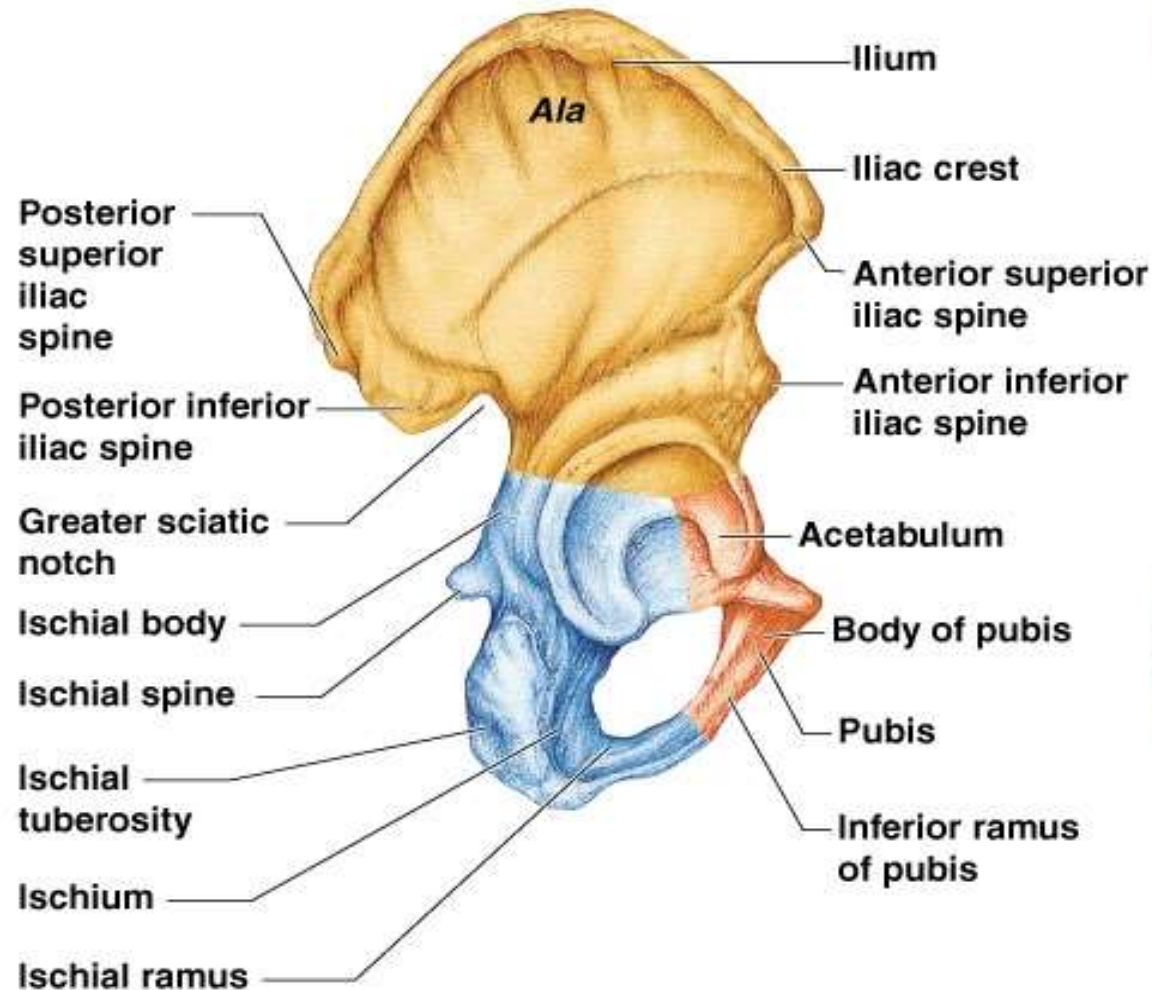




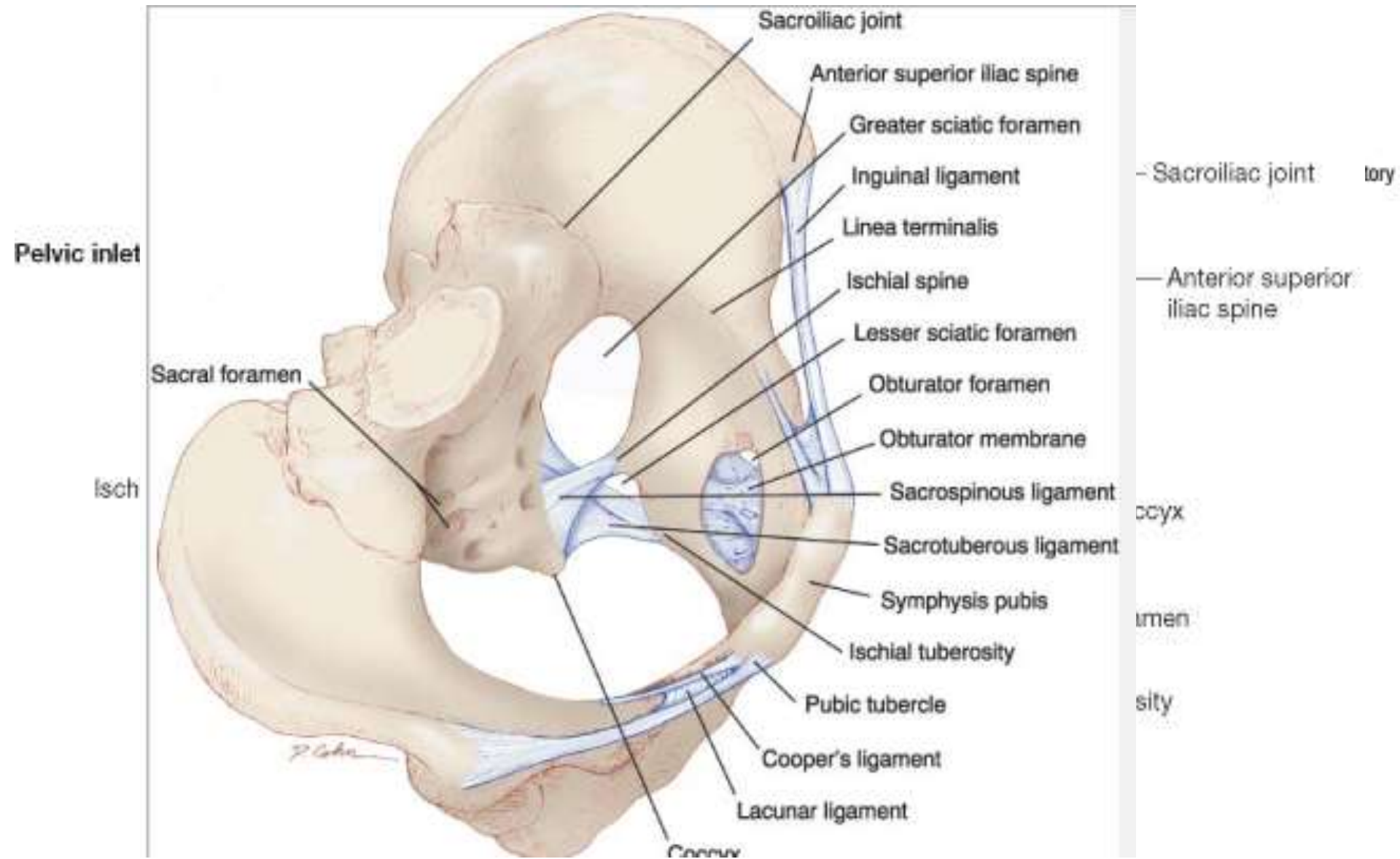
Bones of the lower limb



(Pelvic Girdle)



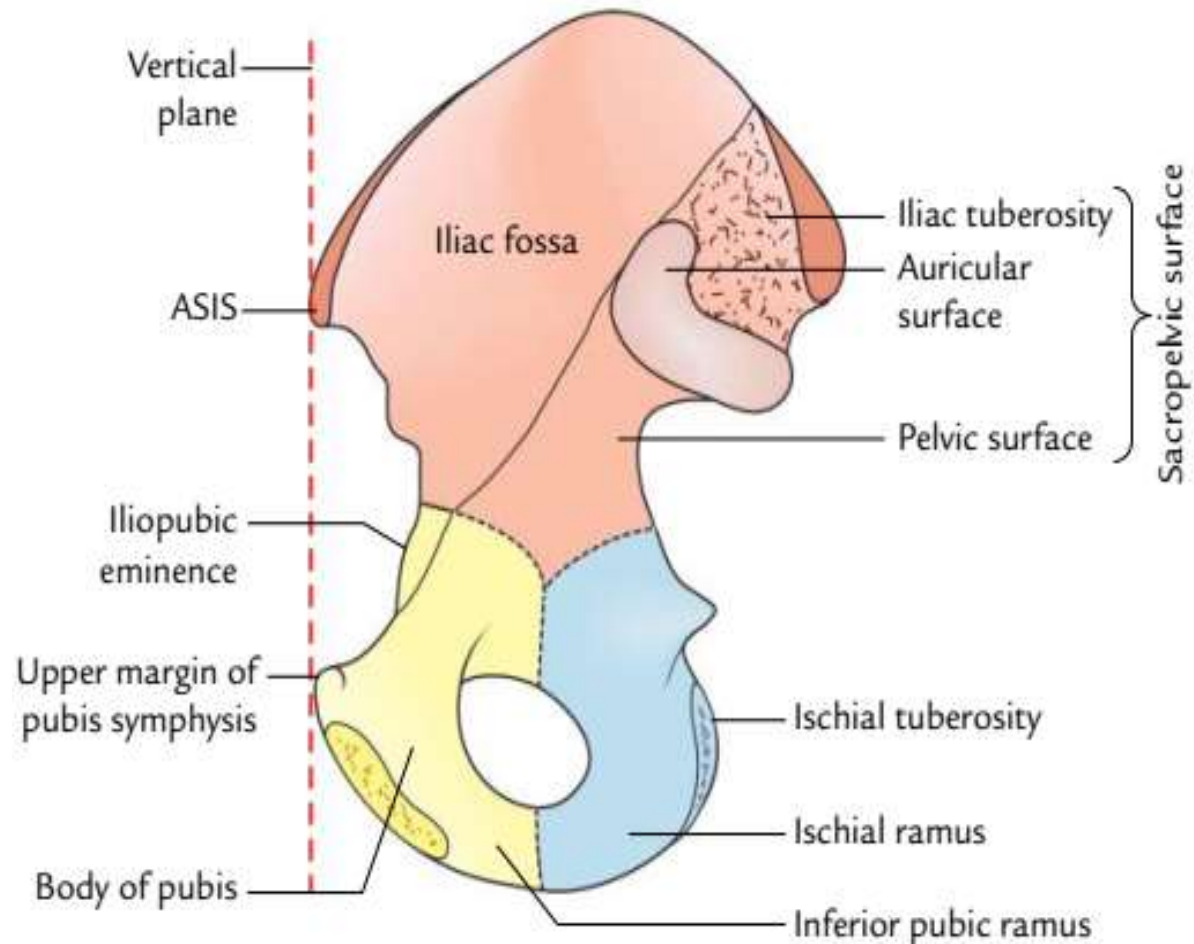
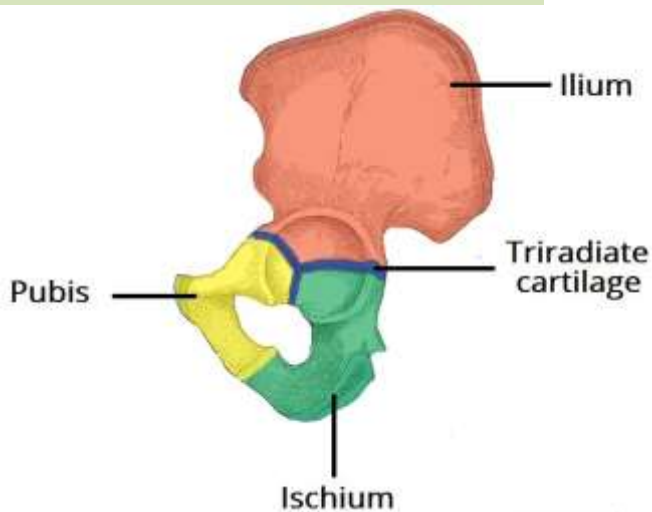
Bony pelvis & Joints



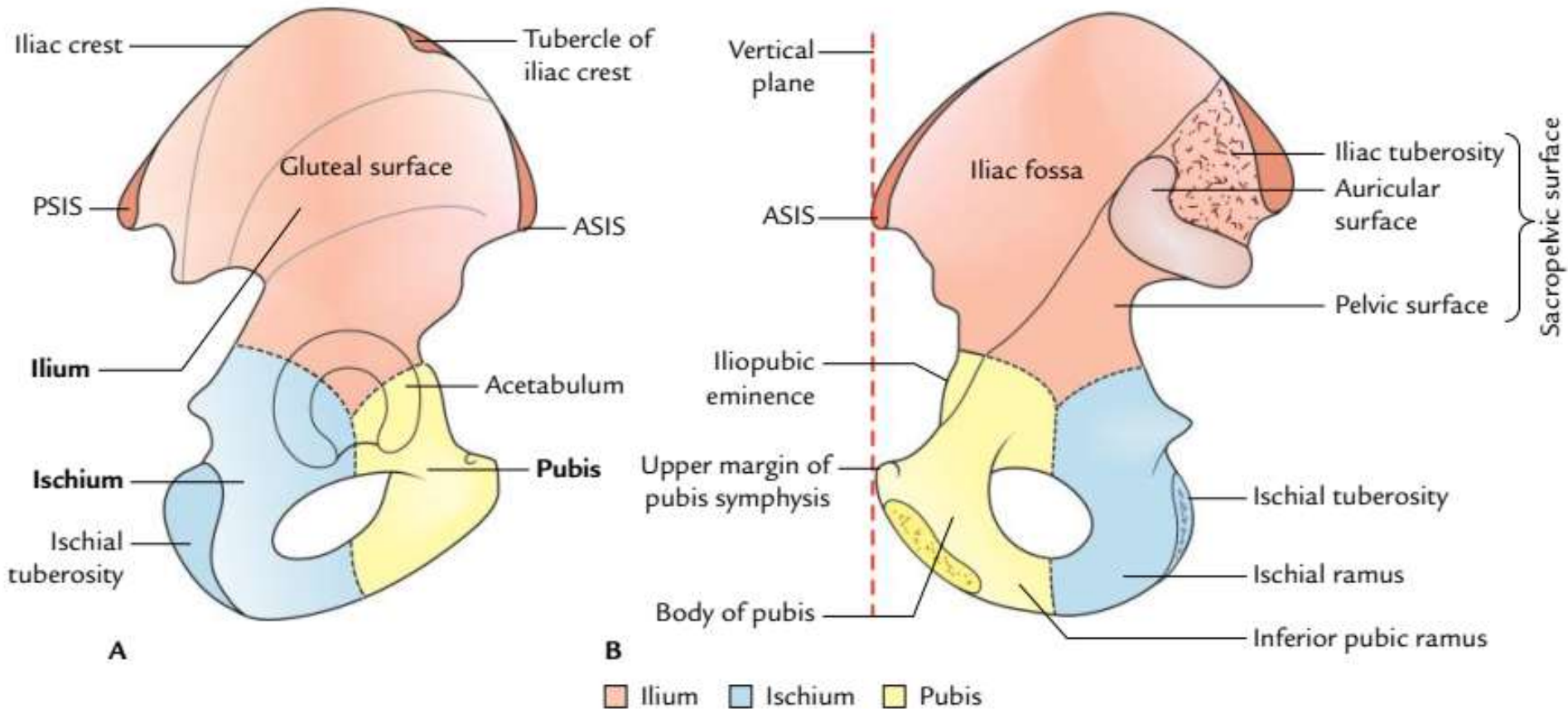
Bony pelvis

□ Four

- Two hip bone
- Sacrum
- Coccyx



Hip bone



Hip bone cont...

1. Ilium

- Part (body & ala)
- Unique features
- Surface

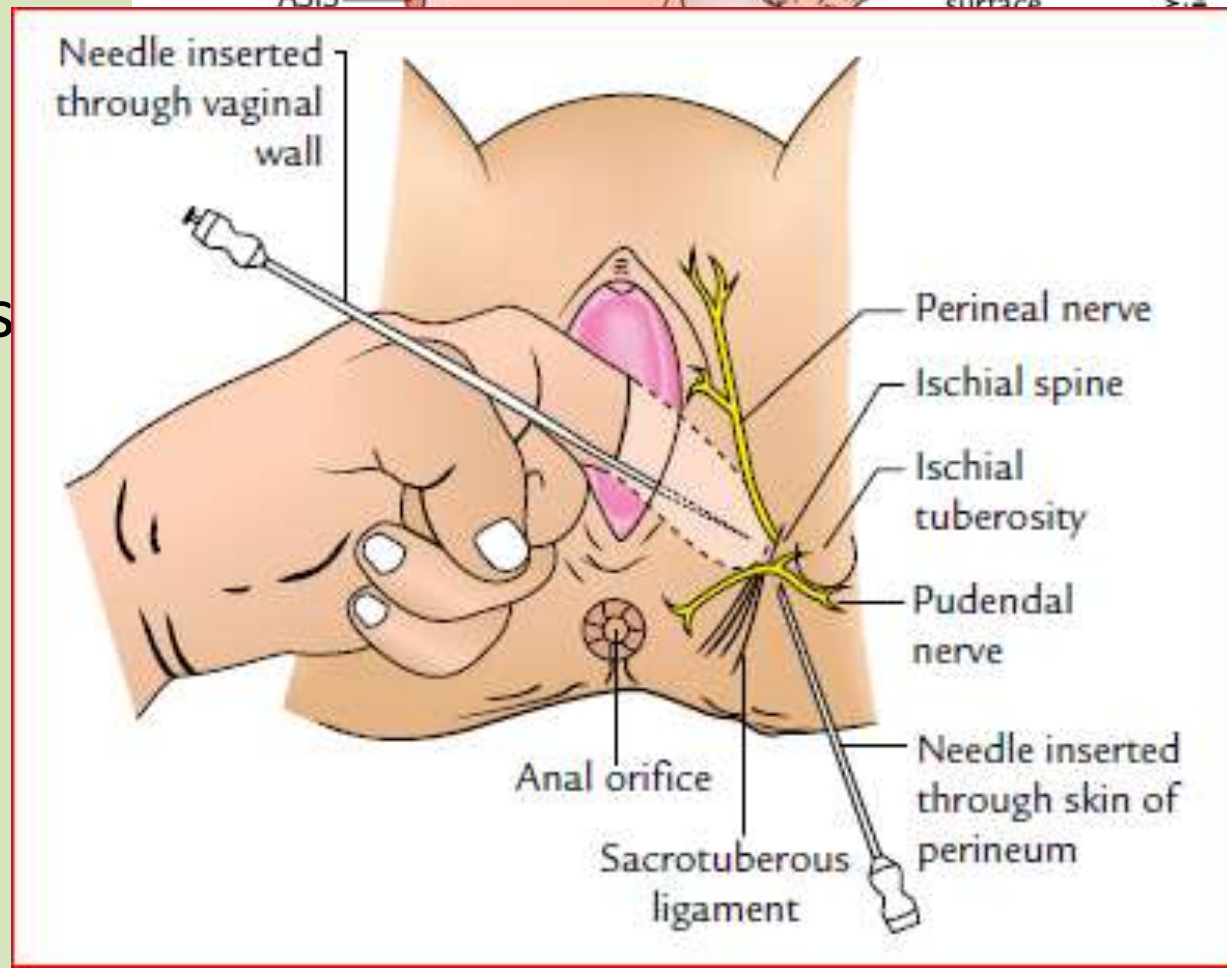


2. Ischium

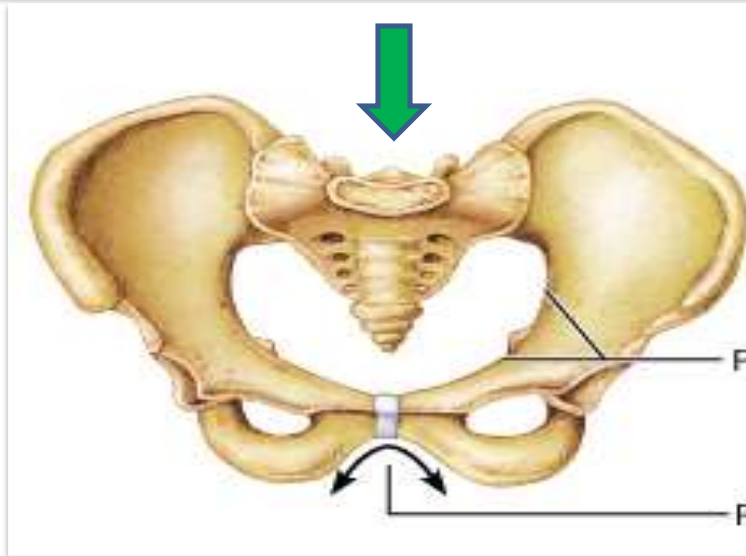
- ✓ Part (body & ramus)
- ✓ Features

3. Pubic bone

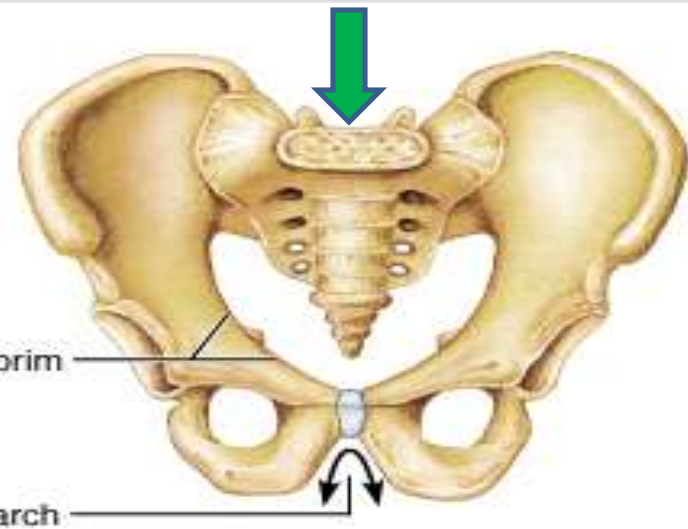
- Part
- Features



Female



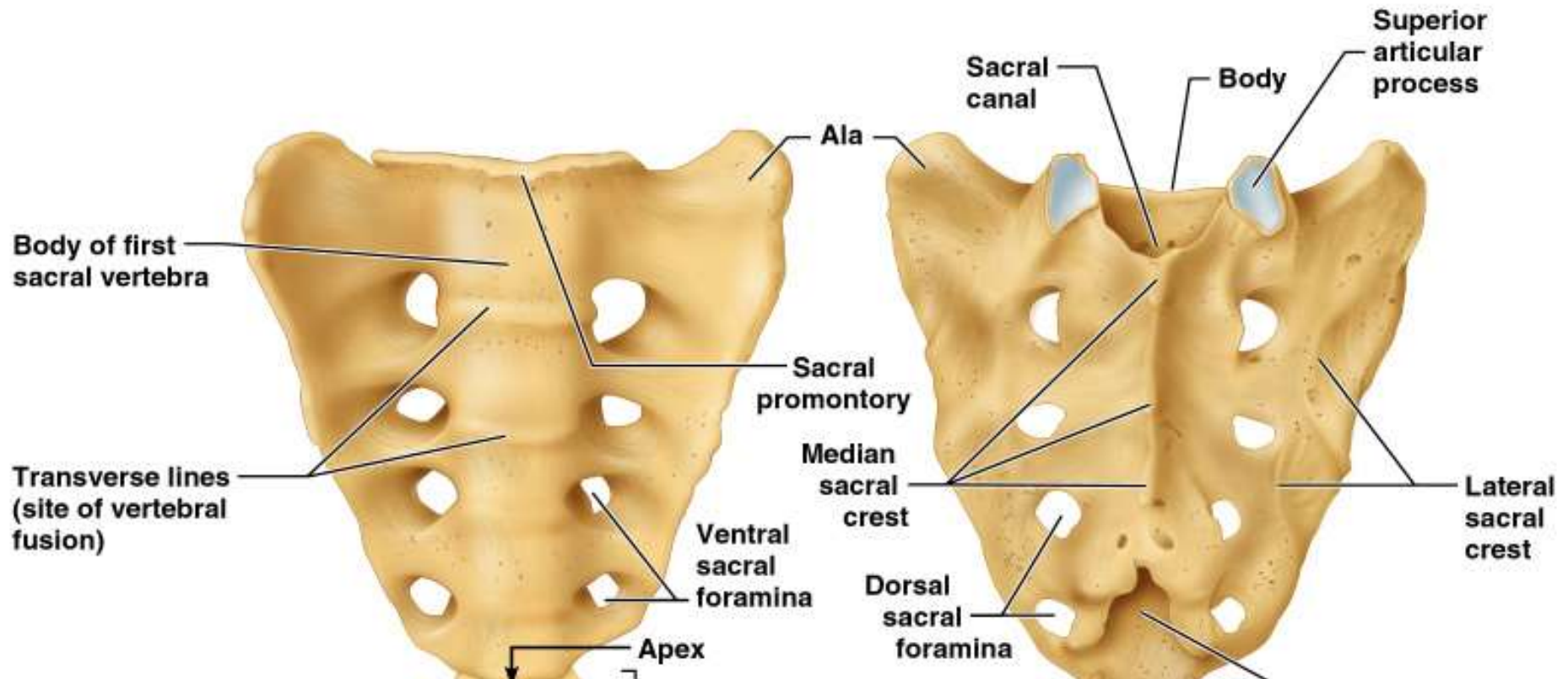
male



- Bones are lighter, thinner
- False pelvis is shallow
- Pelvic cavity is wide & shallow
- Pelvic inlet round/oval
- Pelvic outlet comparatively large
- Sub pubic angle large
- **Coccyx** more flexible, straighter
- **Ischial tuberosities more everted**

- Bones heavier, thicker
- False pelvis is deep
- Pelvic cavity is narrow & deep
- Pelvic inlet heart-shaped + smaller
- Pelvic outlet comparatively small
- Sub pubic angle more acute
- **Coccyx** less flexible, more curved
- **Ischial tuberosities longer, face more medially**

Sacrum & Coccyx



Clinical insight ...

Caudal or Sacral Analgesia (Fig. 54.12)

- (a) The anesthetic solution is injected into the extradural space in the sacral canal via the sacral hiatus. This method is used for painless labor in obstetric practice.

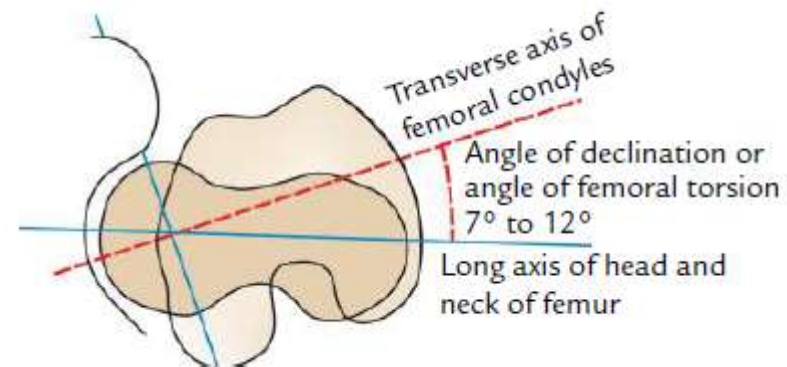
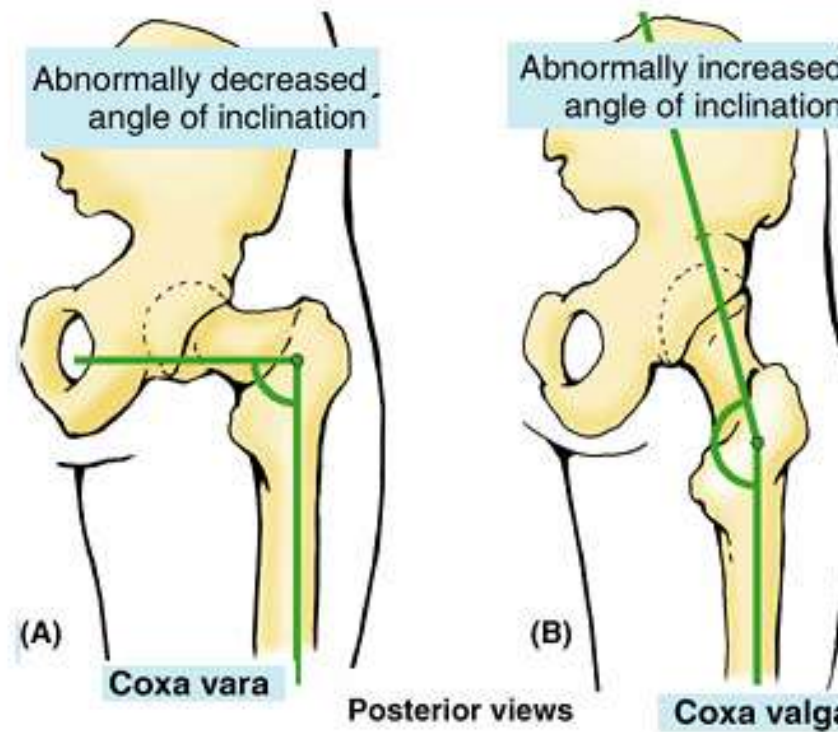
Femur

- The longest, heaviest & strongest bone in the body
- parts
- **Proximal part**
 - ❑ **Head**
 - Rounded
 - Covered with articular cartilage except a medial pit, **fovea** for ligament of head (ligamentum teres)
 - **Articulates with acetabulum**
 - ❑ **Neck**
 - Trapezoidal, narrow end supports head and broad base continuous with shaft



Angle of inclination & torsion of Femur

- **Angle of inclination:** It is an angle between the **long axes of neck and shaft of the femur**
- The normal neck–shaft angle is 125° in adults is 160° in children
- **It is less in female.**
- **Angle of torsion:** It is an angle between long axis of the **head and neck of femur**, and transverse axis of the femoral condyles.
- It measures about **7° in males** and **12° in females**



Anatomy of Femur cont..

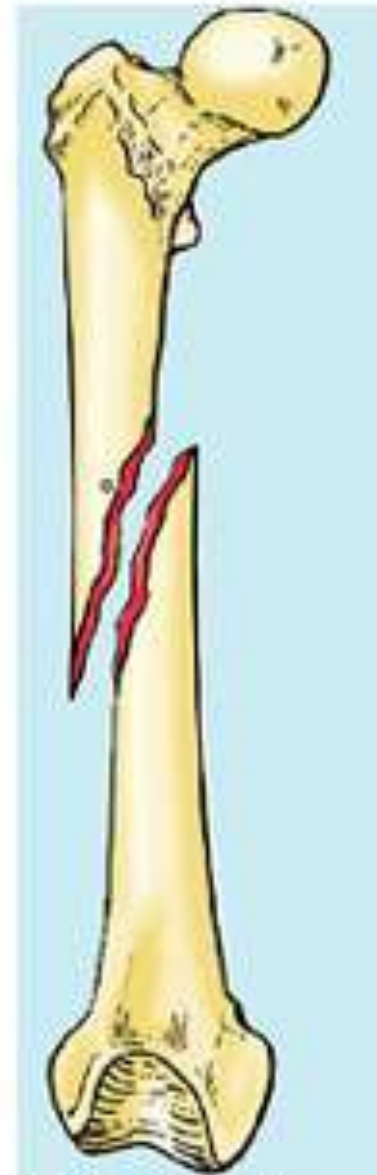
- ❑ Trochanter (Lesser & greater)
 - Intertrochanteric line
 - Intertrochanteric crest
- ❑ Shaft
- ❑ Distal end
 - Medial condyle
 - Lateral condyle
- Femoral fractures
- Neck is the most fracture site



A Transcervical fracture of femoral neck



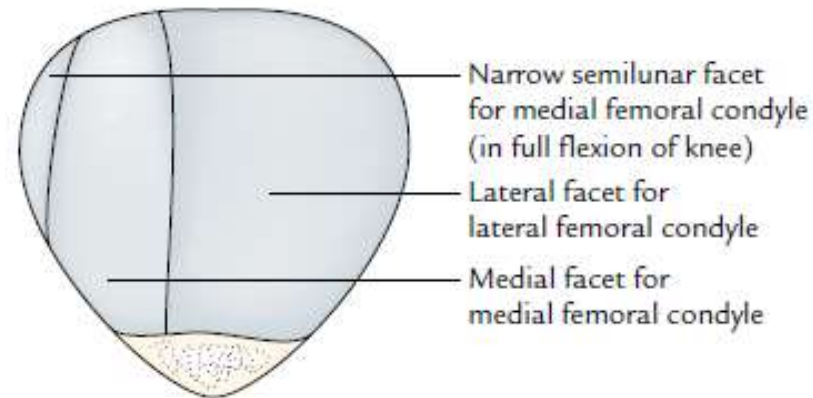
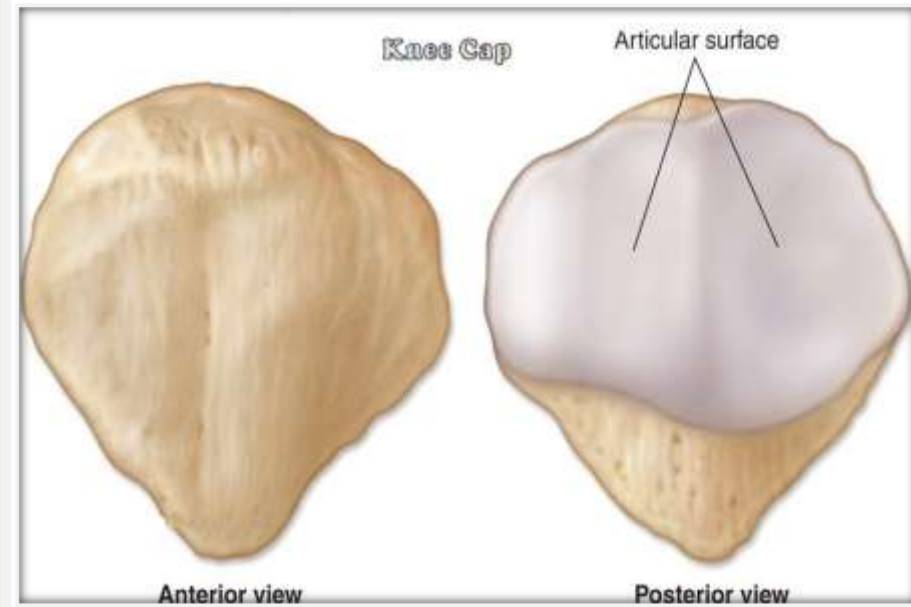
B Intertrochanteric fracture



Spiral fracture

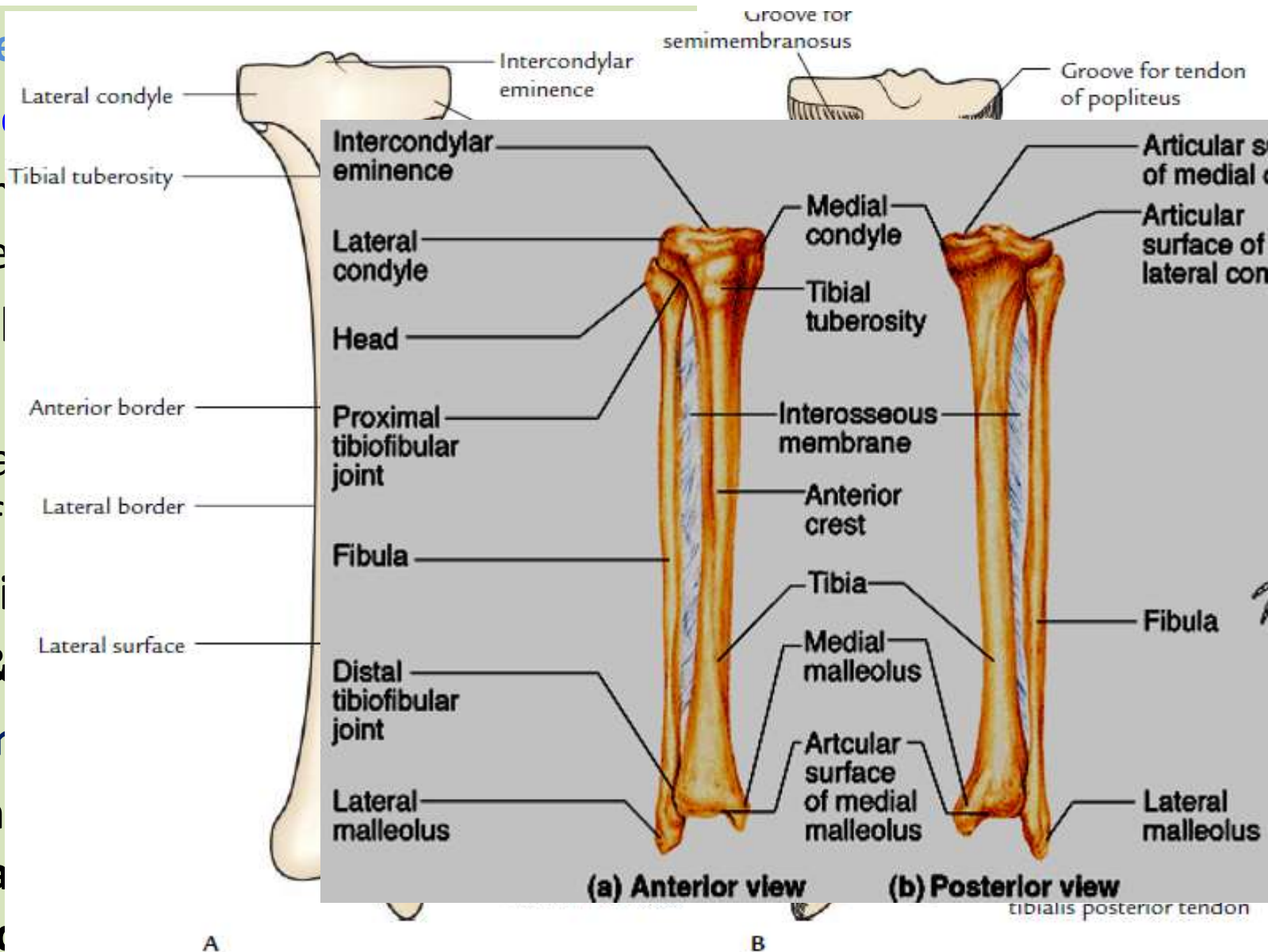
Patella (knee cap)

- Large sesamoid bone, formed intratendinously
- Located anterior to knee joint
- Triangular in shape
- Articulate with patellar surface of femur
- **Anterior surface** is convex
- **Posterior surface** is concave = 2 (medial & lateral)
- **Base** – superior border
- **Lateral(larger)** and **medial border** converge to form **apex**



Tibia

- Large, weight-bearing bone
- Articulates with the femur at the knee joint
 - ❖ Proximal end = knee joint
 - ❖ Distal end = ankle joint
 - ❖ Lateral surface with fibula
- Has medial condyle
- Medial & lateral malleoli
- Shaft is not as thick as femur
- Inferior end is most free



Fibula

- Long, slender bone on the lateral side of leg
- Serves mainly for **muscle attachment**
- **It doesn't bear body weight)**
- has head and neck
- Distal end enlarges to form **lateral malleolus**
- Fractures commonly occur just **proximal to lateral malleolus** and often are associated with dislocation of ankle joint
- At **neck of the fibula** is common site of fracture
- Fibula is a common source of **for bone grafting**



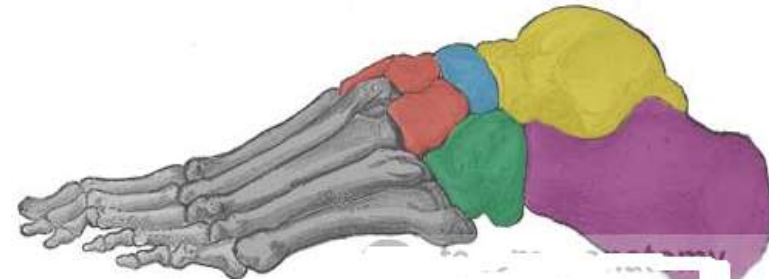
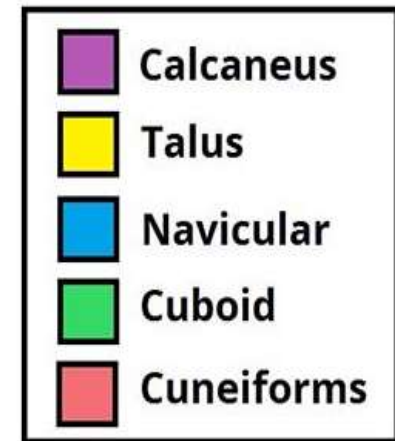
Bones of Foot

□ Tarsal Bones

- **7 bones** which form the ankle joint; **talus, calcaneus, navicular, 3 cuneiforms & cuboid**
- The Only **talus** articulate with leg bones

□ Calcaneus

- Heel bone
- Largest and strongest bone in the foot
- Articulate with talus and cuboid



Bones of Foot...

❑ Talus

- Rests on anterior 2/3 of calcaneus
- Superior surface, trochlea of talus, bears weight of the **body** and articulates with two malleoli

❑ Navicular

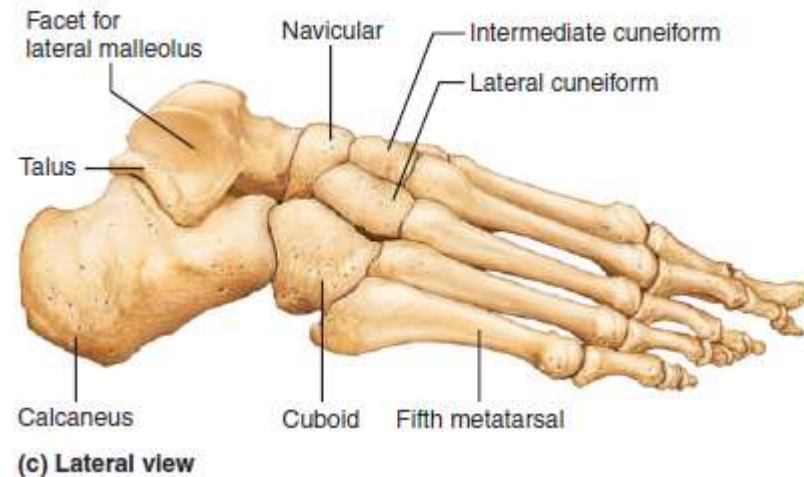
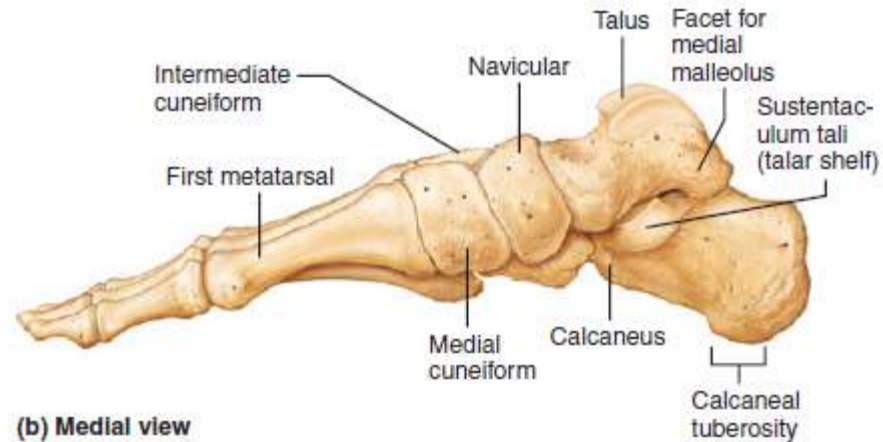
- Flattened, boat shaped
- Located between talar head and cuneiforms

❑ Cuboid

- Most lateral

❑ Cuneiforms: medial, intermediate and lateral

- Wedge shaped
- Each articulate with navicular posteriorly and metatarsal anteriorly



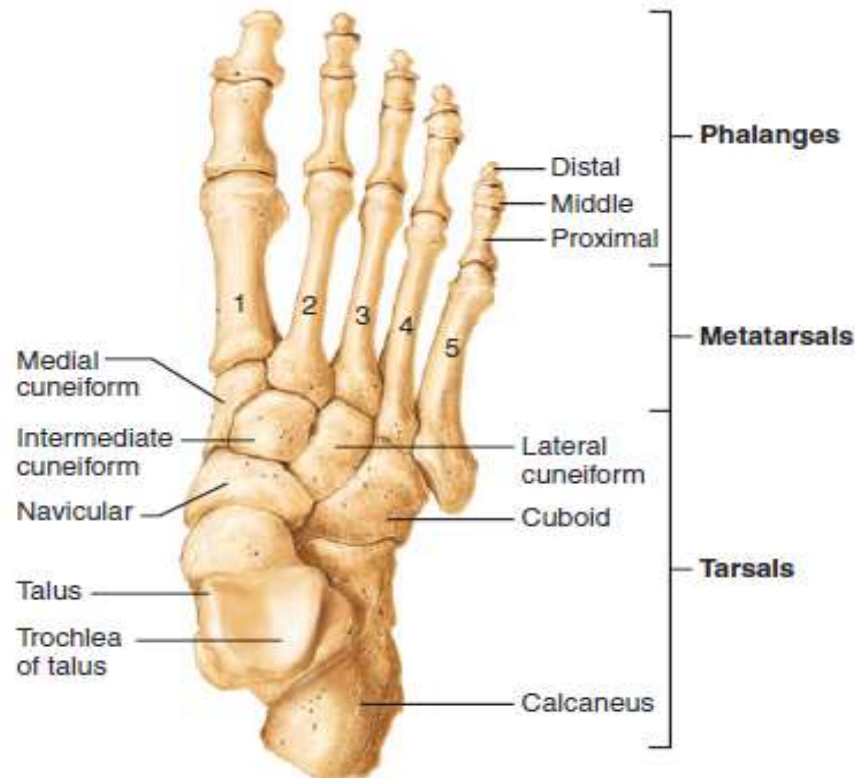
Bones of Foot...

☐ Metatarsal Bones

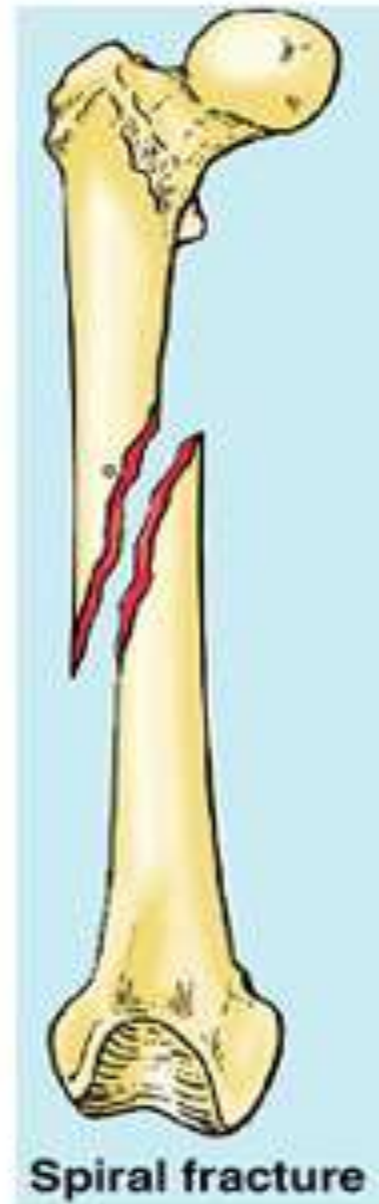
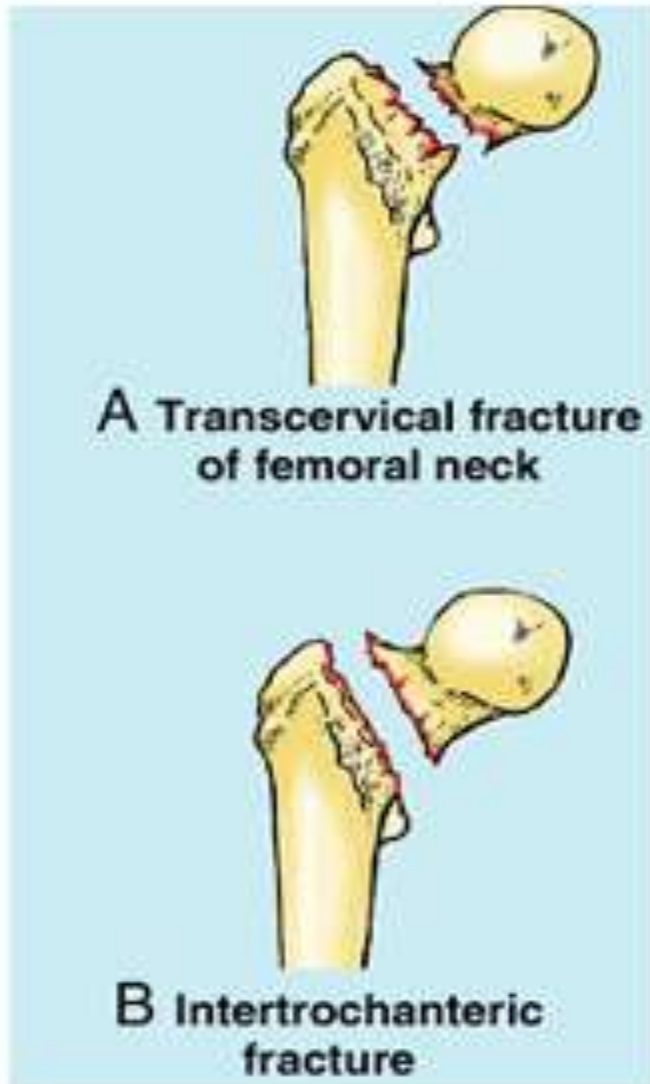
- 5 bones which form middle part of the foot
- Each bone has base, body & head
- Bases articulate with cuneiform and cuboid bones
- Heads articulate with proximal phalanges

☐ Phalanges

- 14 bones
- Lateral four digits have proximal, middle & distal phalanx
- Great toe (hallux) has only proximal & distal phalanx
- Each phalanx has base, body & head

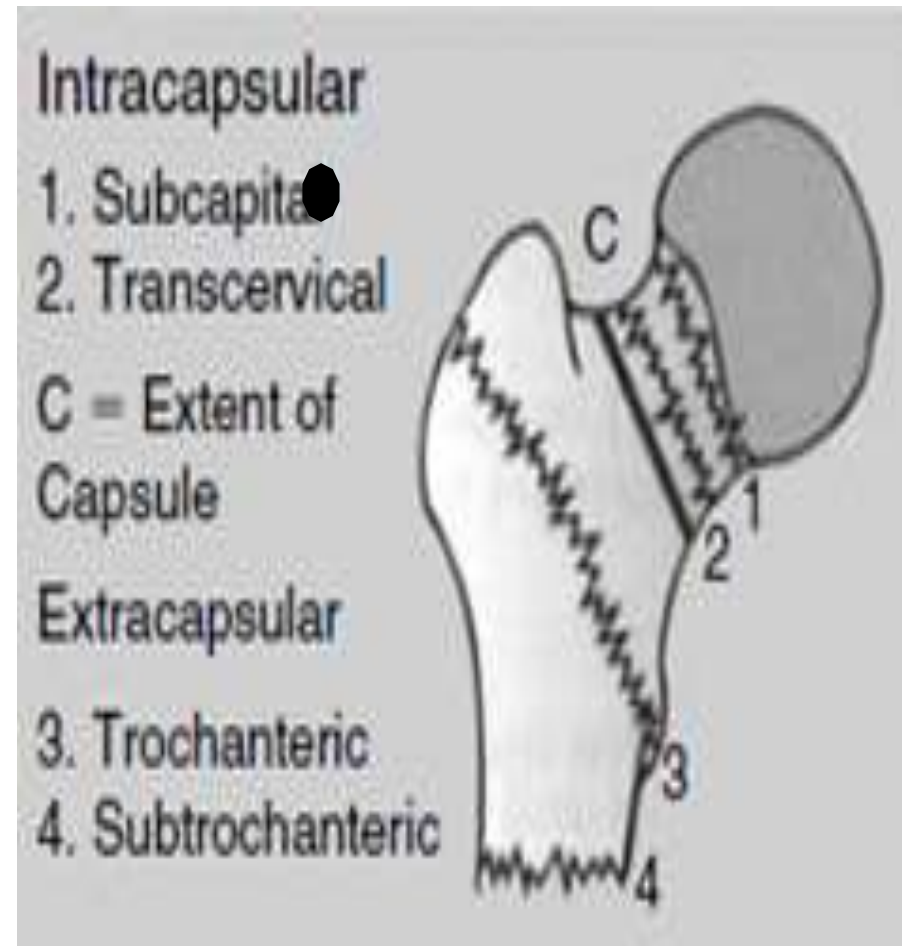


Bone Fracture



Fracture of the Femur

- The upper end of the femur
- Neck fracture:
 - immediately beneath the head
 - near its midpoint
 - adjacent to the trochanters,
 - along or just below the trochanters.
- Fractures of femoral neck
 - interrupt blood supply from the diaphysis.



Fracture of the Femur

Subcapital neck fracture



Transcervical neck fracture



Intertrochanteric fracture



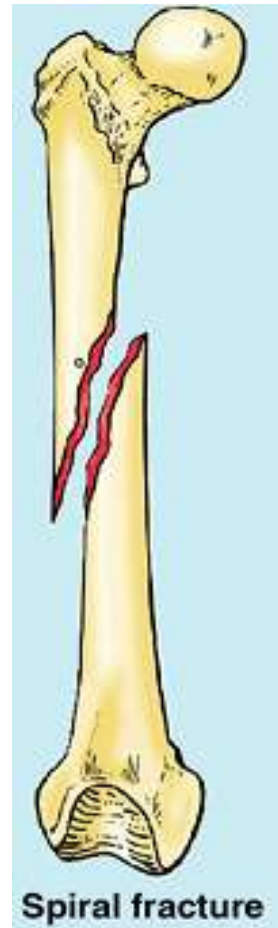
Subtrochanteric fracture



Fracture of the greater trochanter

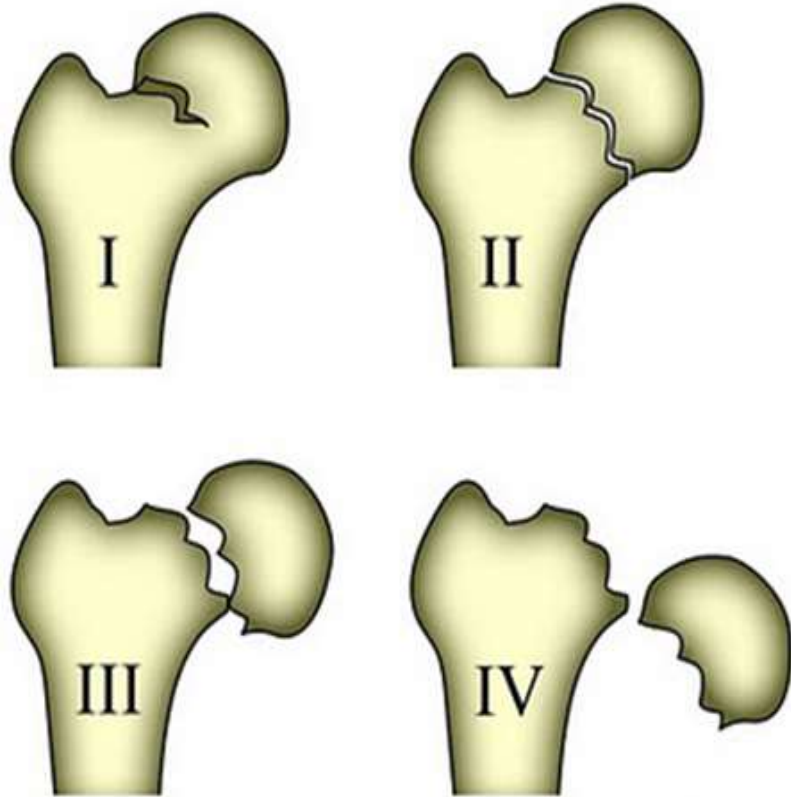


Fracture of the lesser trochanter

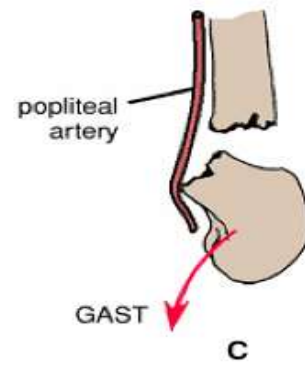
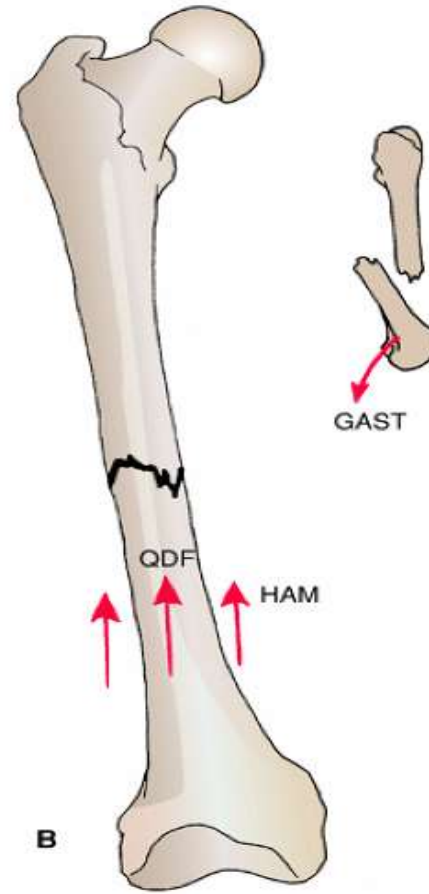
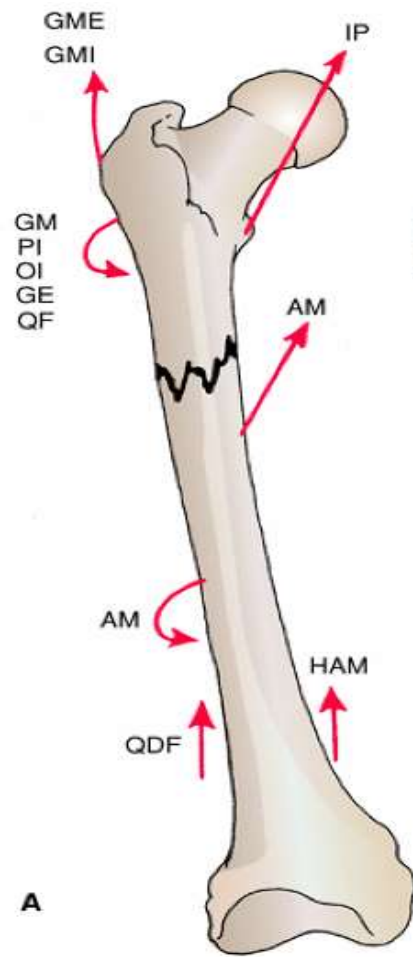


Spiral fracture

Garden classification

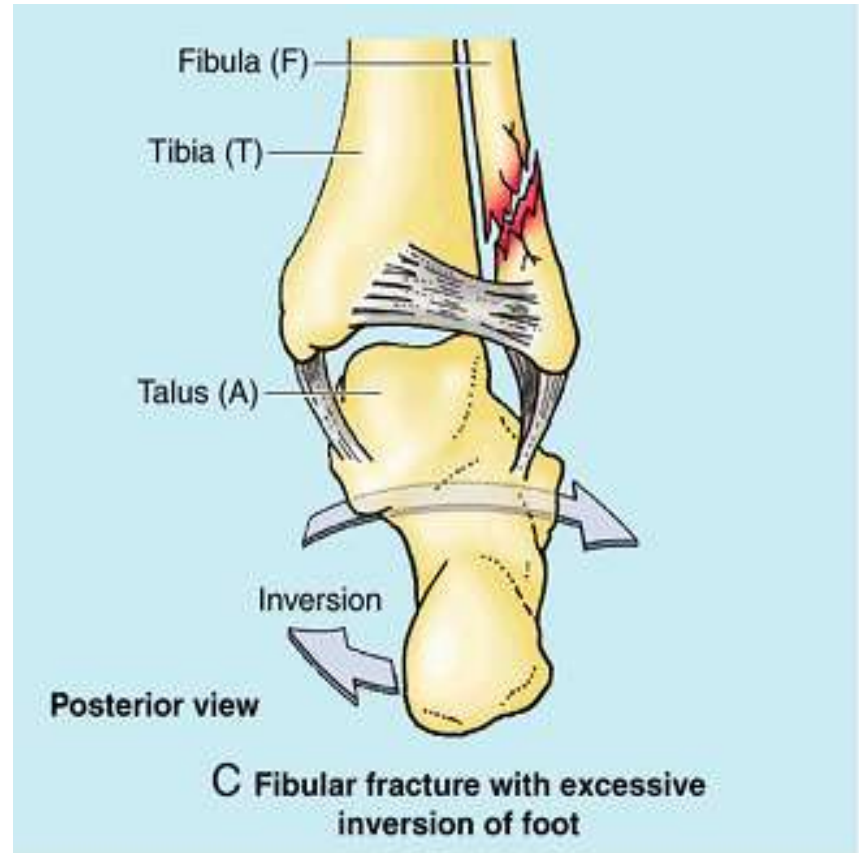


- . Incomplete impacted
- . Complete un-displaced
- . Displaced capsule intact
- . Displaced



Dupuytren's Fracture /Pott's fracture

- Is a fracture of the **lower end of the fibula**.
 - accompanied by a fracture of the medial malleolus or
 - rupture of the **deltoid ligament**.
- Caused by forced eversion of the foot.

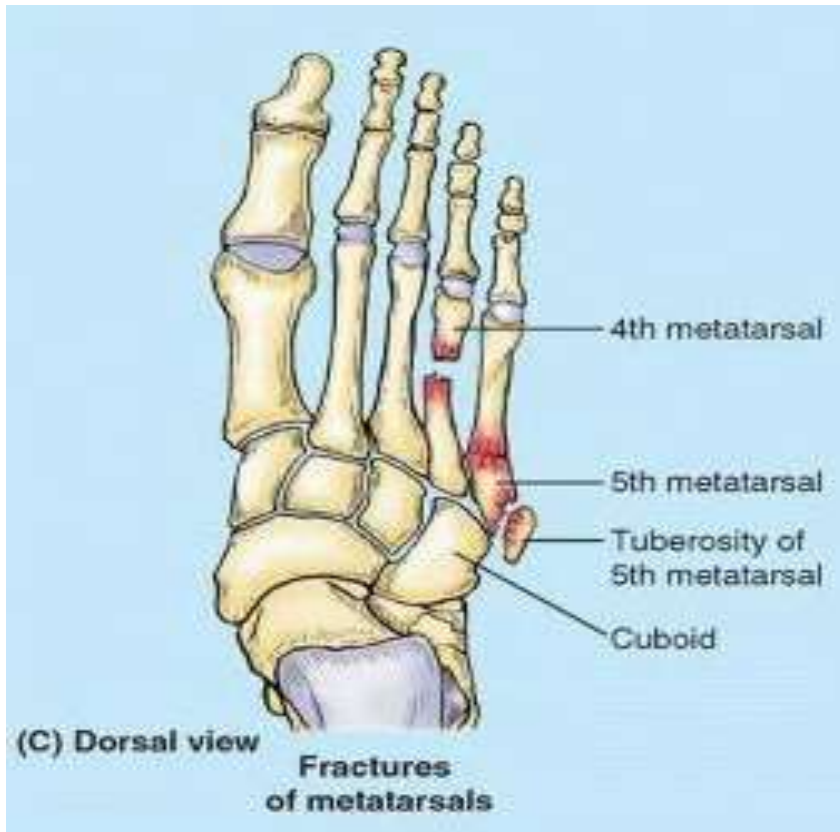


Pilon Fracture:

- distal metaphysis of the tibia

March Fracture / Stress Fracture:

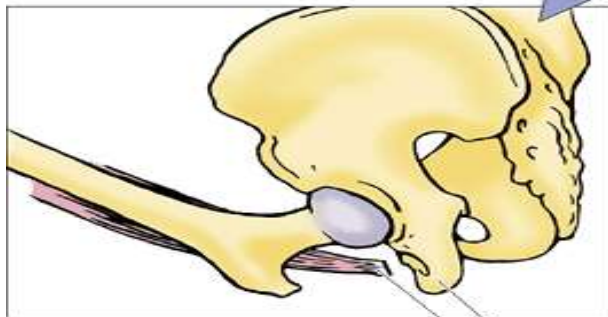
- Is a fatigue fracture of one of the metatarsals



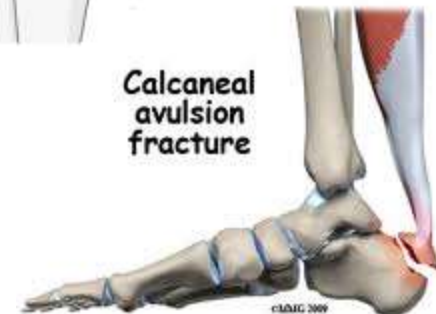
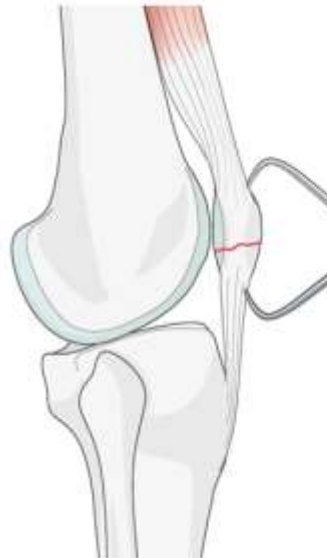
- **Flat foot (Pes Planus)**
 - weakened ligaments
- **Claw foot (Pes Cavus)**
 - medial arch is too elevated



Avulsion fractures



Ischial tuberosity
Hamstrings tendon
(torn and avulsed
from tuberosity)

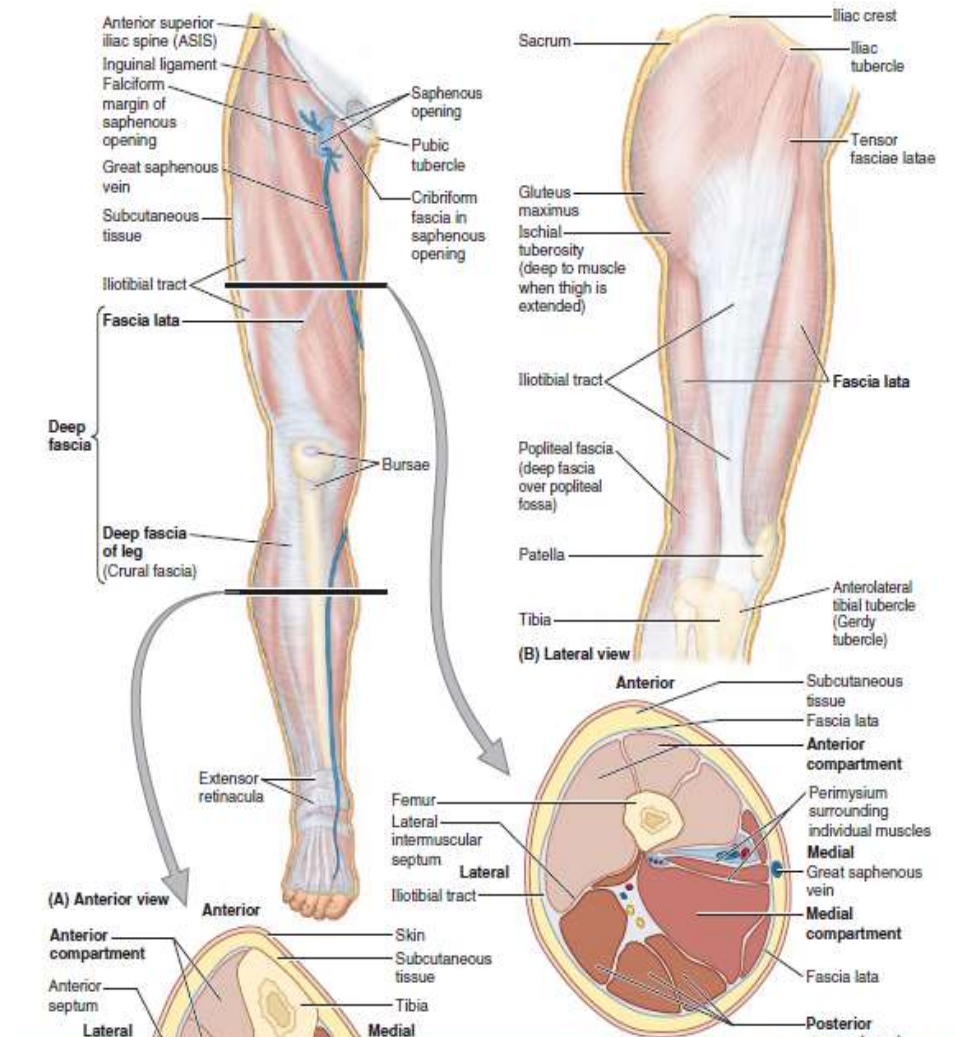


SUPERFICIAL STRUCTURE OF LOWER LIMB

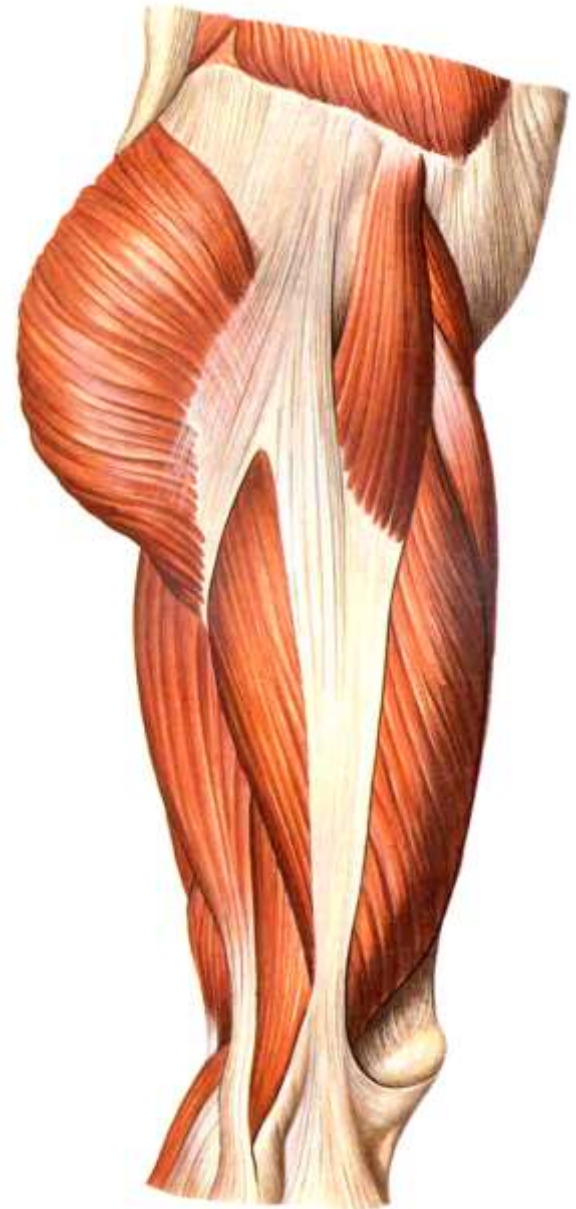
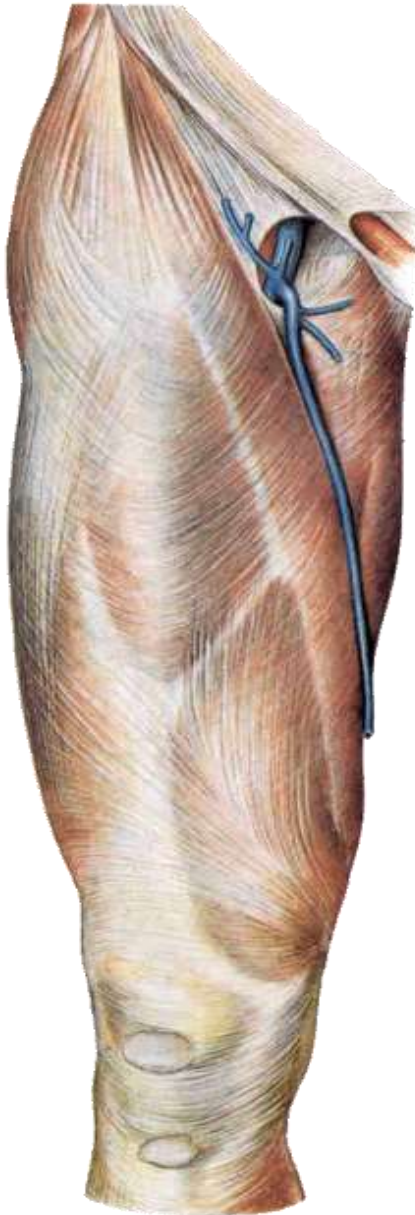
SUPERFICIAL STRUCTURE OF LOWER LIMB

□ Fascia of lower limb

- Subcutaneous tissue (**superficial fascia**) – lie deep to the skin
- **Deep fascia**- strong investing the limb like **elasting stocking**
- **Fascia lata**-deep fascia of thigh
- **Crural fascia**- deep fascia of leg
- Intermuscular septa(anterior & posterior)



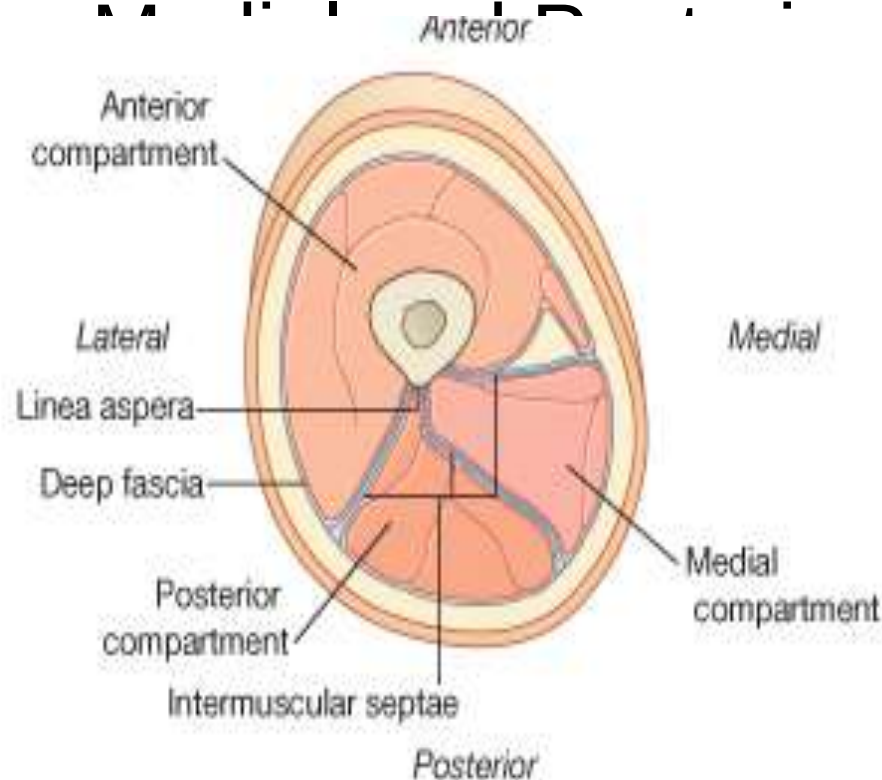
Fascia lata & Iliotibial tract

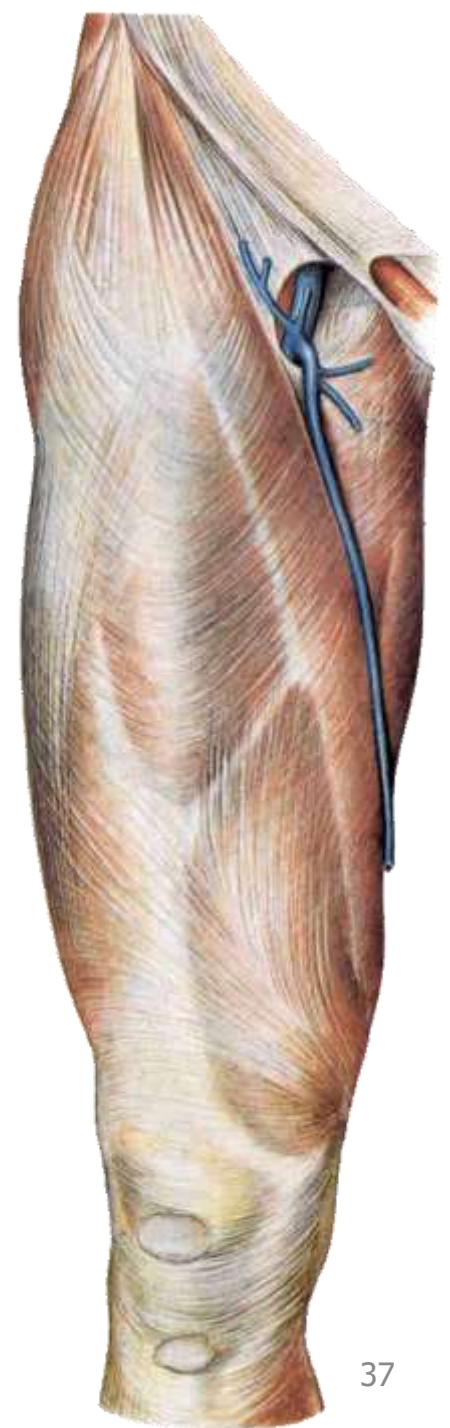
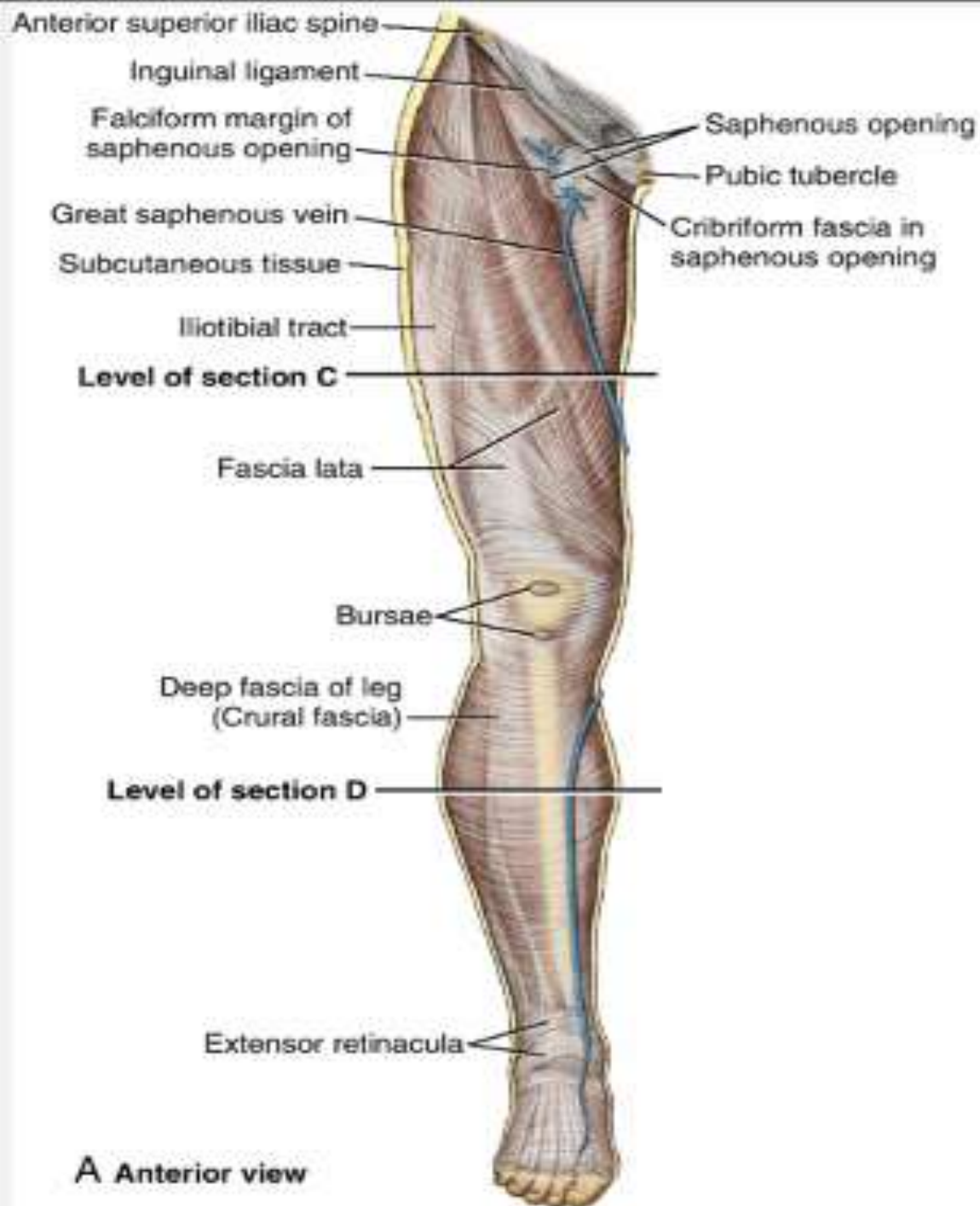


Fascial compartments

- **Thee** intermuscular septa **arise from** fascia lata **and attach to** linea aspera.
- **Thigh muscles are separated into 3 compartments:**

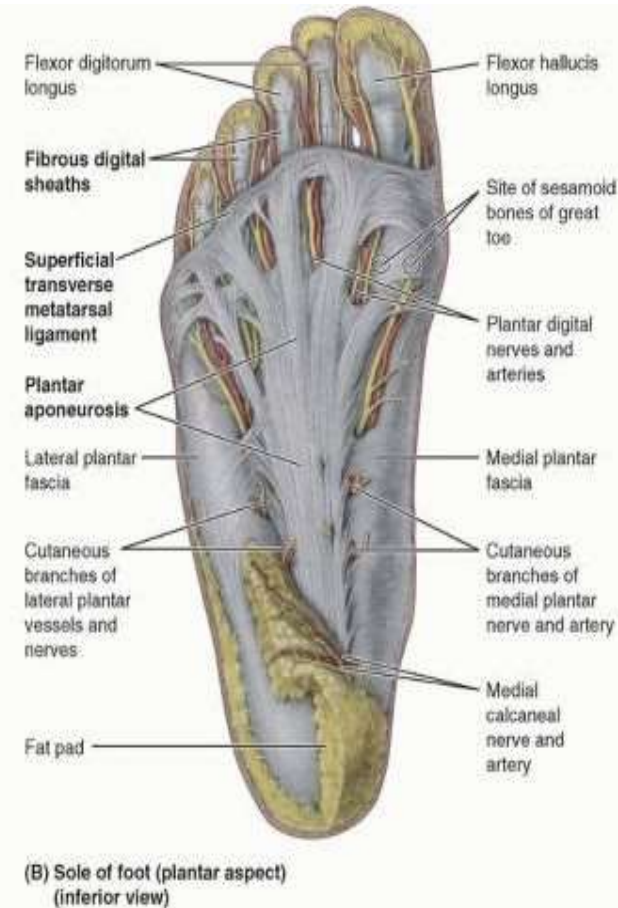
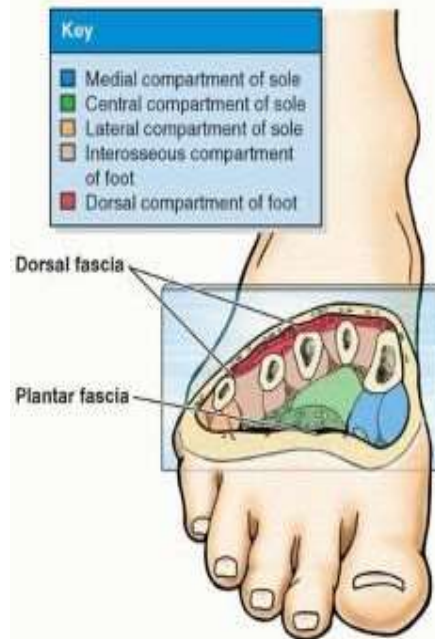
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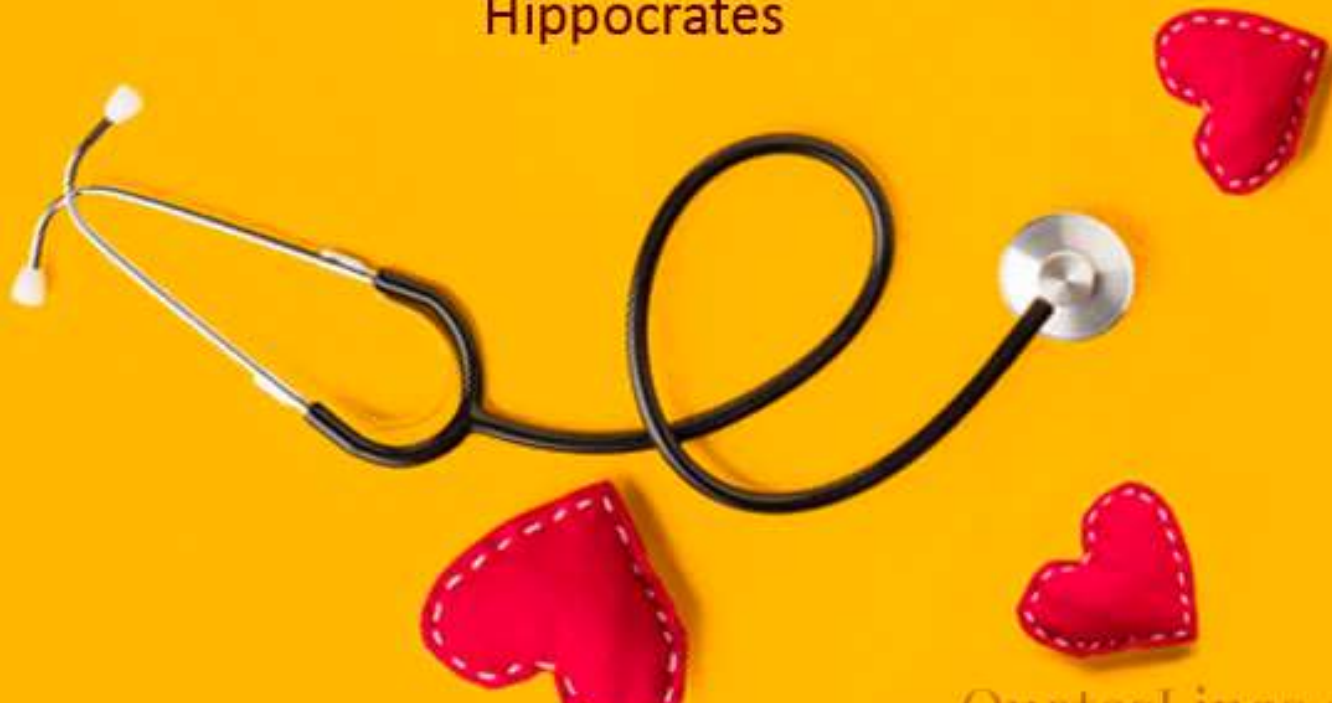
Plantar fascia

- Attached :
 - **posteriorly to the** medial process of the calcaneus
 - **anteriorly to the heads of metatarsals** by 5 bands
- Thin **in the** medial **and** lateral portions
- Thick **centrally** as the plantar aponeurosis.



**“Wherever the art of
medicine is loved, there is
also a love of humanity.,,”**

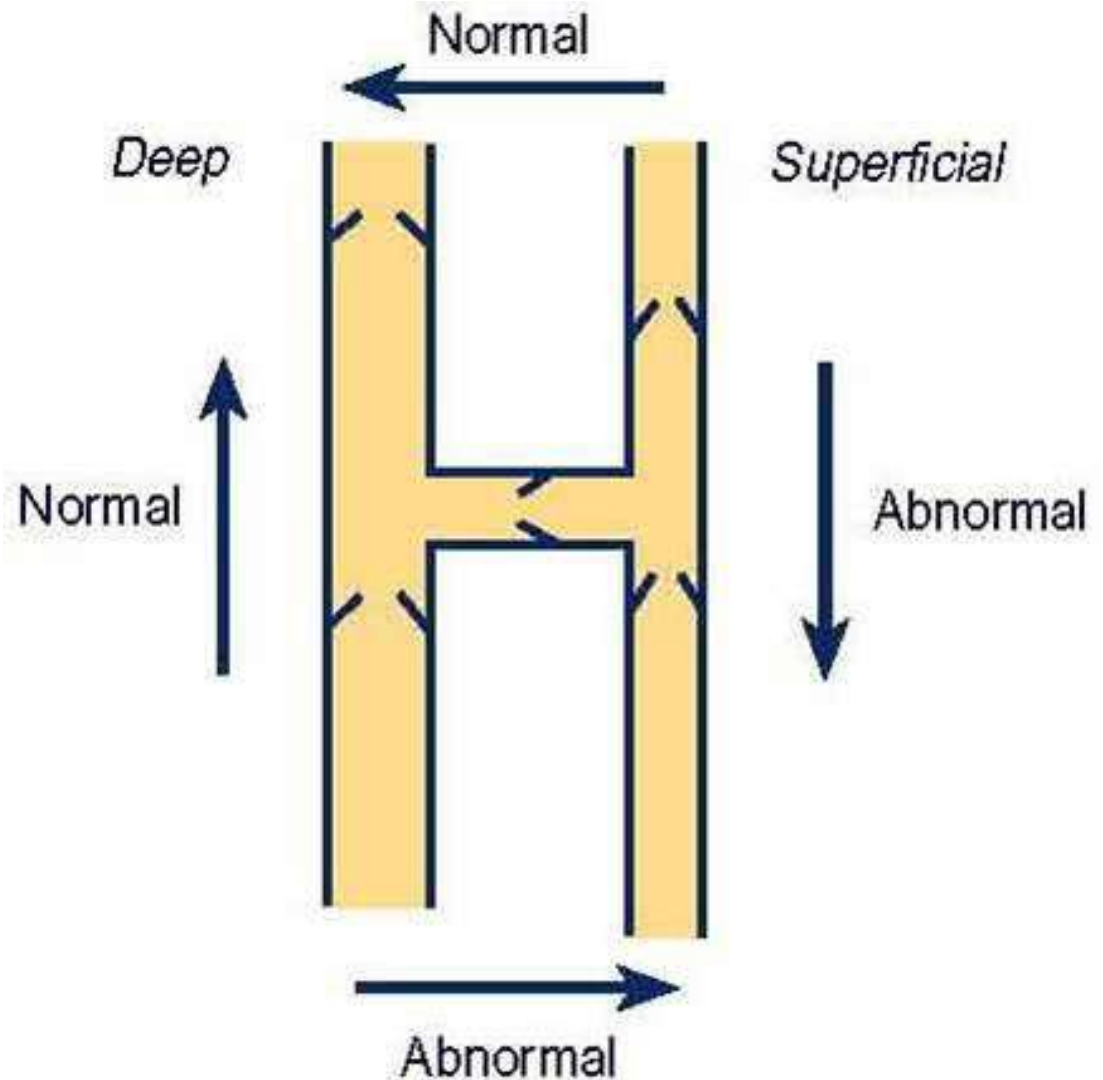
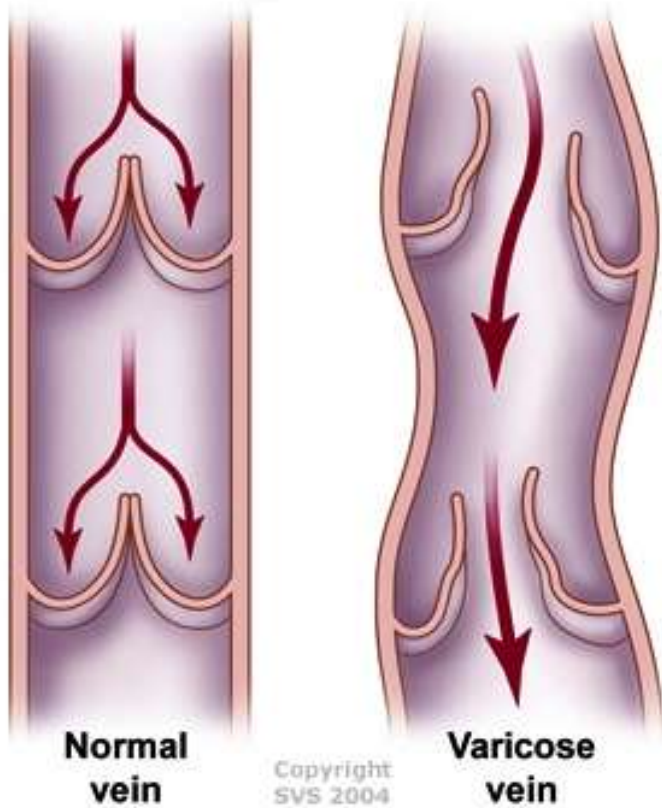
Hippocrates



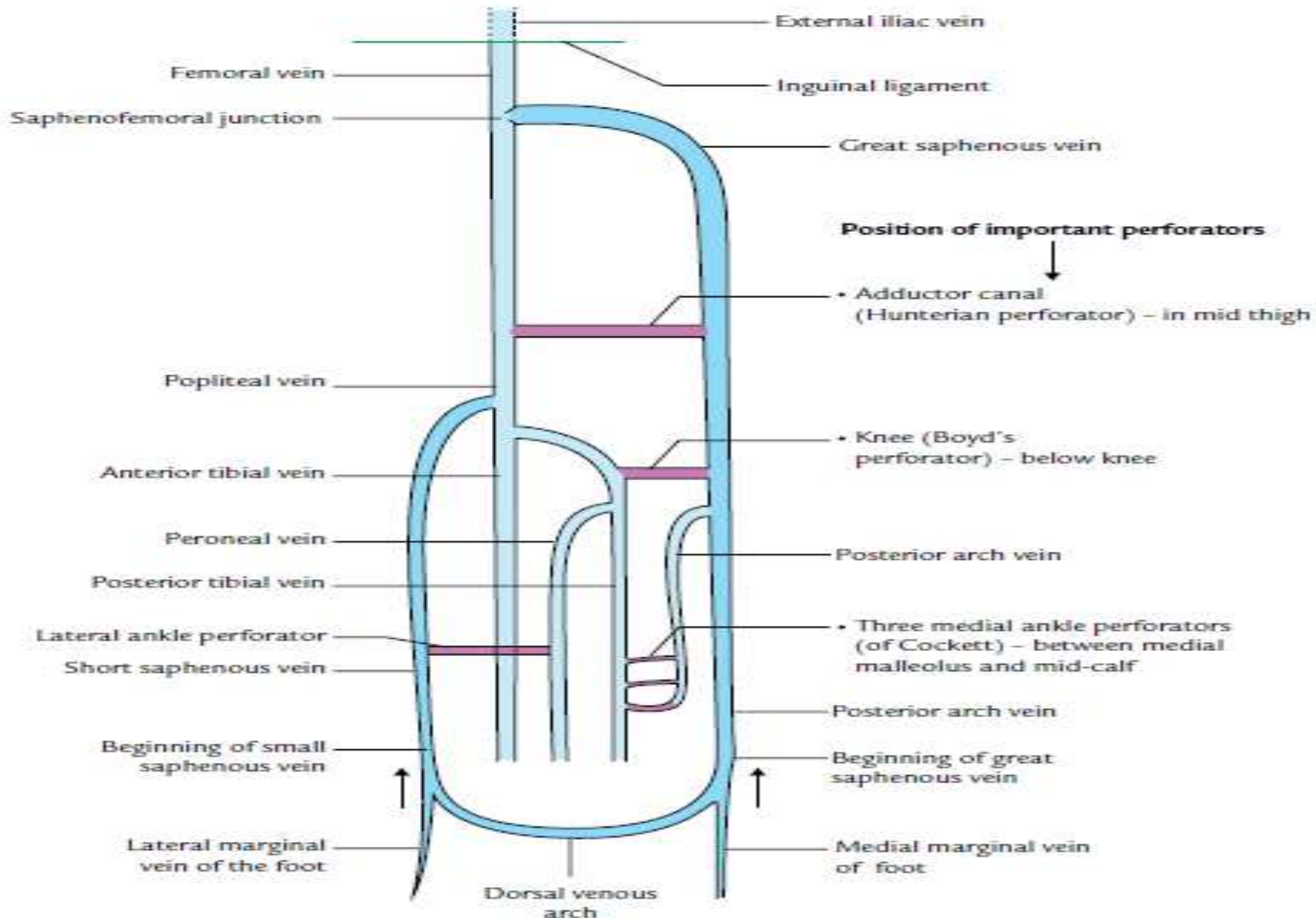
The background of the slide is a photograph showing the lower legs and feet of several people standing in shallow water, likely a swimming pool. The water is clear and blue, and the people are wearing blue swimwear. The image is slightly blurred, giving it a soft, artistic feel.

Venous Drainage Of Lower Limb

How do veins work?



SUPERFICIAL VEINS OF LOWER LIMB



Superficial veins of the lower limb

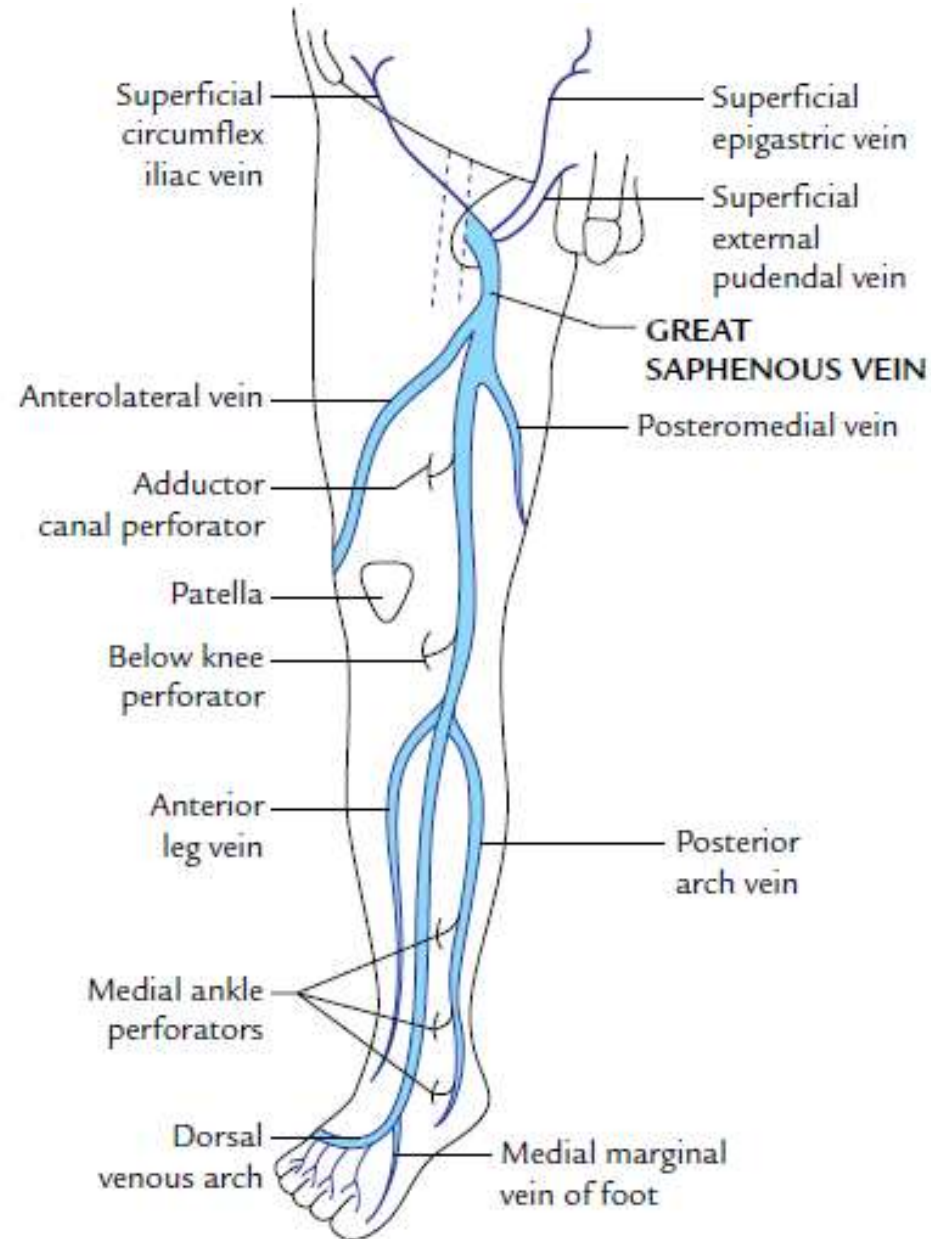
□ Great saphenous vein(long)

• Origin

- It arises from the union of dorsal digital vein of great toe and dorsal venous arch on the dorsum of the foot

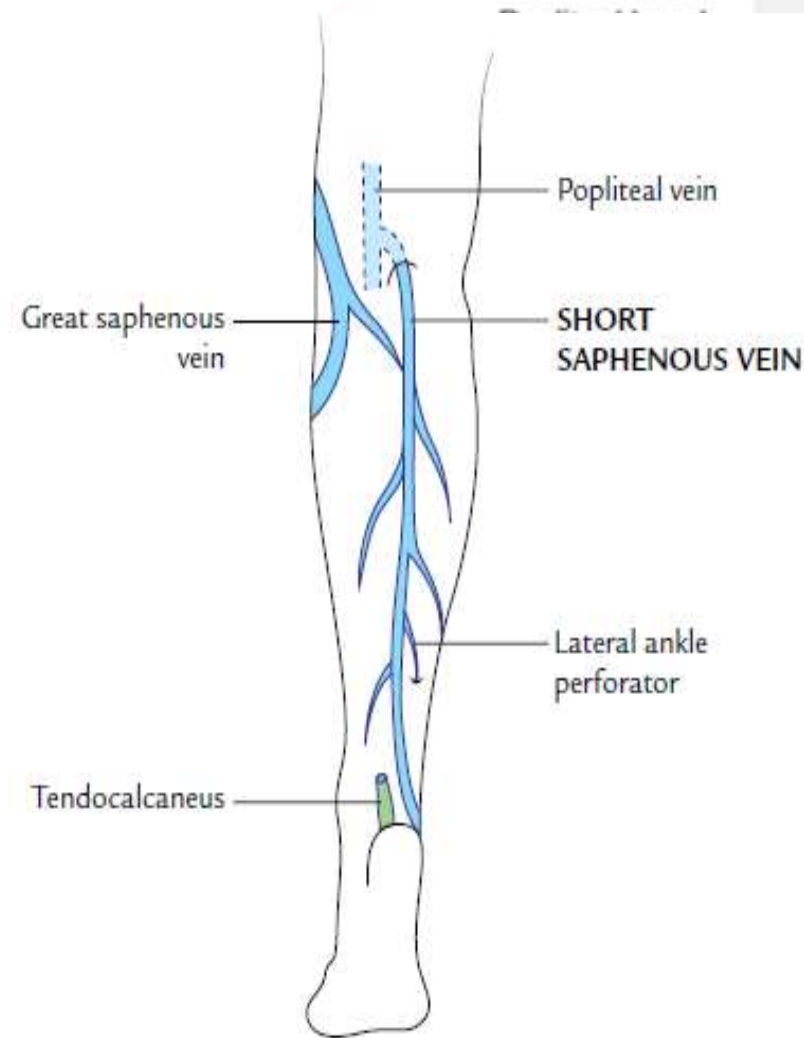
• Course

- Runs anterior to the medial malleolus
- crosses the lower 1/3 of the medial surface of tibia
- runs 1 cm posterior to the medial border of the tibia and **inter in saphenous opening** and join with femoral vein
- It has 10-20 valves



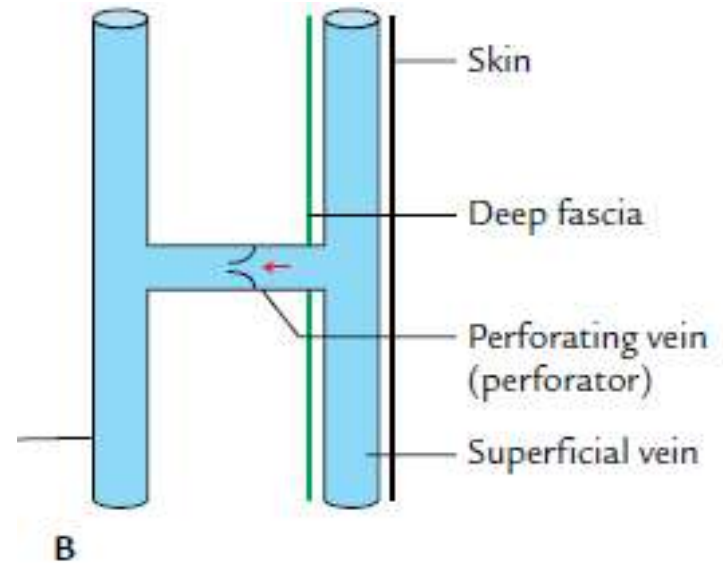
Superficial veins of the lower limb

- **Small saphenous vein (short)**
- **Origin**
 - Arises on lateral side of the foot from the union of dorsal digital vein of little toe with dorsal venous arch
- **Course**
 - Ascend **posterior to lateral malleolus**
 - Pass along lateral border of **calcaneal tendon**
 - Incline to midline of fibula and penetrates deep fascia
 - Empties into **popliteal vein in popliteal fossa**



Perforator veins of the lower limb

- These veins called because they perforates the deep fascia
- They are two types
 - **Direct perforating** - Connect superficial veins with deep veins directly
 - **Indirect perforating** –Connect superficial veins with deep veins thru muscular veins



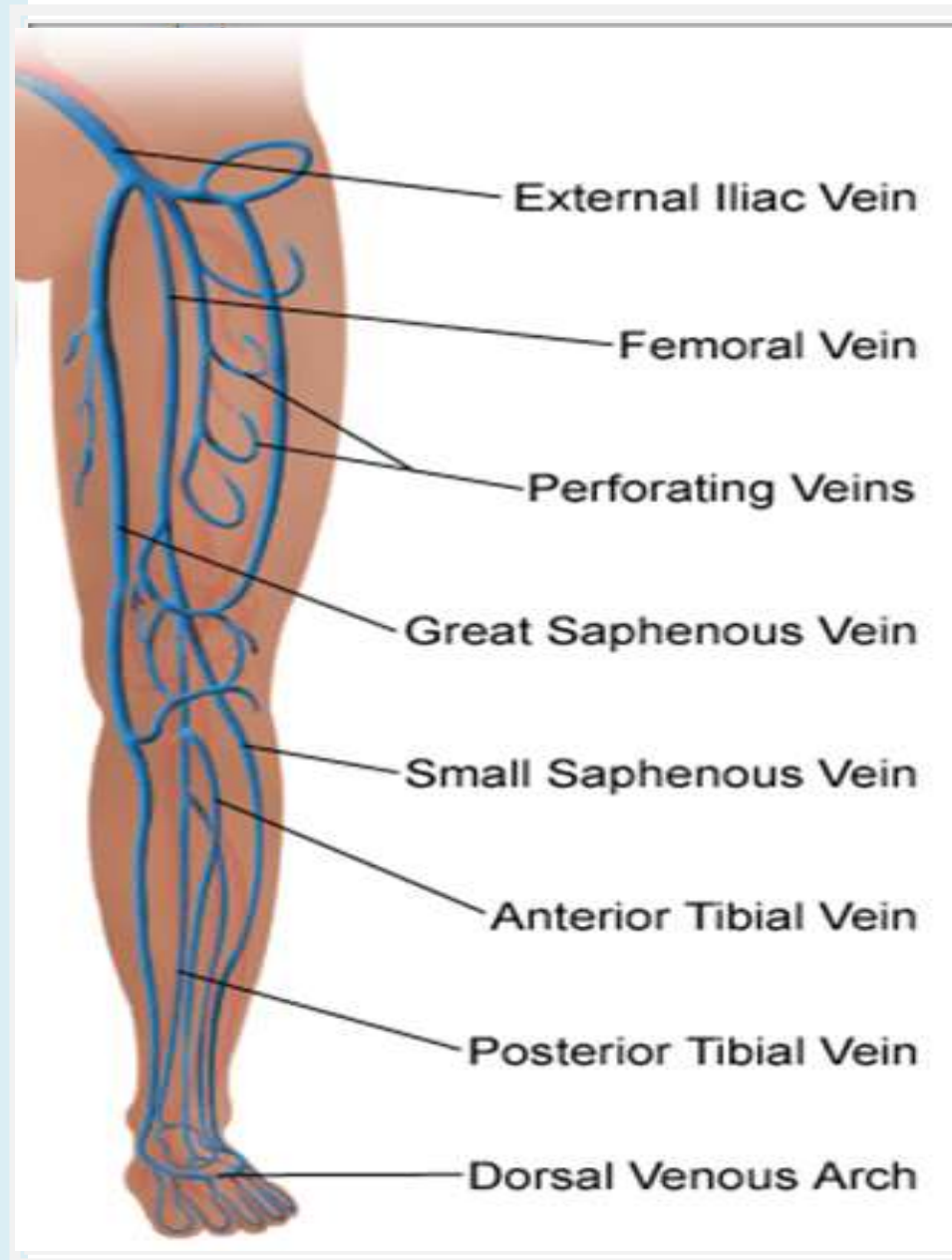
Clinical correlation

Calf pump and peripheral heart: In upright position, the venous return from the lower limb occurs against gravity and depends largely on the contraction of calf muscles. Therefore, these muscles are termed **calf pump**.

The soleus muscle contains venous sinuses filled with blood. When soleus muscle contracts, it pumps the blood from its large venous sinuses into the deep veins, and when it is relaxed it sucks the blood from the superficial veins, and the venous sinuses within it are refilled. The unidirectional blood flow is maintained by the valves in the perforating veins. Hence, the soleus is sometimes termed **peripheral heart**.

Deep veins of the lower limb

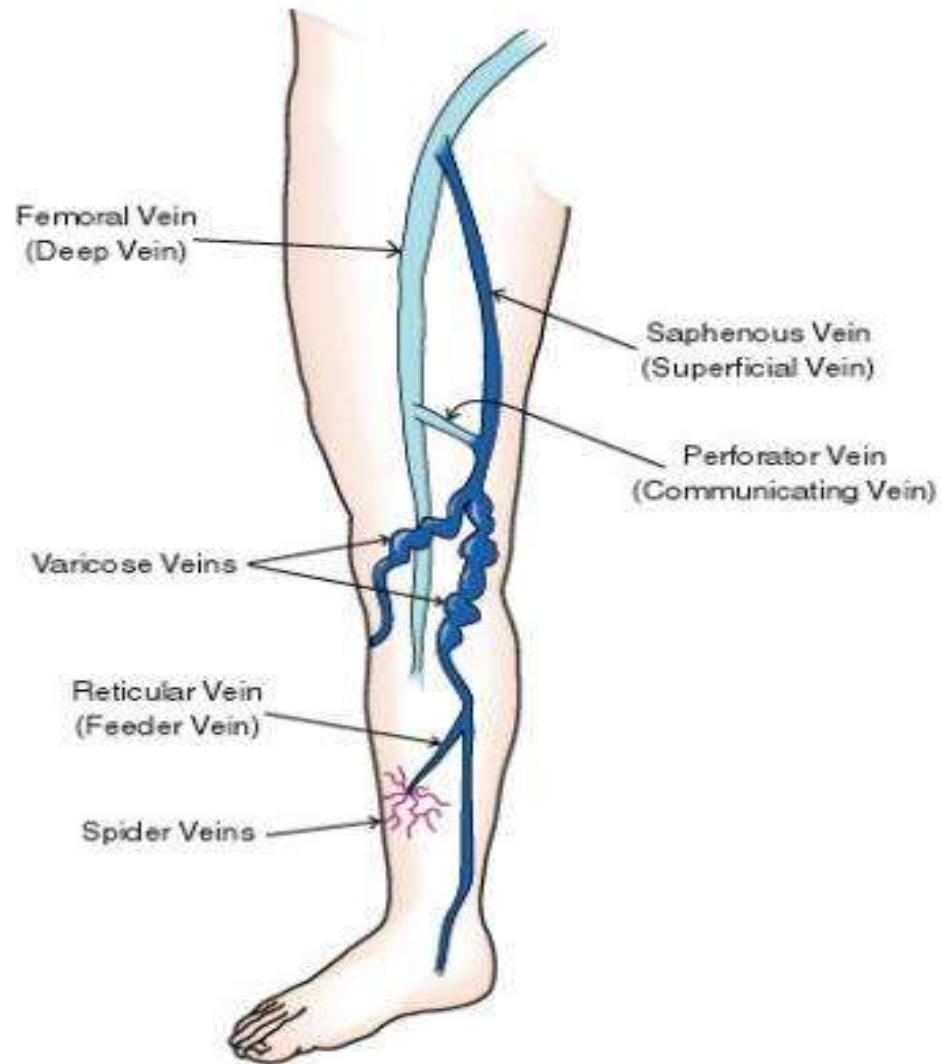
- **Accompany an arteries**
- Usually paired, frequently interconnecting veins
- Contained within vascular sheath with the artery
 - **arterial pulsations help move the blood**
- In the leg they flow to **popliteal vein**, which become **femoral vein** in thigh which become **external iliac vein** in pelvis



Varicose vein

- Larger diameter than normal, elongated & tortuous
- Occurs in superficial veins of lower limb
- Caused by:
 - ✓ Hereditary
 - ✓ Weakness of vein wall
 - ✓ Incompetent valves
 - ✓ Increase intrabdominal pressures

Varicose vein





Great Saphenous Vein Cut Down

- Usually performed at the ankle.

Thus, it would be called Ankle Vein Cut Down.

The saphenous nerve branches are blocked with anaesthesia.

The great saphenous vein can be entered through the ankle.

Groin Vein Cut Down

The branches of Ilioinguinal and intermediate cutaneous nerves of the thigh

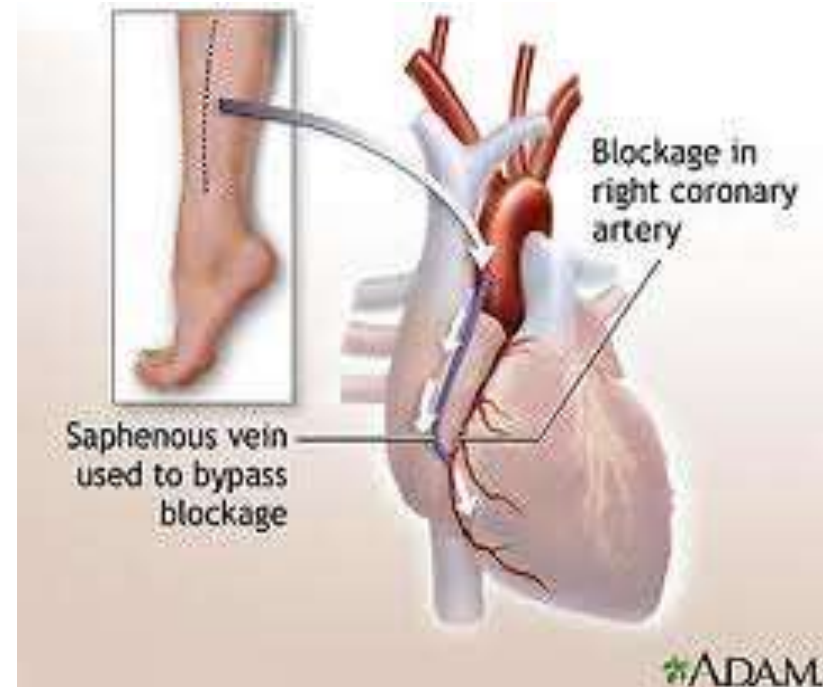
Clinical correlation

- **Venesection of the great saphenous vein:** The great saphenous vein in front of medial malleolus at ankle is the most preferred site of venesection (cut-down) in emergency situation when the superficial veins elsewhere in the body are collapsed and invisible, to insert the canula for prolonged administration of intravenous fluids.

Note that, in front of medial malleolus, saphenous nerve lies in front of the vein. Hence, during cut-down procedure, the saphenous nerve should be recognized to avoid its injury.

Great Saphenous Vein Graft

- In patients with occlusive coronary disease caused by atherosclerosis, the diseased arterial segment can be bypassed by inserting a graft consisting of a portion of Great saphenous vein.
- Vein segment is reversed to prevent obstruction by valves



Deep Vein Thrombosis



Painful Swelling:

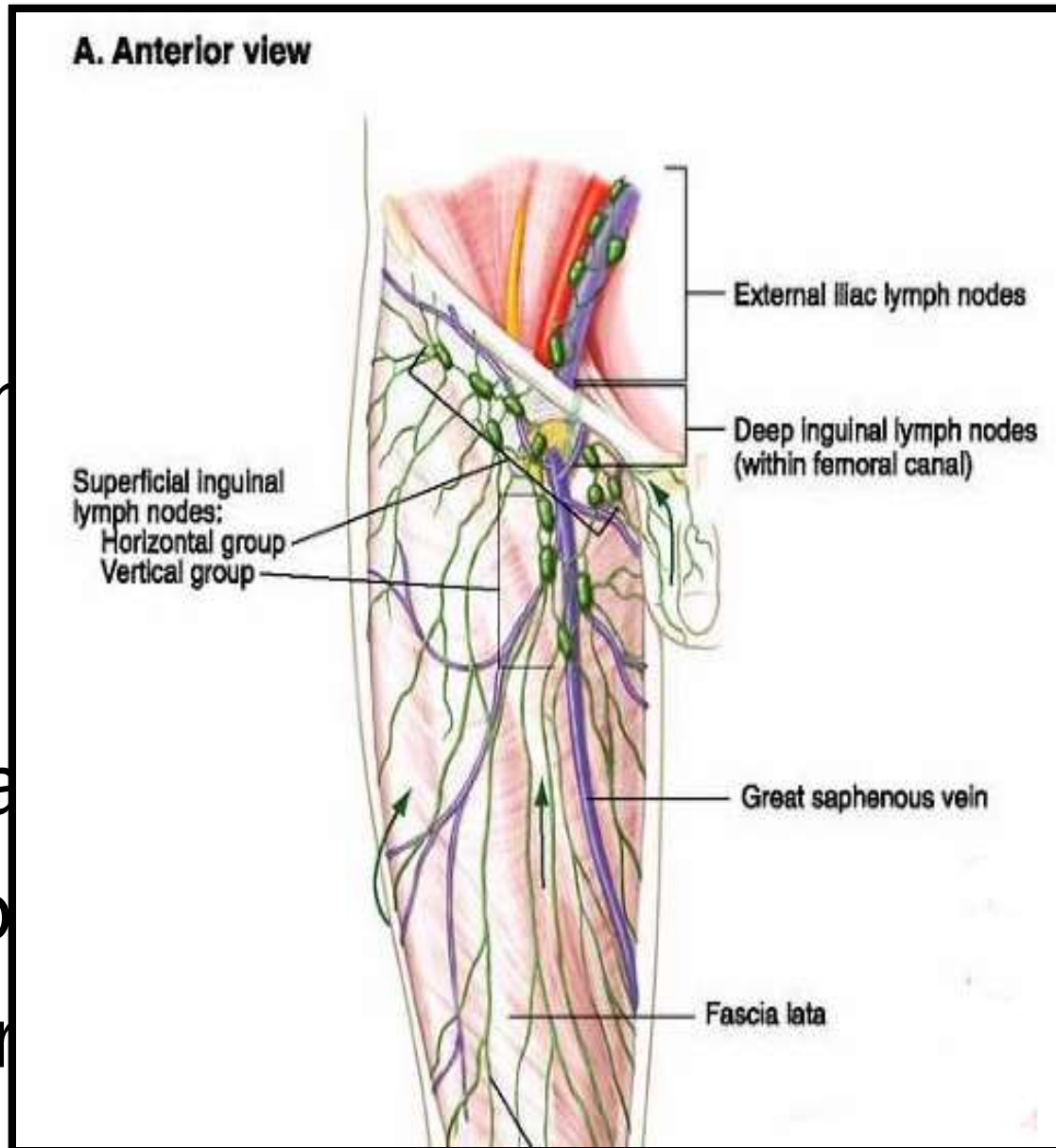
Sudden, intense swelling in your leg could signal a potential blood clot.

HEALTHY FACTS



Lymph nodes & lymphatic vessels

- Superficial &
- Deep lymphatic
- Inguinal lymph nodes
 - palpable
 - divided into:
 - superficial
 - » Superior
 - » Inferior
 - deep group



- The lymphatic drainage of the lower limb is **clinically important**.
 - inflammatory lesion cause pain full enlargement of the limb



- It may blocked the lymphatic drainage
- There are two lymph node

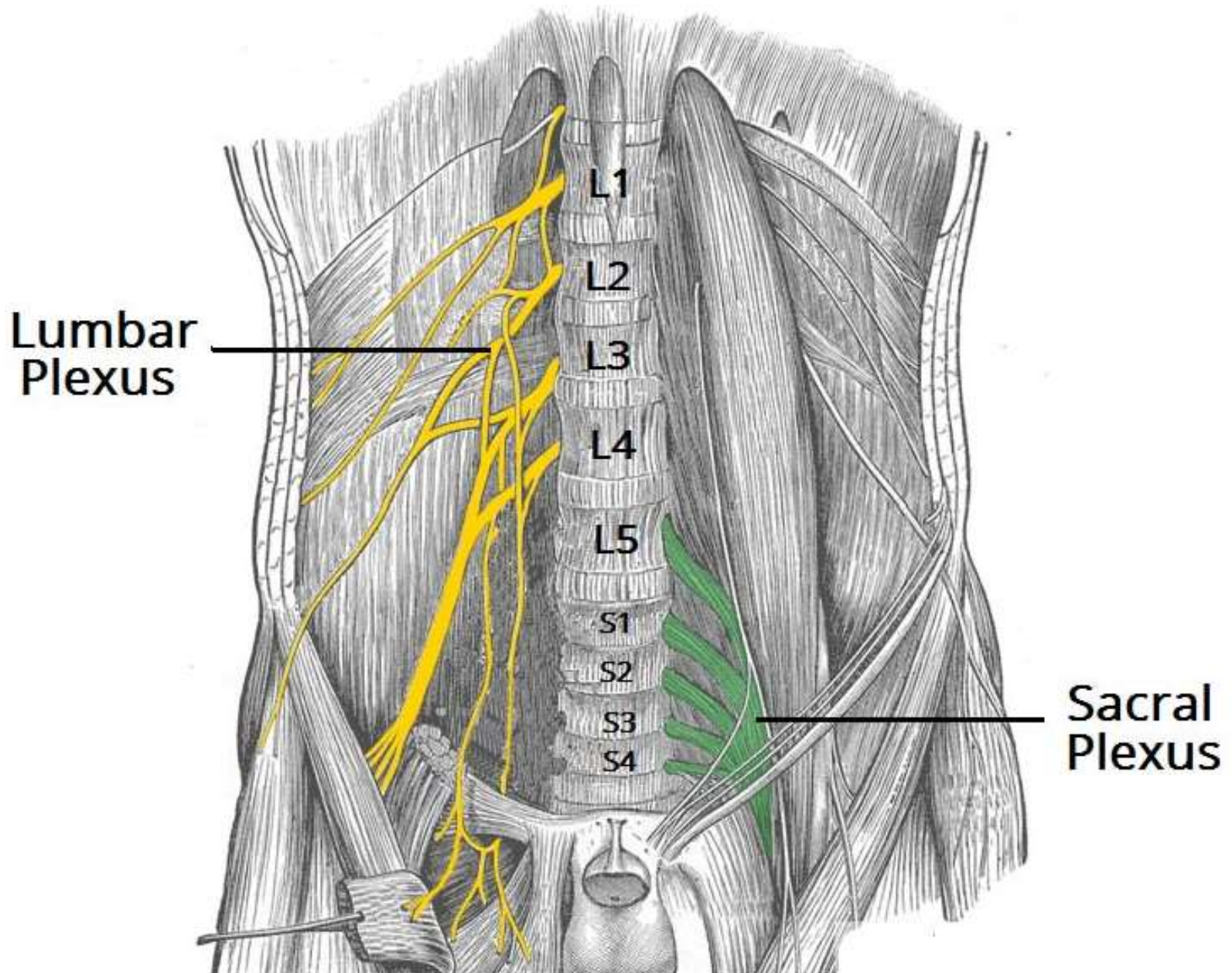
1. **Superficial lymph node= T -Shape**
2. **Deep lymph node**

Clinical correlation

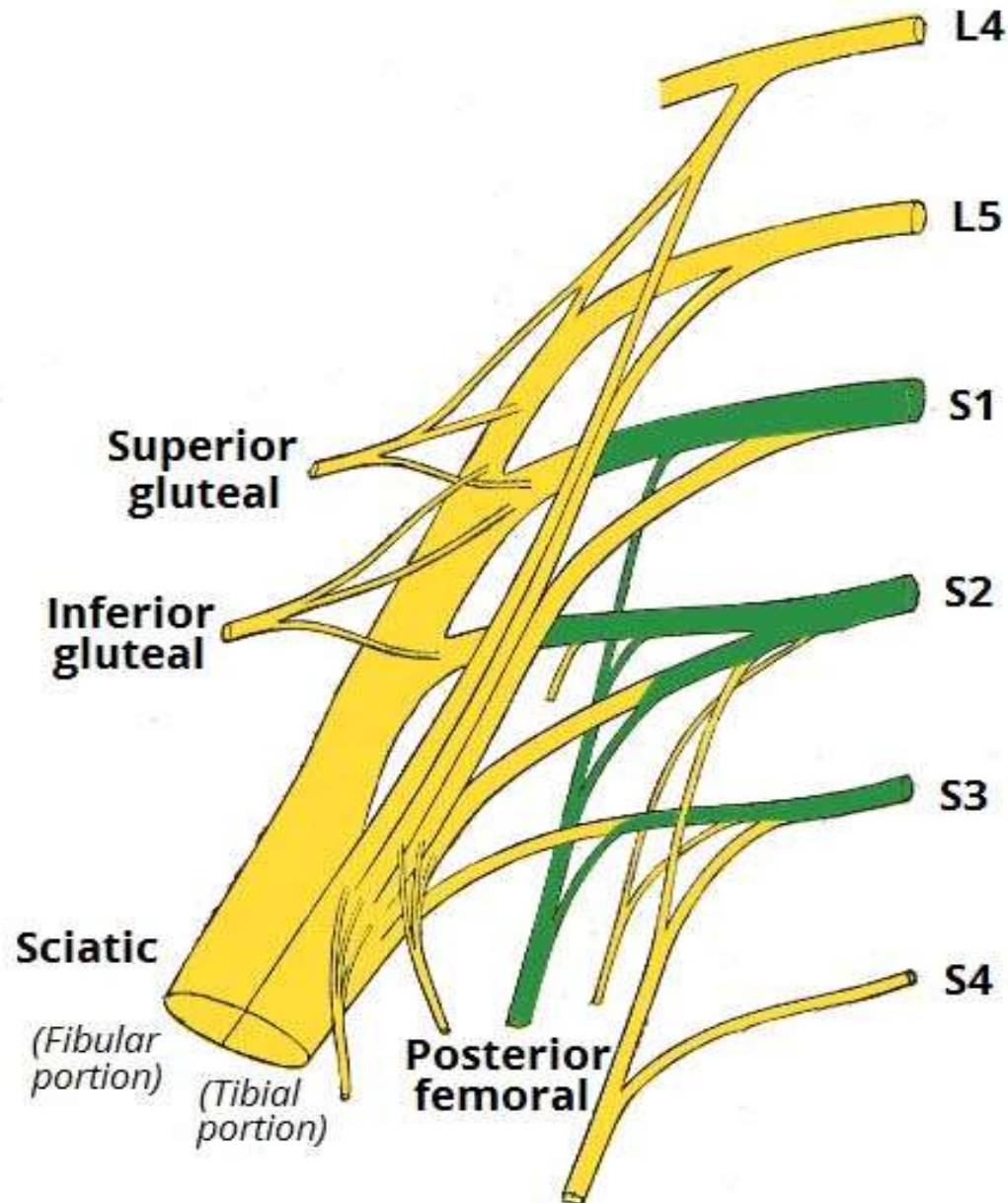
- **Elephantiasis (Fig. 33.7):** The lymph vessels of the lower limb are often blocked, particularly in the endemic areas, by the microfilarial parasites (*Wuchereria bancrofti*). This causes massive edema of the lower limb producing a clinical condition called **elephantiasis**. In this condition, there is hypertrophy of the skin and subcutaneous tissue to an unusual proportion.
- **Enlargement of the inguinal lymph nodes:** It is the commonest cause of swelling in the subinguinal region. The common causes of enlargement of these lymph nodes is infection, boil or abscess in the drainage area. The other causes are filariasis, Hodgkin's disease, etc.
The enlarged inguinal nodes should be differentiated from *ectopic testis*, *femoral hernia*, and *psoas abscess*.

Cutaneous nerves of the Lower Limb

Lumbar plexus

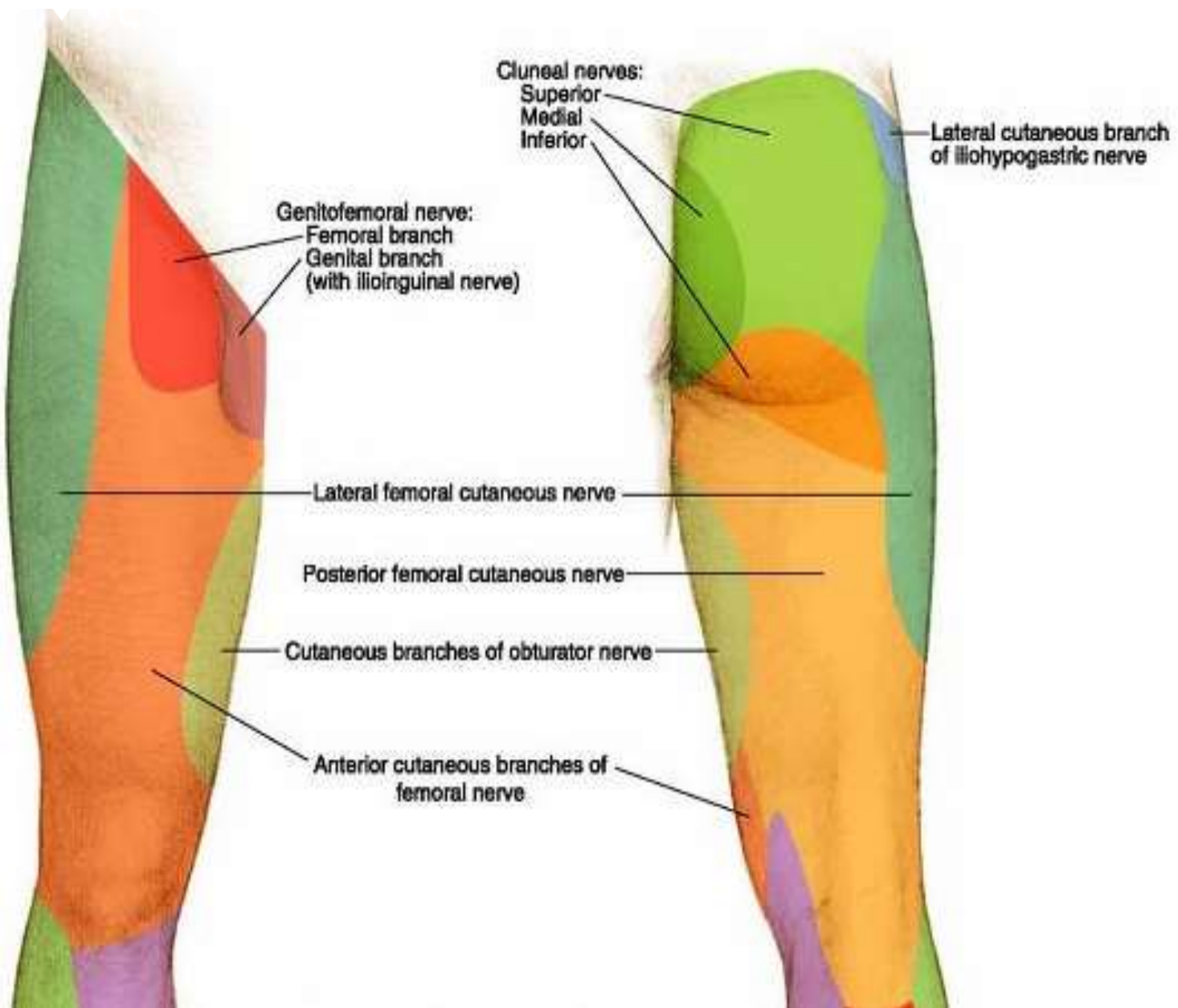


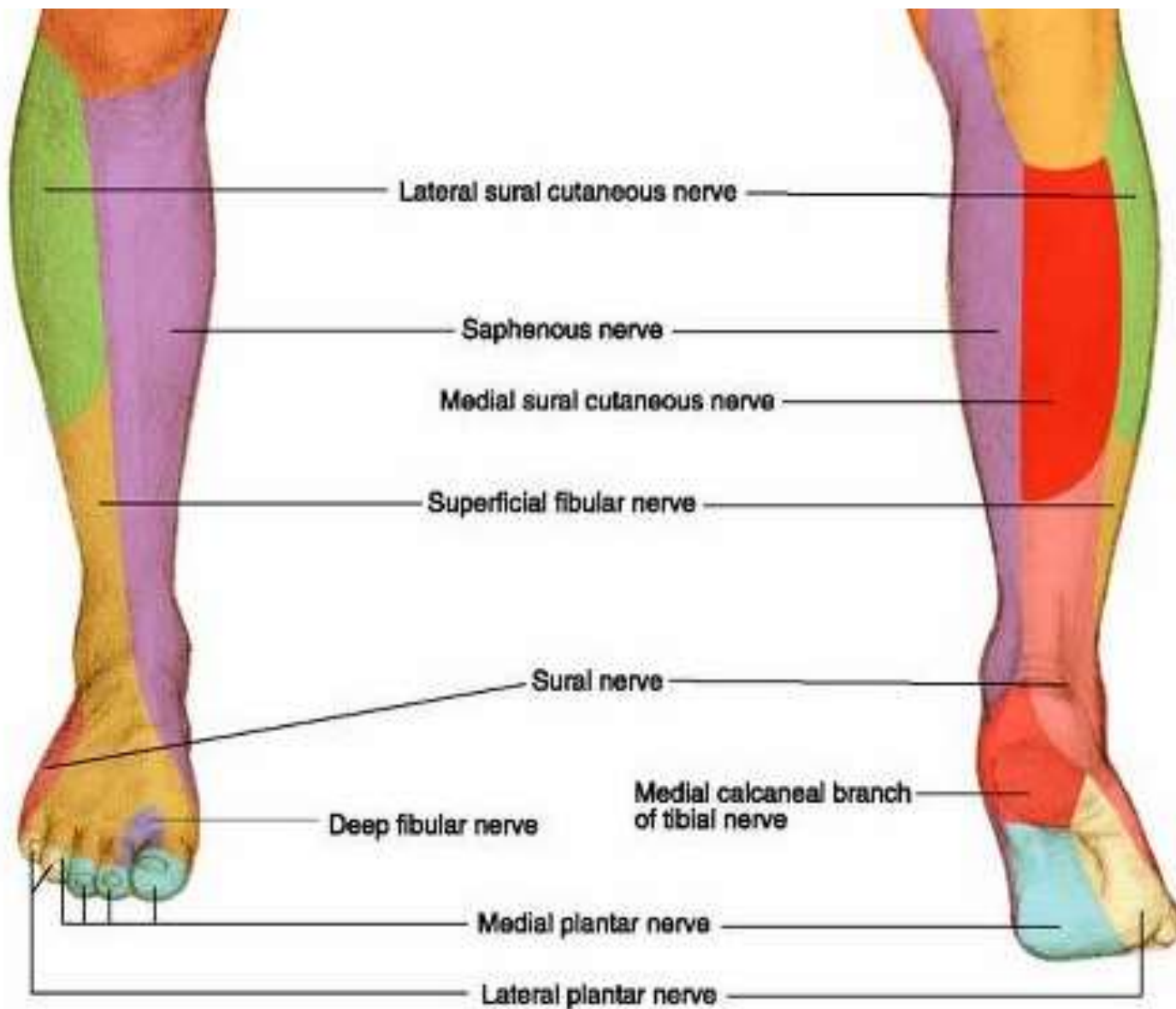
Sacral plexus

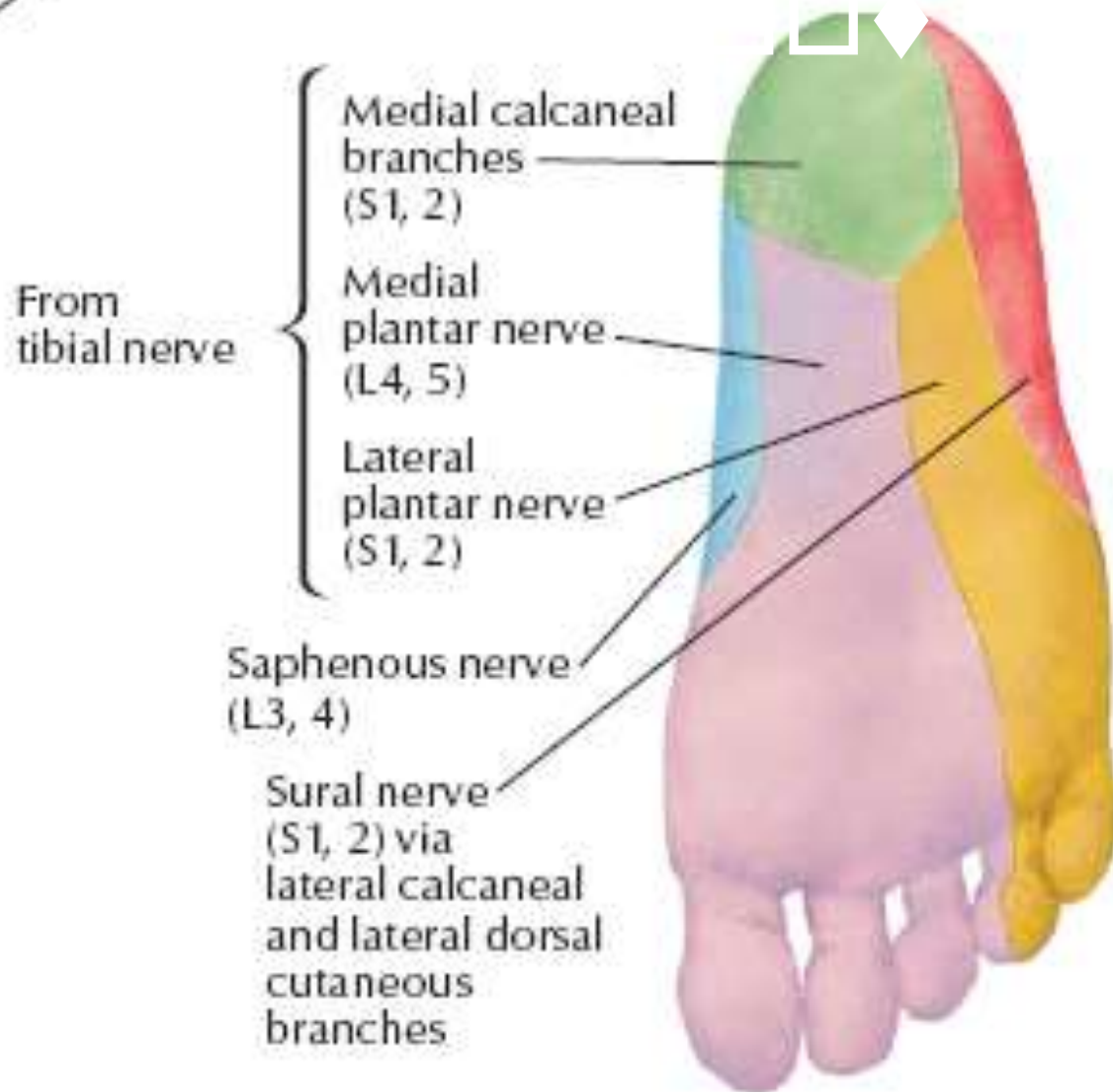


Arise from 10 spinal cord segments – T12- S4

- Upper and lateral gluteal region - **T12 & L1, L2**
- Front, medial and lateral thigh - **L1, L2, L3**
- Medial and front of the leg - **L 4**
- Lateral and front of the leg and intermediate of the dorsum of the foot - **L5**
- Lateral border of the foot - **S1**
- Sole of the foot medial part - **L4**
- Sole of the foot intermediate part - **L5**
- Sole of the foot lateral part - **S1**
- Back of the leg and thigh - **S2**
- Inferior and medial part of gluteal region - **S3, S4**







Cutaneous innervation of sole



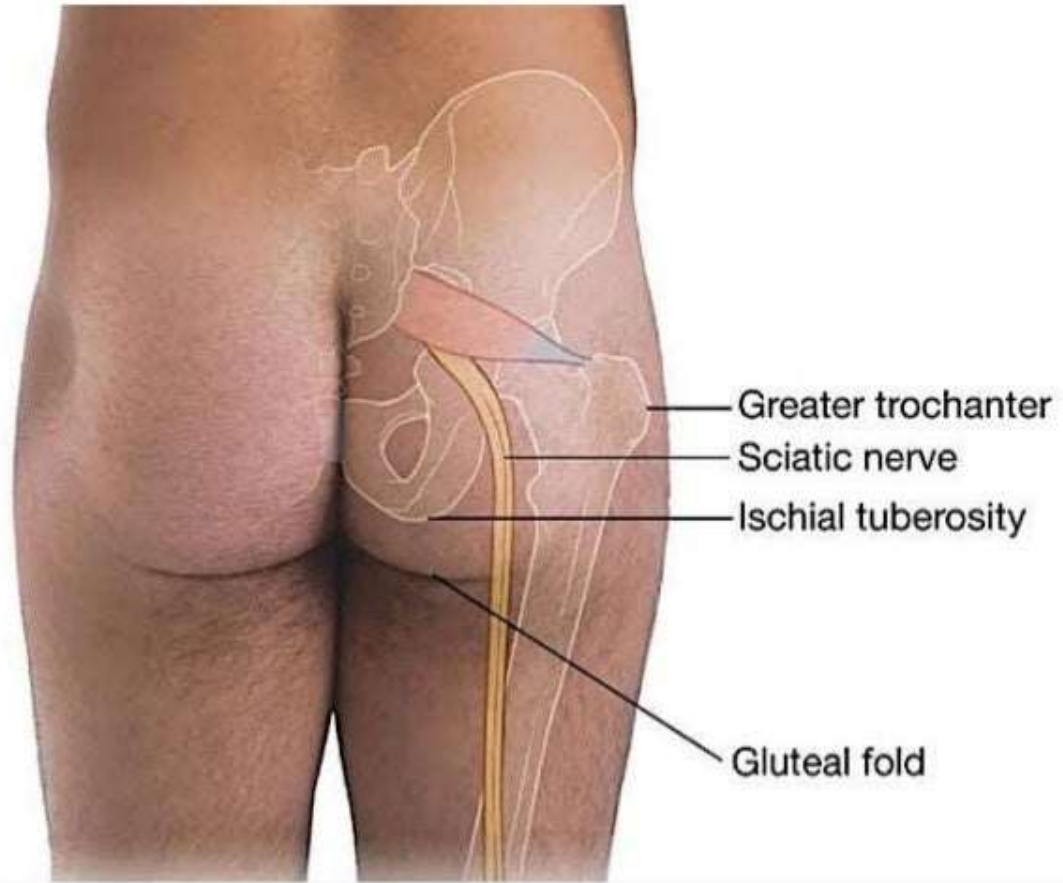
Medical education is not just a program for building knowledge and skills in its recipients... it is also an experience which creates attitudes and expectations.

— *Abraham Flexner* —

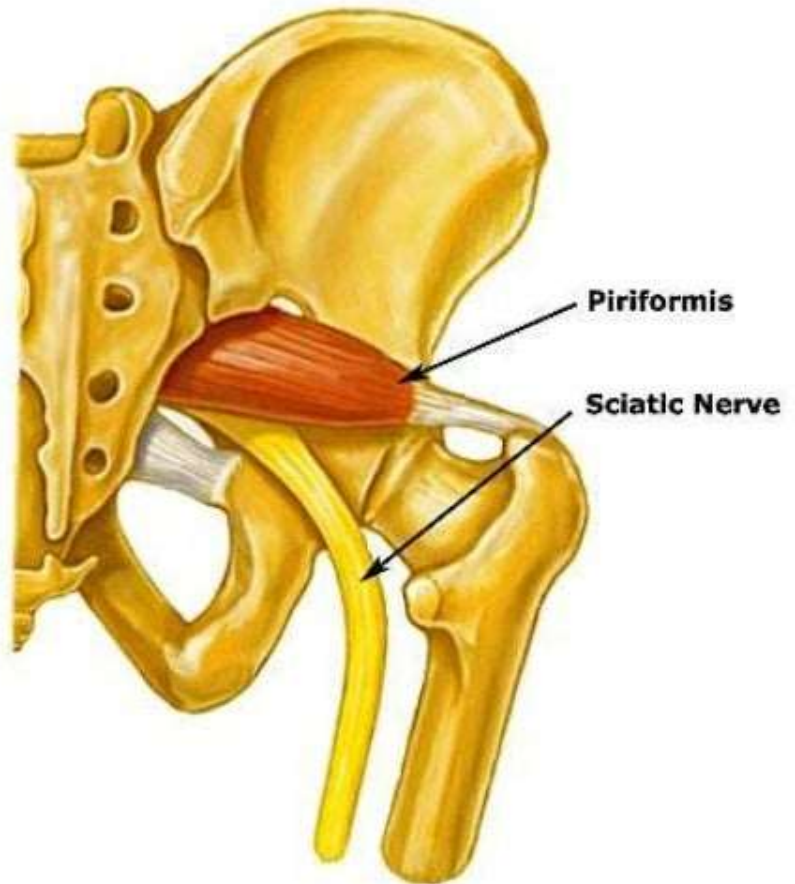
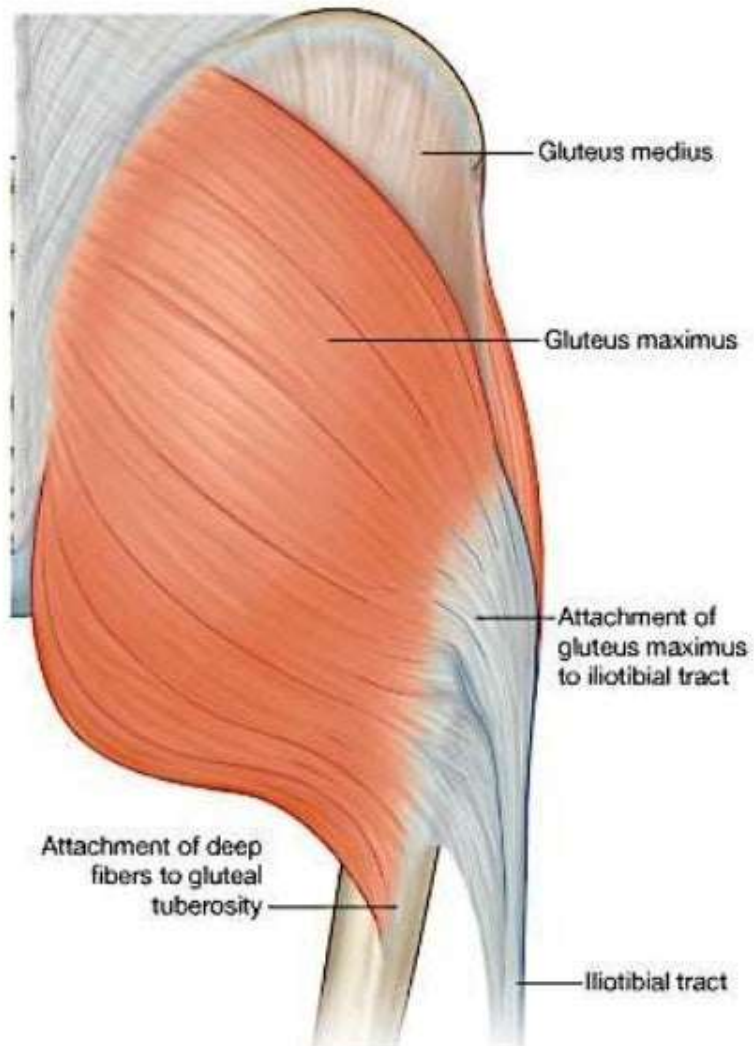
GLUTEAL REGION
&
POSTERIOR COMPARTMENT OF THIGH

Gluteal region

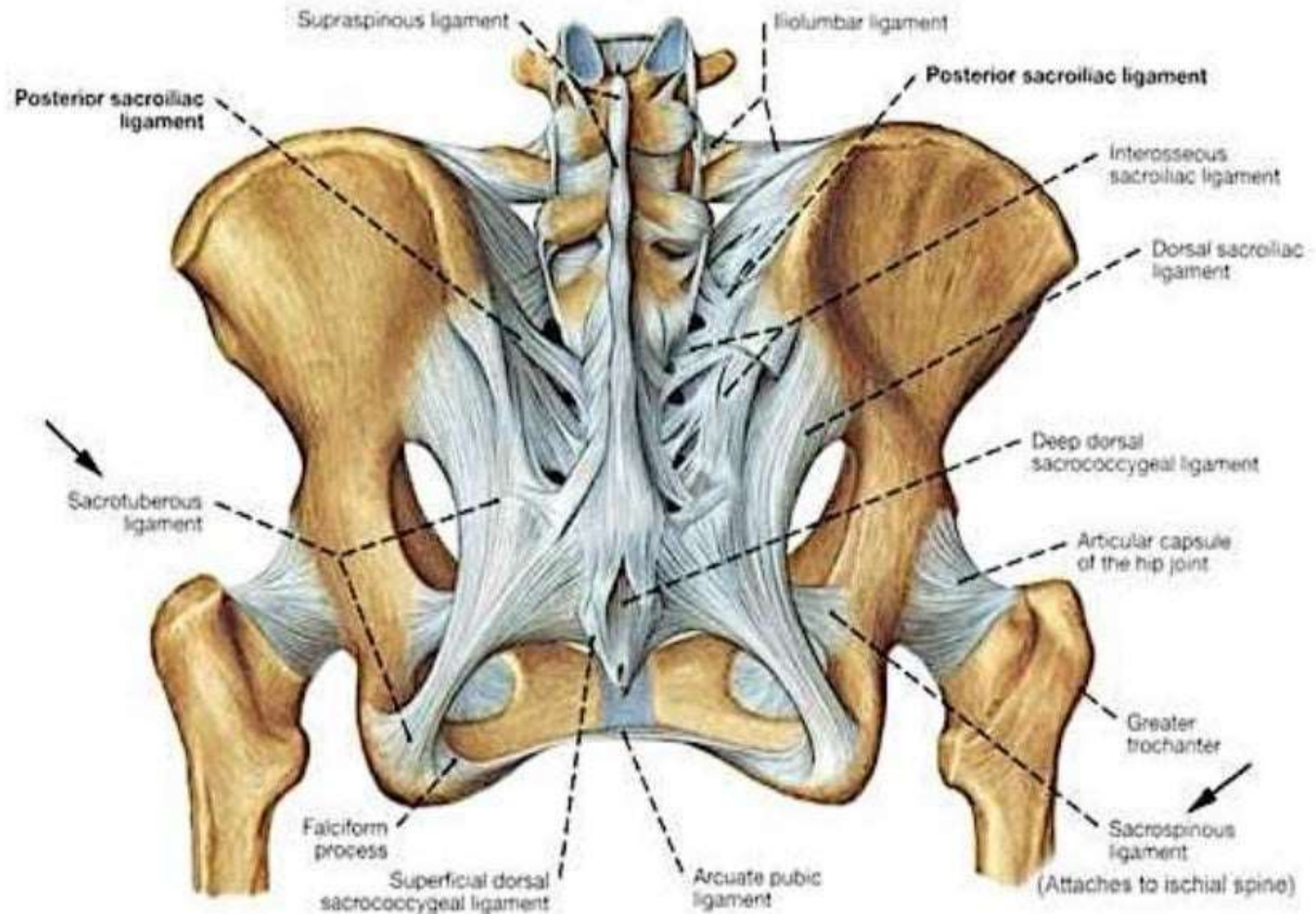
- Transitional between trunk & lower extremity.
- Anatomically – trunk
- Functionally - lower extremity
- Includes rounded posterior buttocks & lateral hip



Key muscles

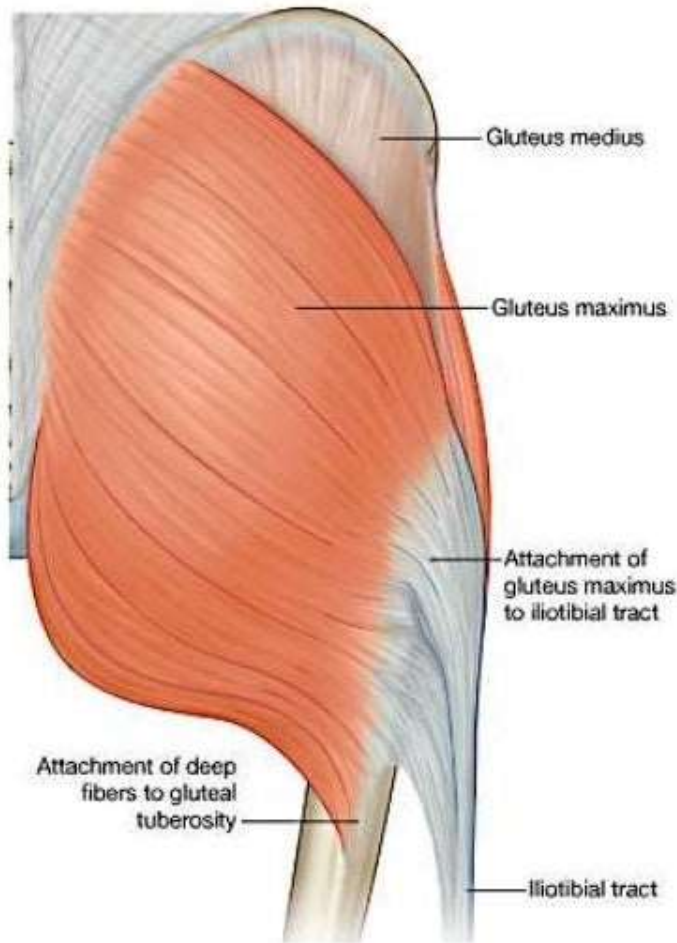


Key ligament



Gluteus maximus

Largest muscle responsible for the prominence of the 'buttocks'



Origin

- Outer surface of ilium behind posterior gluteal line
- Posterior surface of sacrum, coccyx and sacrotuberous ligament.

Insertion

- 75% of fibres into ilio-tibial tract – 25%(deep) into gluteal tuberosity

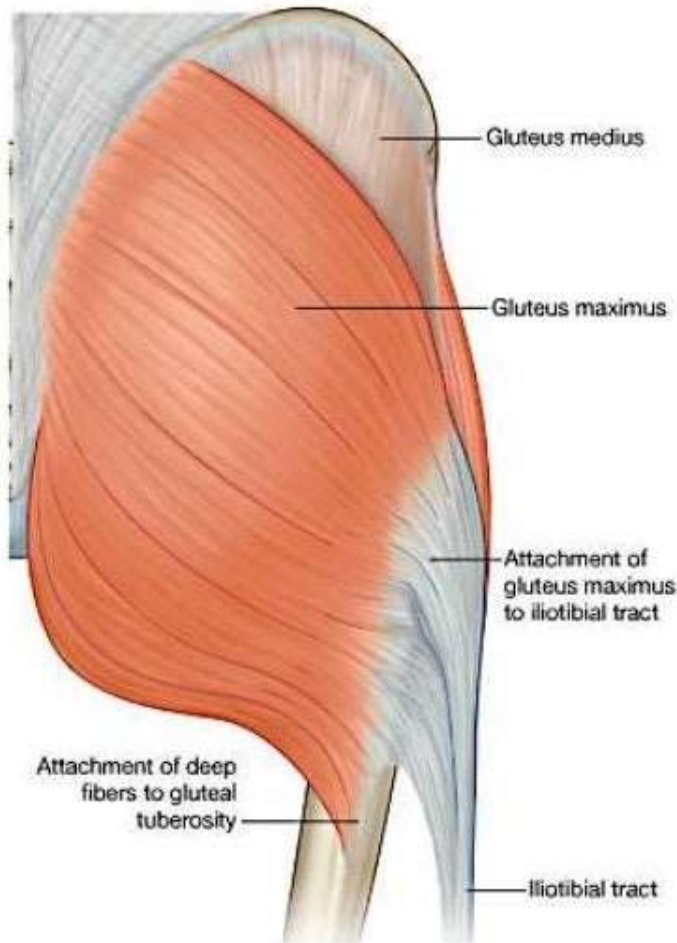
Gluteus maximus

Largest muscle responsible for the prominence of the 'buttocks'

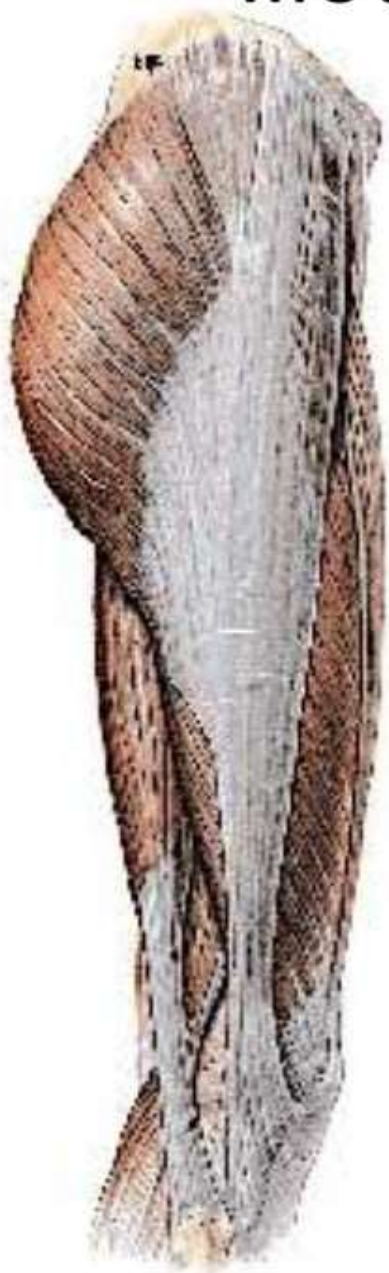
Nerve supply - inferior gluteal nerve

Action

- Extensor & lateral rotator of thigh at hip joint.
- Helps to maintain extension at knee through the ilio-tibial tract
- Paralysis of muscle – inability to raise the trunk from sitting or stooping positions.



Iliotibial tract & Tensor fascia lata

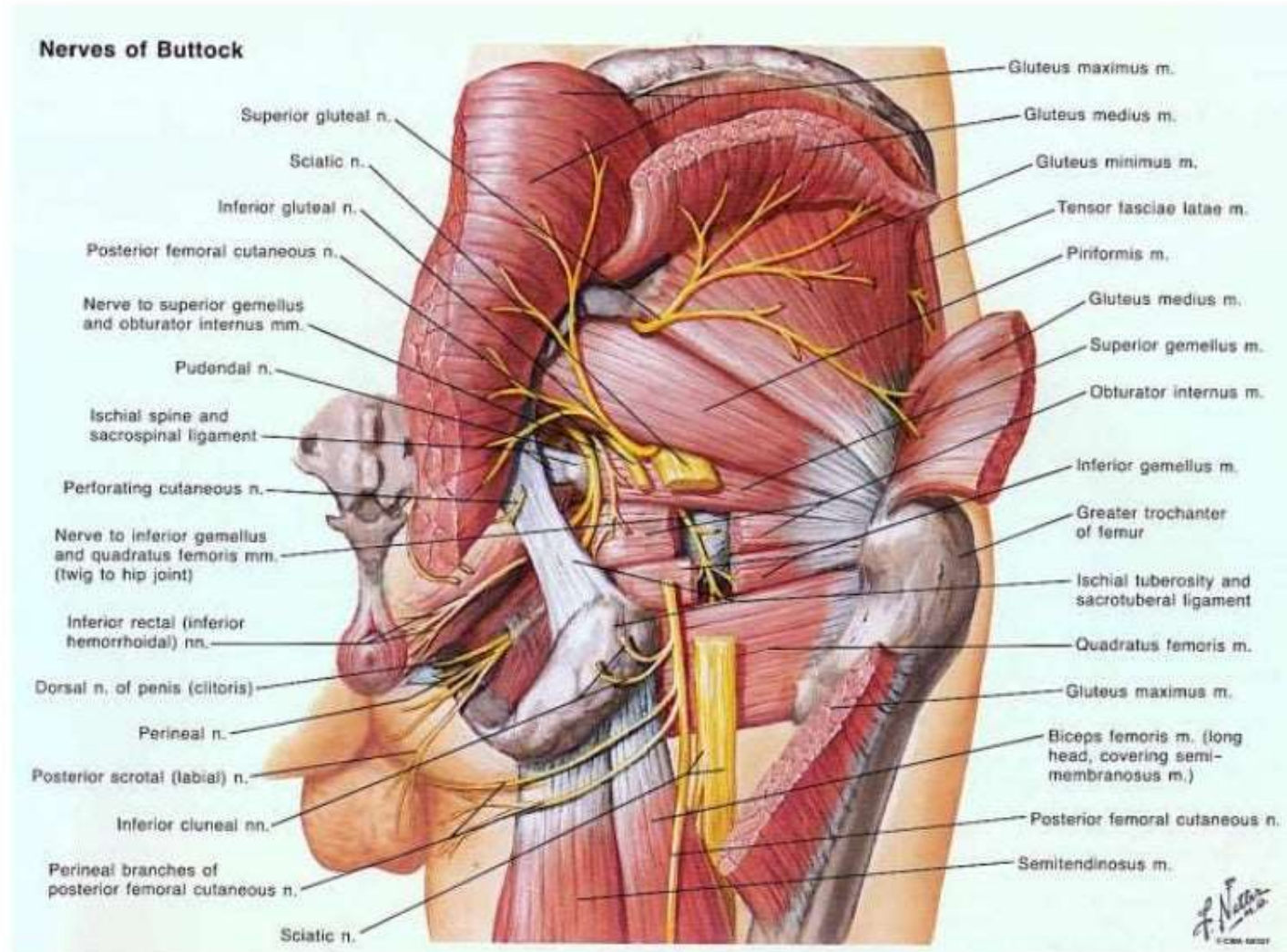


Tensor fascia lata

- **Origin:** anterior part of iliac crest, ASIS and notch below
- **Insertion:** inserts into the iliotibial tract.
- **Nerve:** Superior gluteal nerve
- **Action:** Assists gluteus maximus in keeping knee in extended position

Structures undercover of gluteus maximus

- Muscles
- Vessels
- Nerves
- Bones & Joints
- Ligaments
- Bursae



Gluteus medius

Posterior part covered by gluteus maximus

Origin

Outer surface of ilium b/w iliac crest above & posterior gluteal line behind and middle gluteal line below.

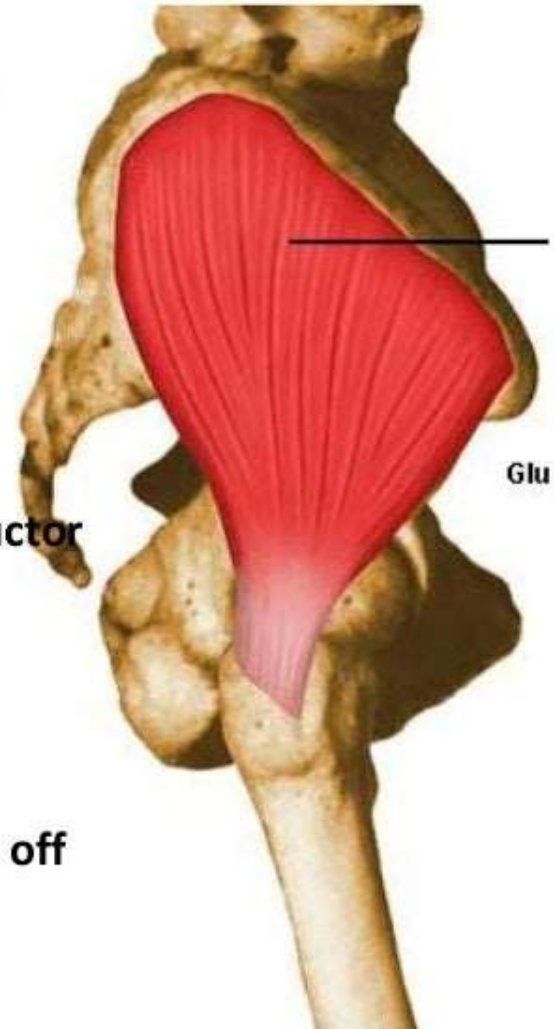
Insertion

Lateral surface of greater trochanter

Nerve – superior gluteal nerve

Action

- With G minimus and tensor fascia lata -powerful abductor at hip.
- Anterior fibres medially rotate the thigh
- Walking and running
- Holds opposite side of pelvis horizontally when foot is off the ground.



Gluteus minimus

Origin

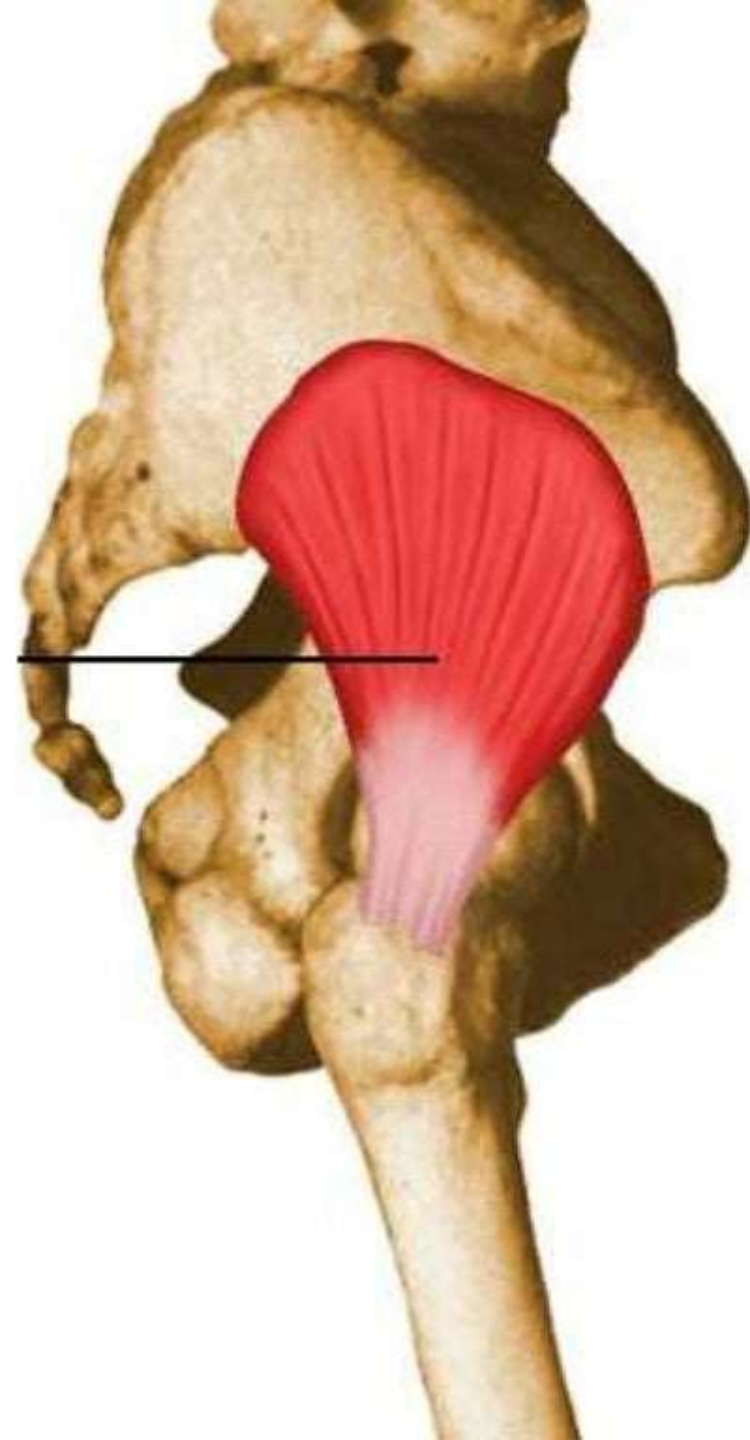
Outer surface of ilium b/w middle & inferior gluteal lines

Insertion - Anterior surface of greater trochanter

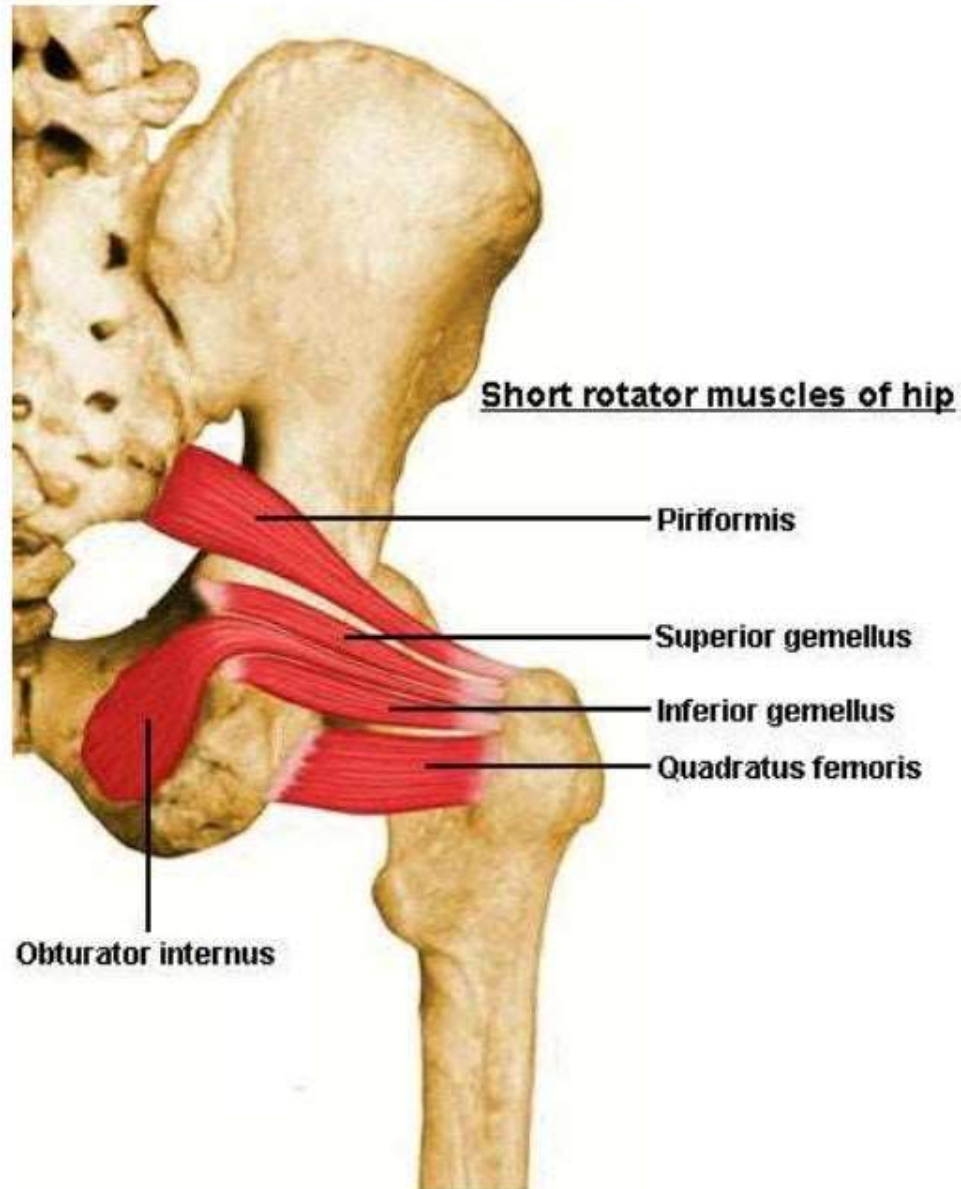
Nerve-superior gluteal nerve

Action

- With G minimus and tensor fascia lata -powerful abductor at hip.
- Anterior fibres medially rotate the thigh



Short Lateral Rotators of Thigh



Short Lateral Rotators of Thigh

Piriformis

- Key muscle of gluteal region

Origin

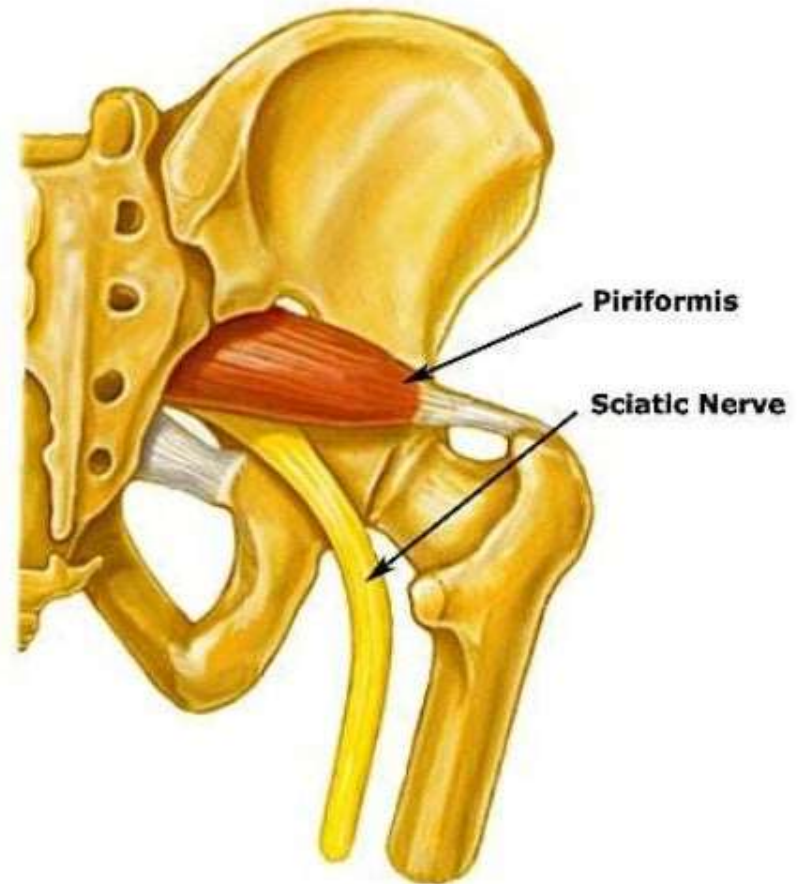
- Pelvic surface of sacrum 2nd, 3rd & 4th pieces
- Sacrotuberous ligament

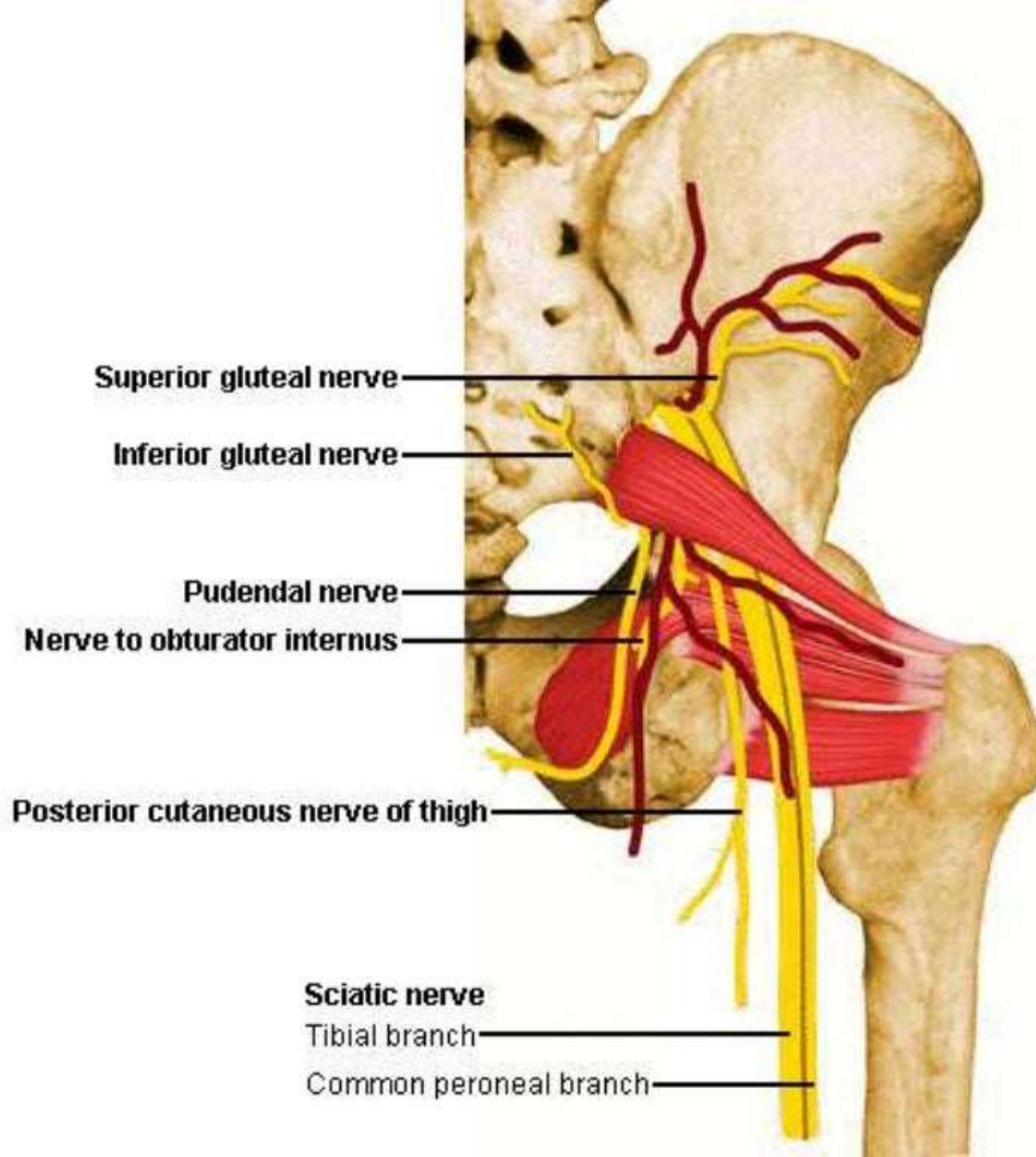
Insertion

- Upper border of greater trochanter
- Leaves the pelvis through greater sciatic foramen and separates gluteal vessels and nerves to superior and inferior

Nerve

- S1 2 anterior rami





Short Lateral Rotators of Thigh

Obturator Internus

Origin: from pelvic surfaces of

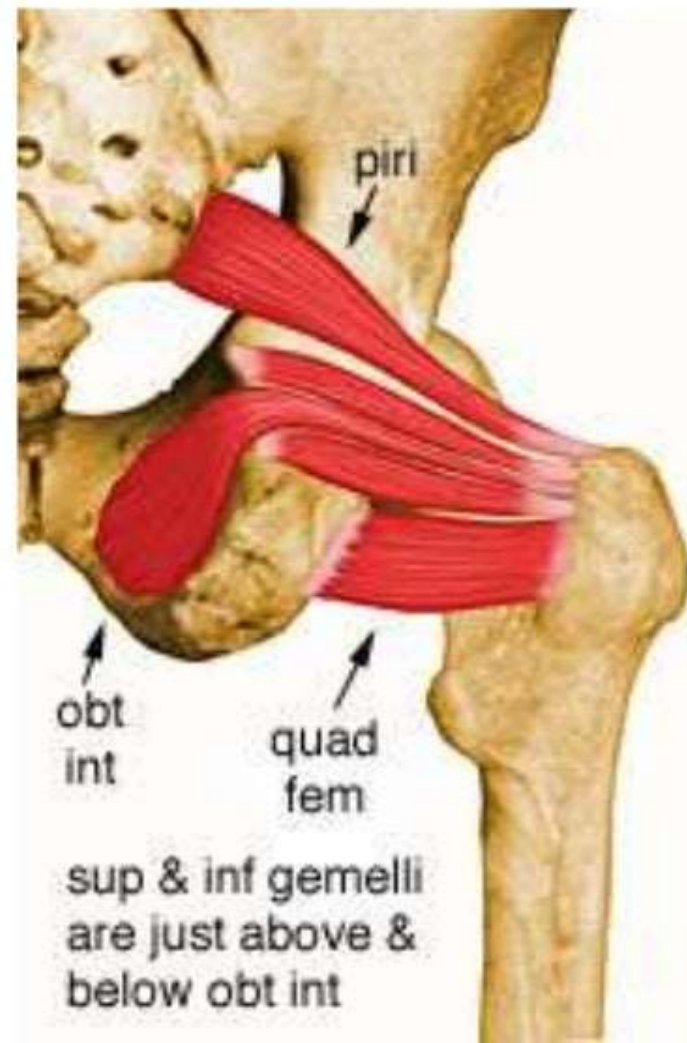
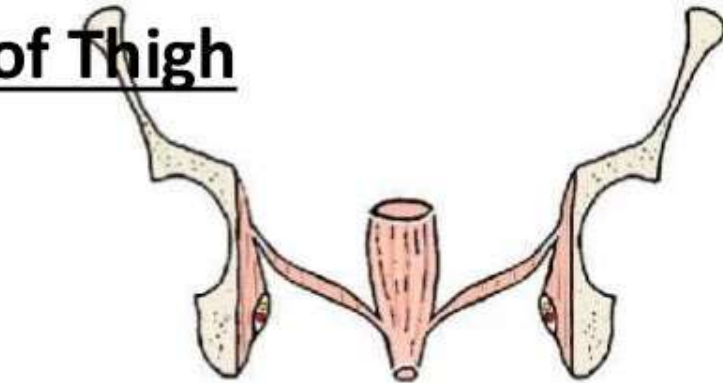
- Body of ischium
- Ischial tuberosity
- Ischio-pubic ramus
- Obturator membrane & fascia.

Insertion: tendon passes out of the pelvis through the lesser sciatic foramen and enters gluteal region
>> upper border of greater trochanter.

*One ½ of muscle in pelvis
other ½ in perineum
Tendon in gluteal region*

Nerve:

Nerve to obturator internus L5 S1 2



Short Lateral Rotators of Thigh

Gemellus superior

Origin-spine of ischium

Insertion-tendon of OBT int

Nerve- to OBT internus

Gemellus inferior

Origin-ischial tuberosity

Insertion-tendon of OBT internus

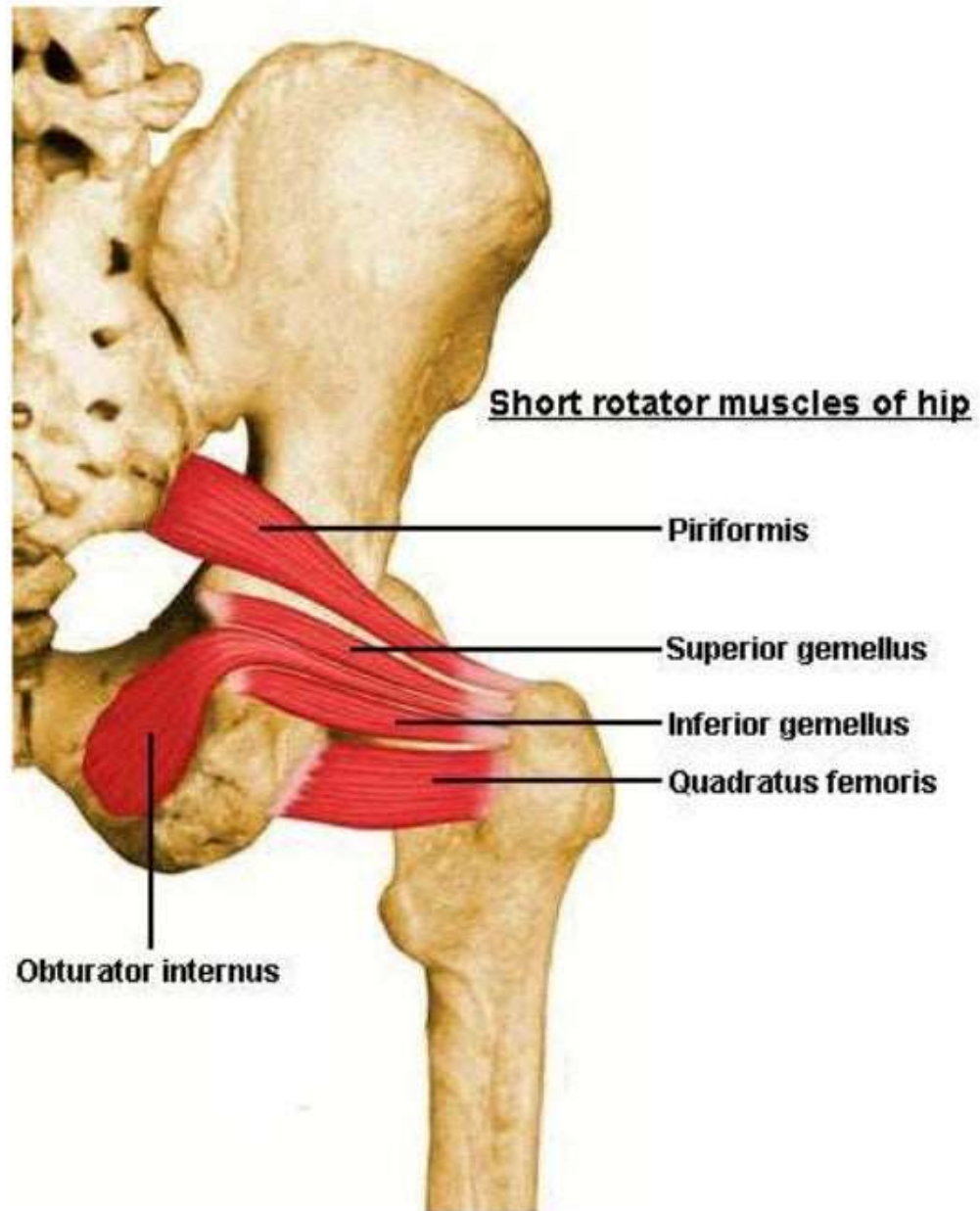
Nerve-to Quadratus femoris

Quadratus femoris

Origin-ischial tuberosity

Insertion-quadrate tubercle

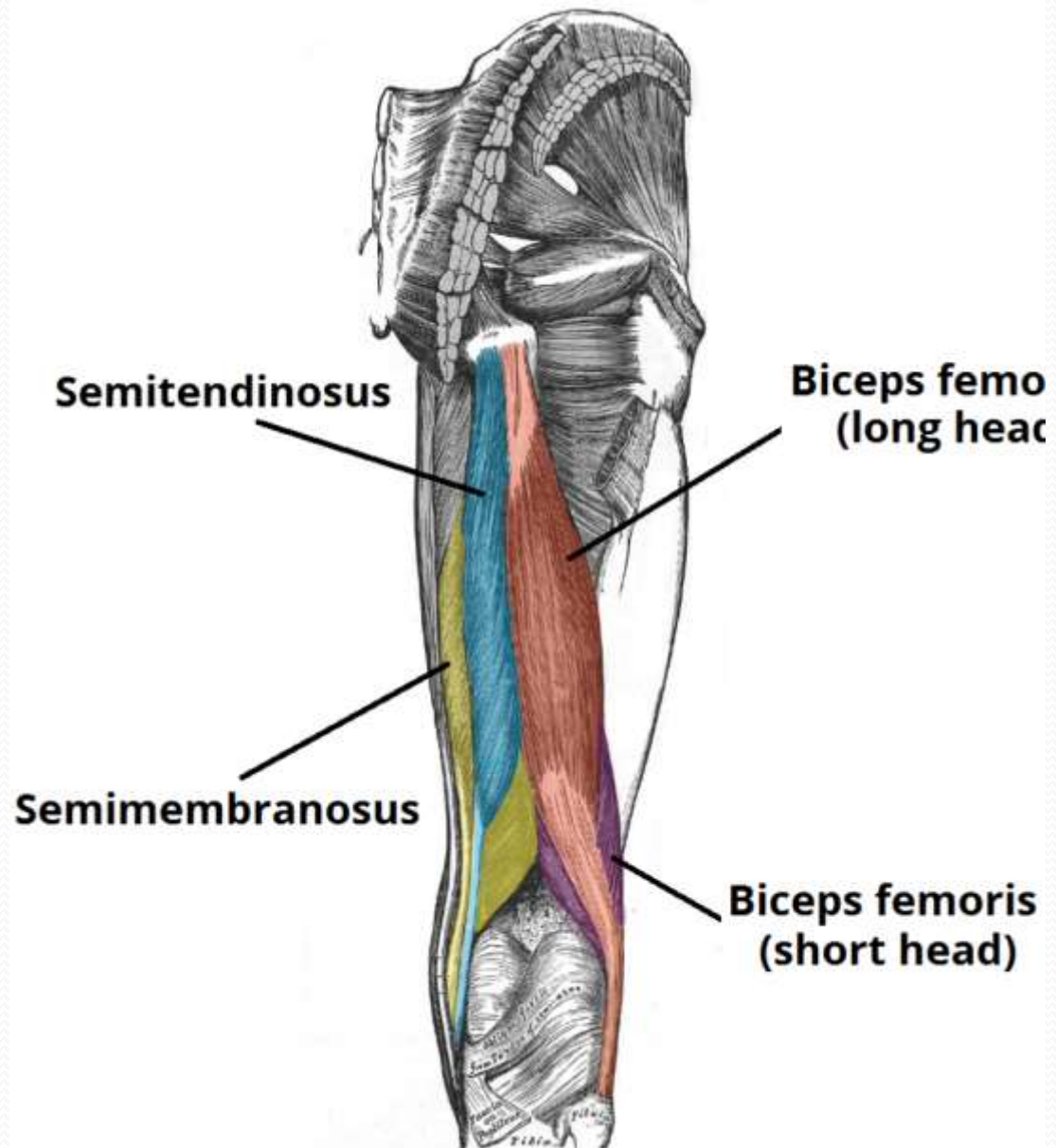
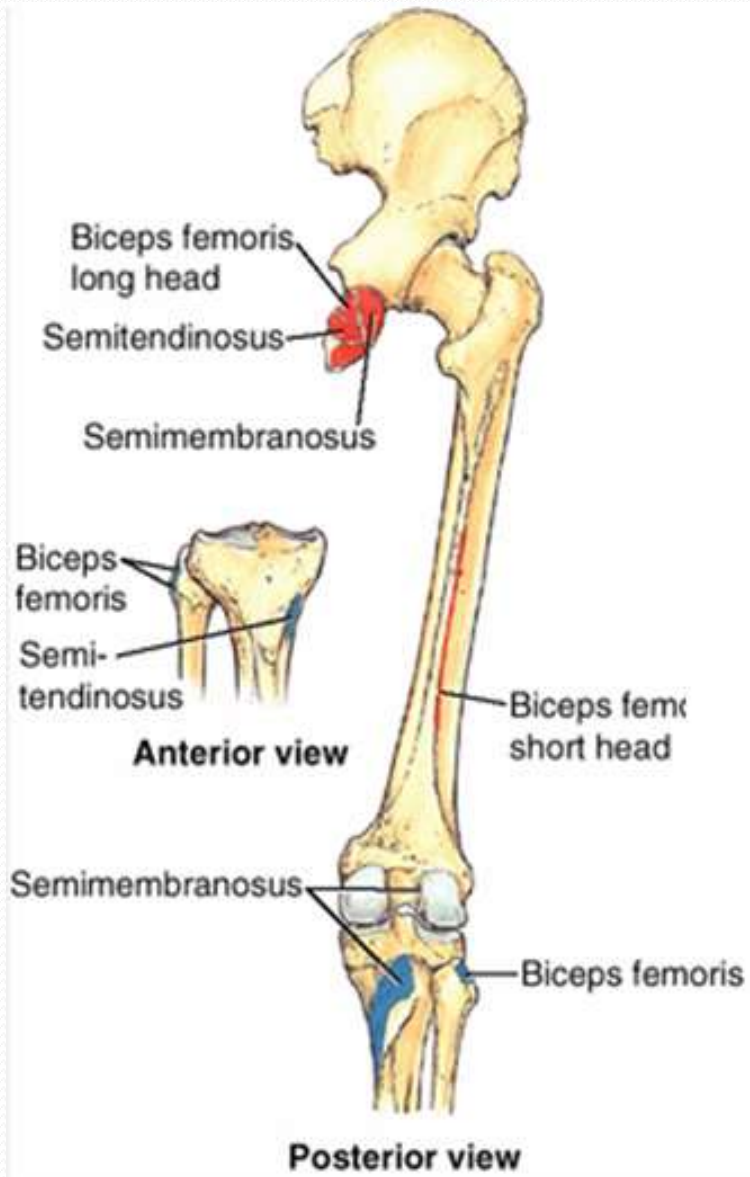
Nerve-sacral plexus



Posterior thigh muscles

- **Three of the four muscles in posterior thigh are hamstrings**
 - **Semitendinosus**
 - **Semimembranosus**
 - **Biceps femoris (long head)**
- **Short head of biceps femoris**
 - **not hamstring as it crosses only knee joint**
 - **innervated by fibular division of sciatic nerve**

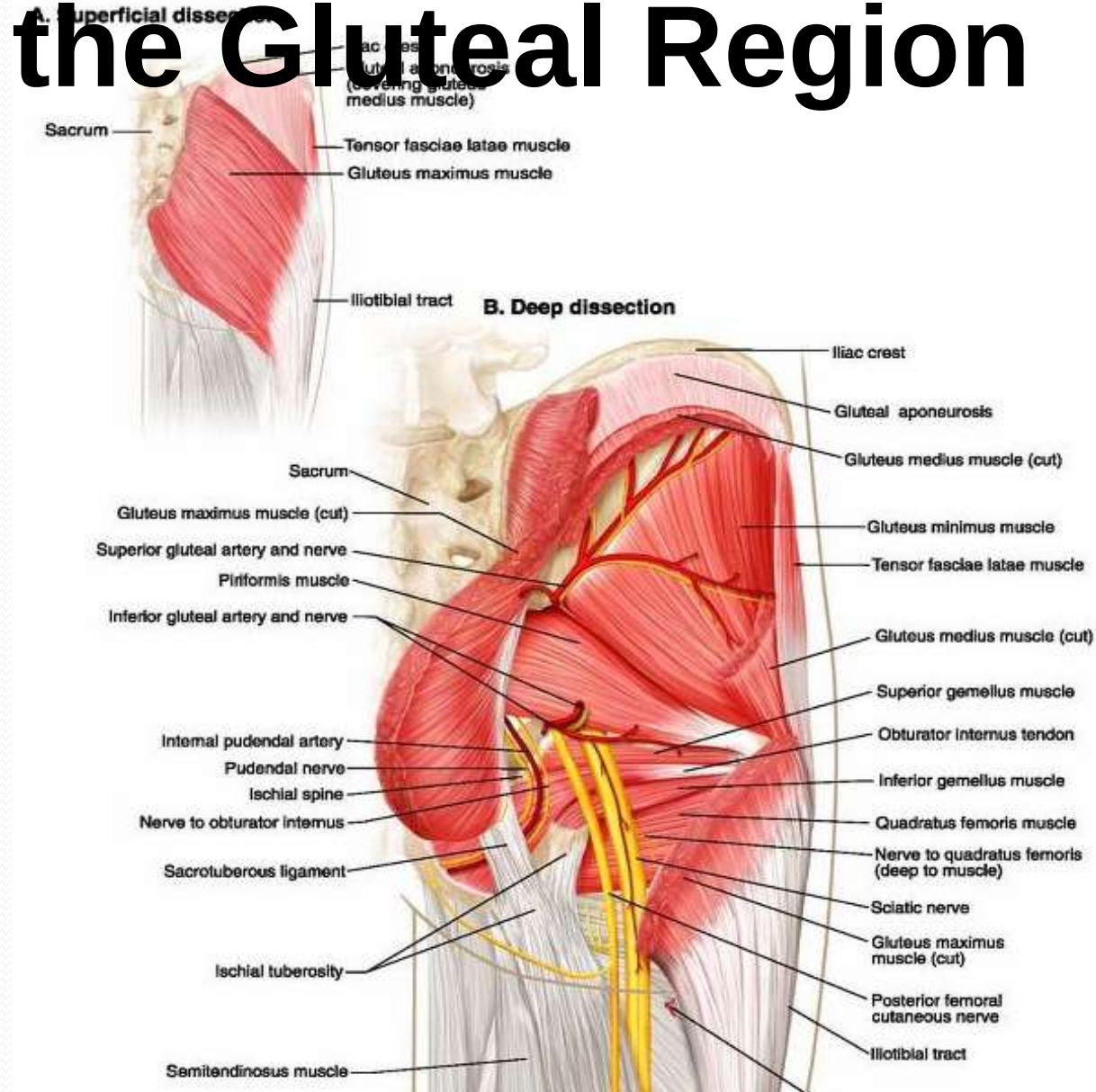
Posterior thigh muscles



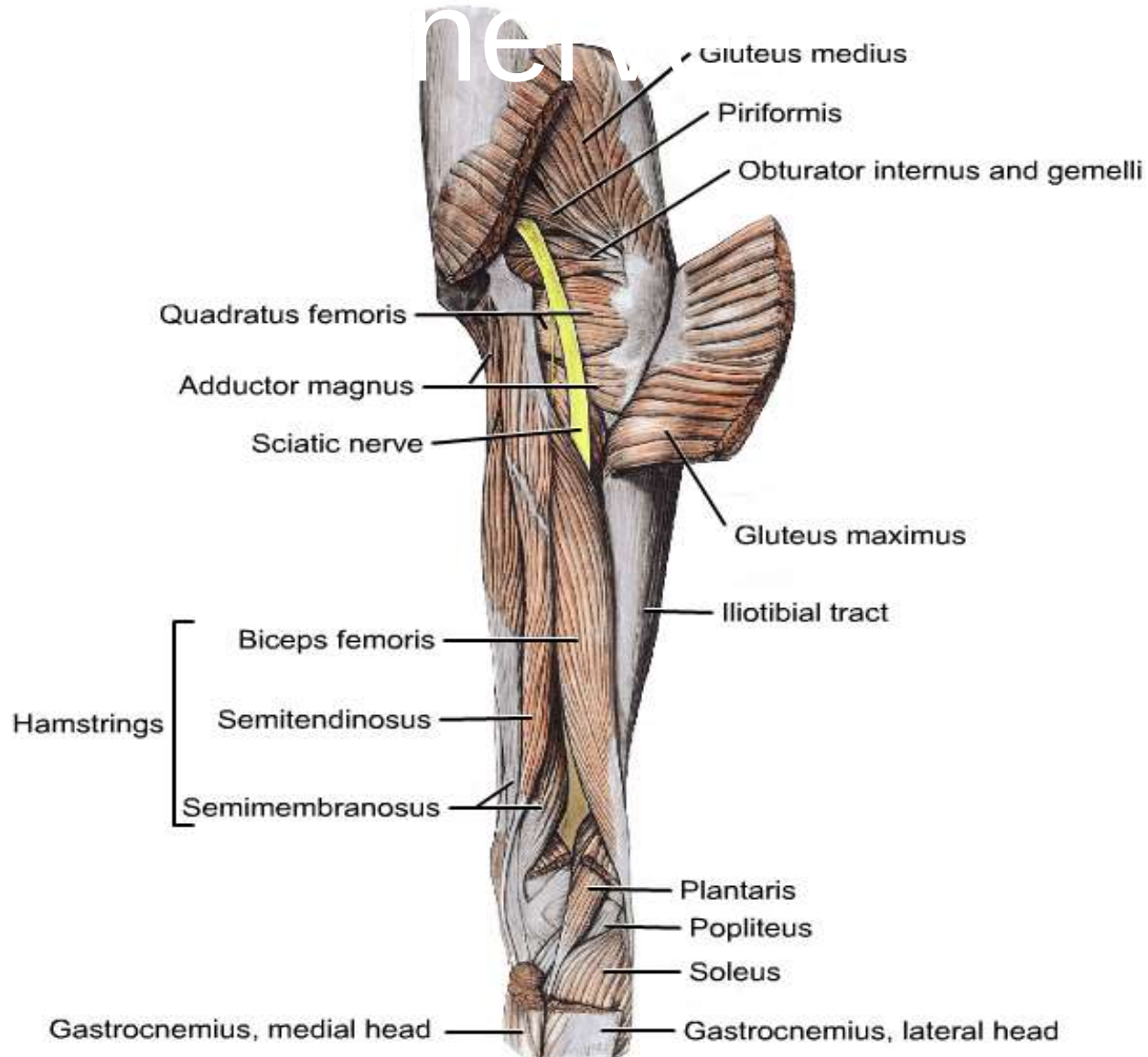


Neurovascular structures of the Gluteal Region & Posterior compartment of Thigh

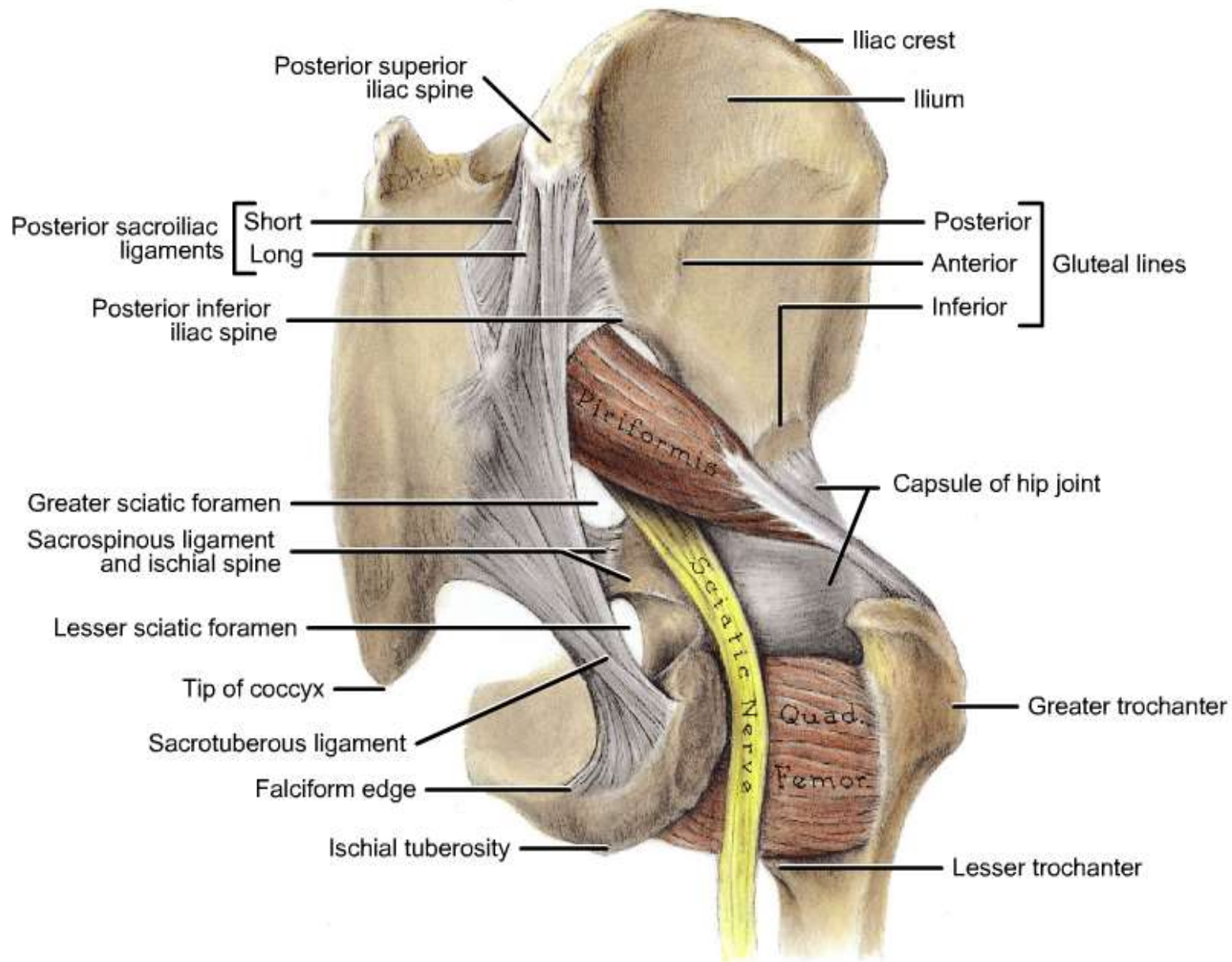
Neurovascular structures of the Gluteal Region



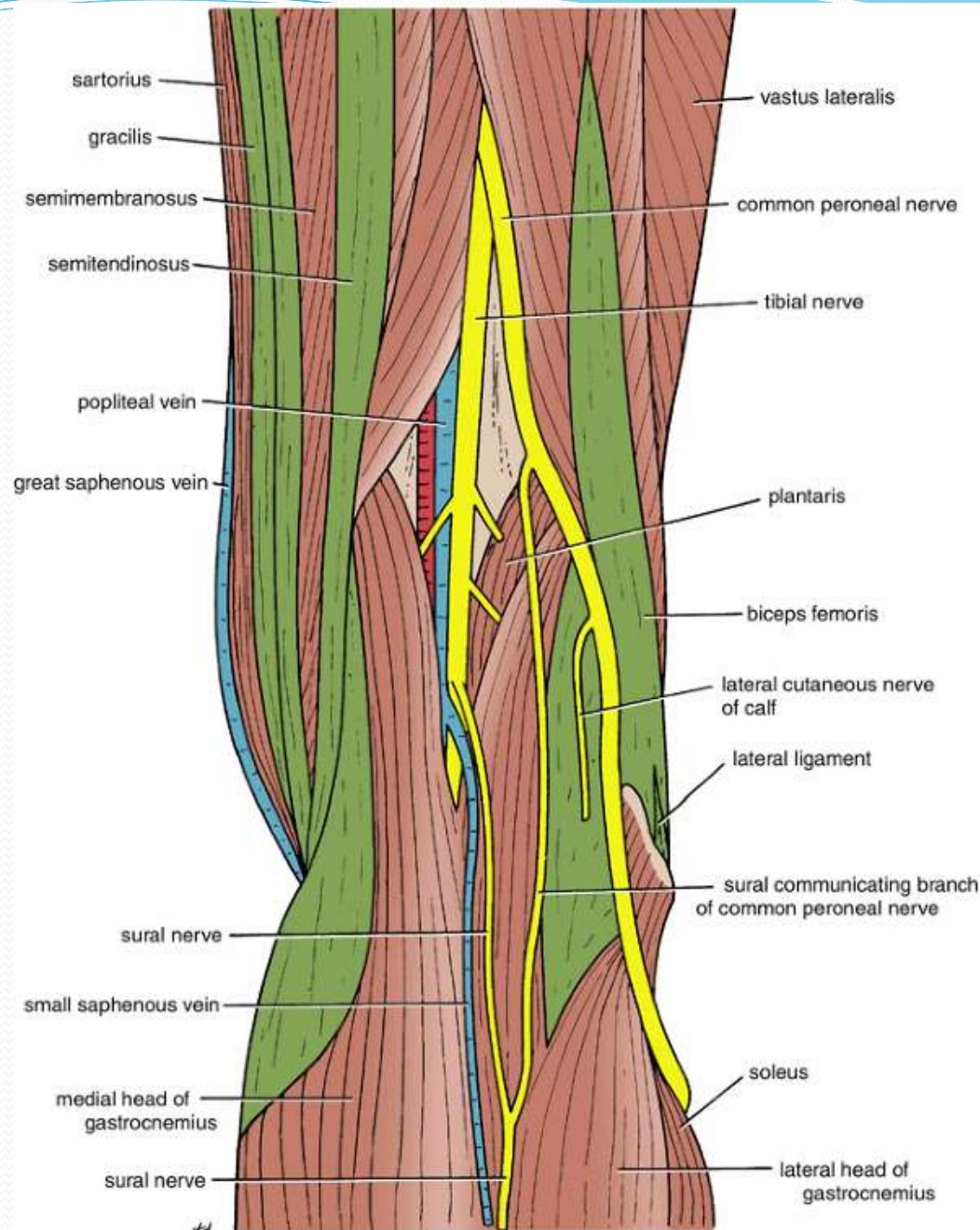
Course of Sciatic



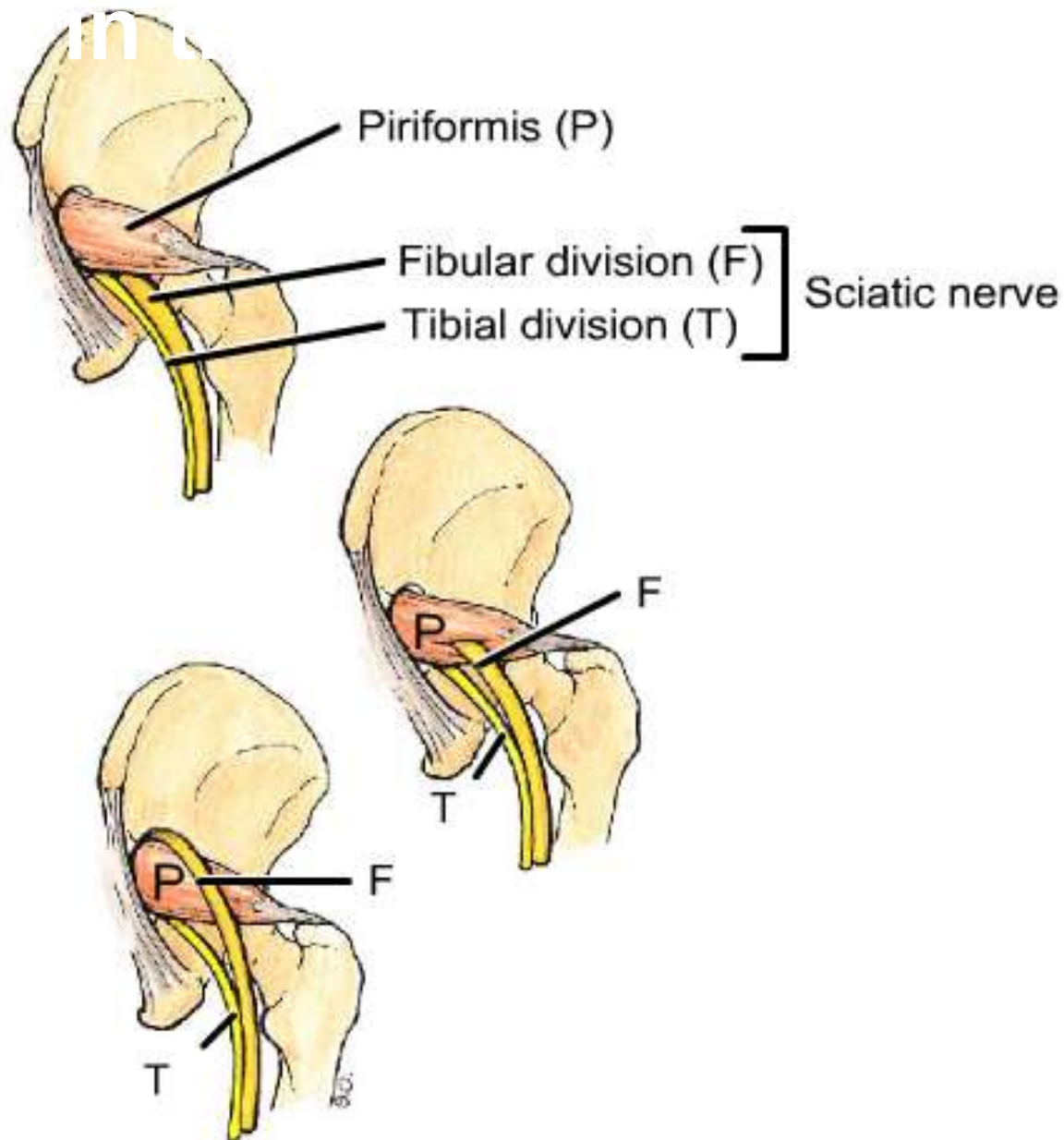
Neurovascular structures of the Gluteal Region



Course of Sciatic nerve and terminal branches

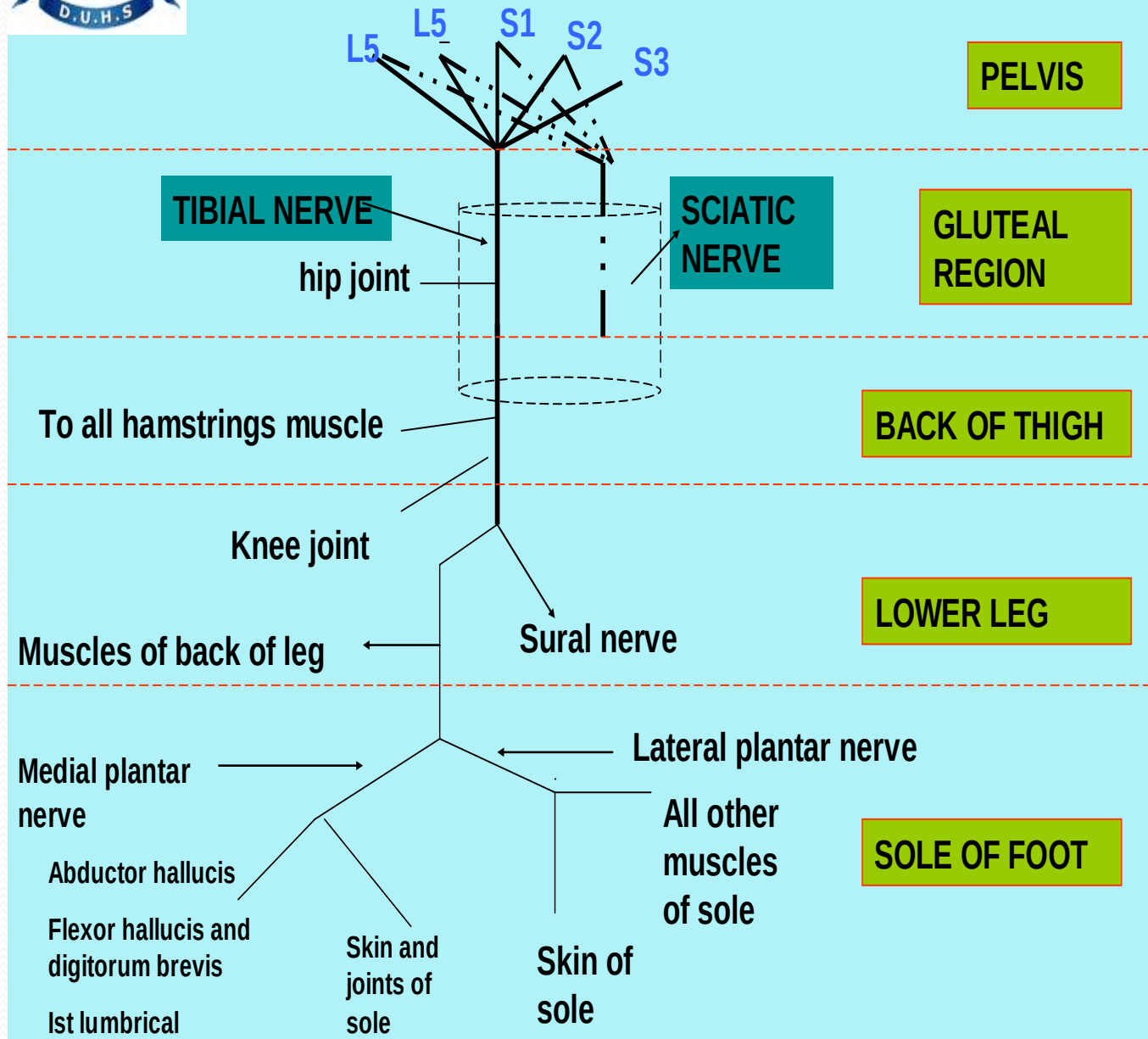


Common Variation of Sciatic nerve



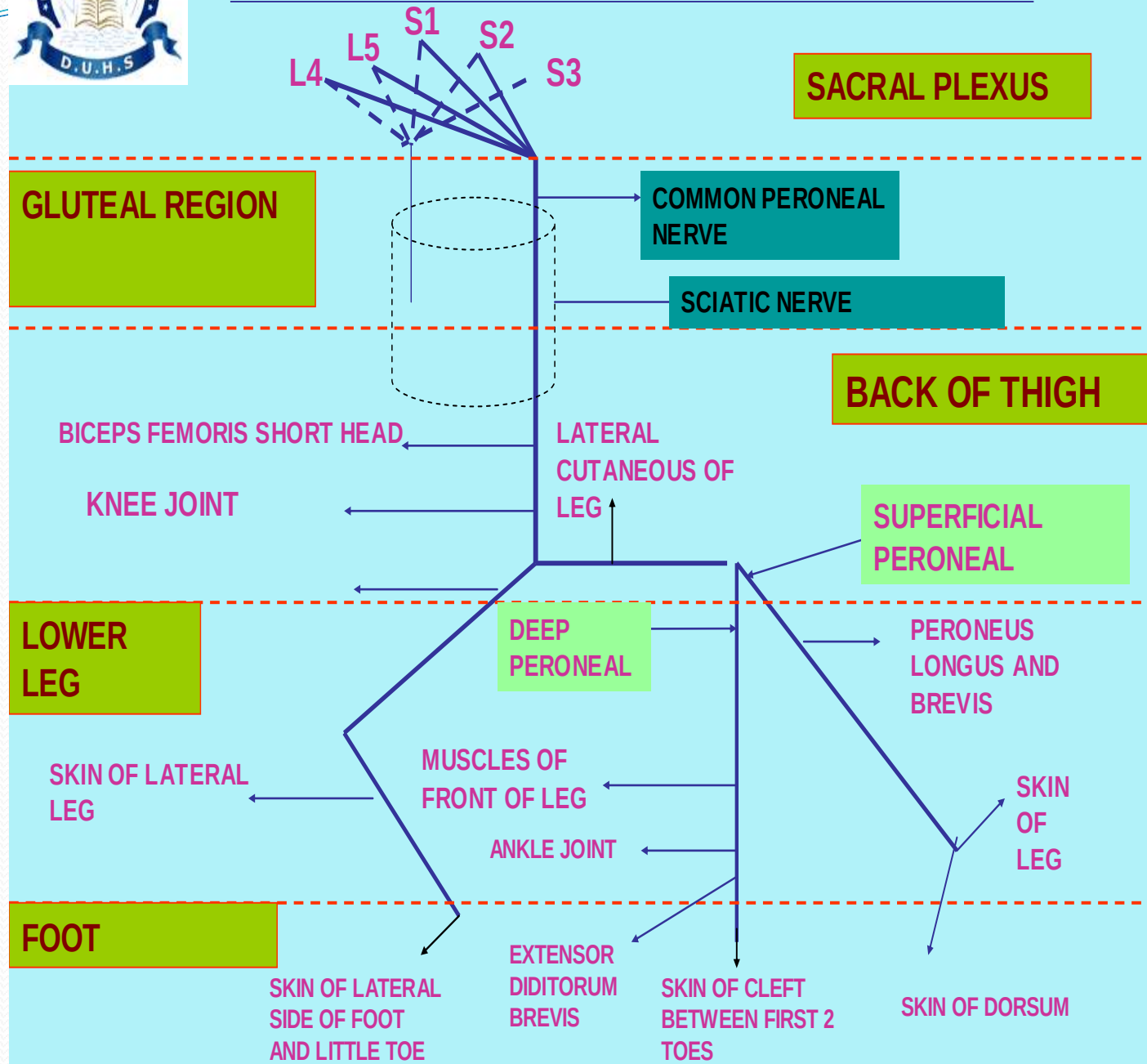


SCIATIC NERVE AND TIBIAL NERVE





SCIATIC AND COMMON PERONEAL NERVE



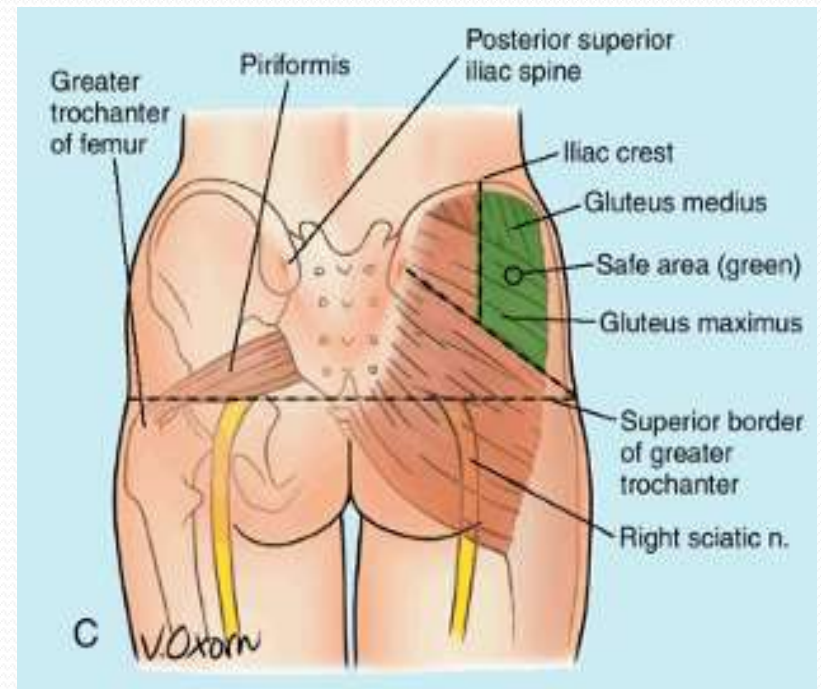
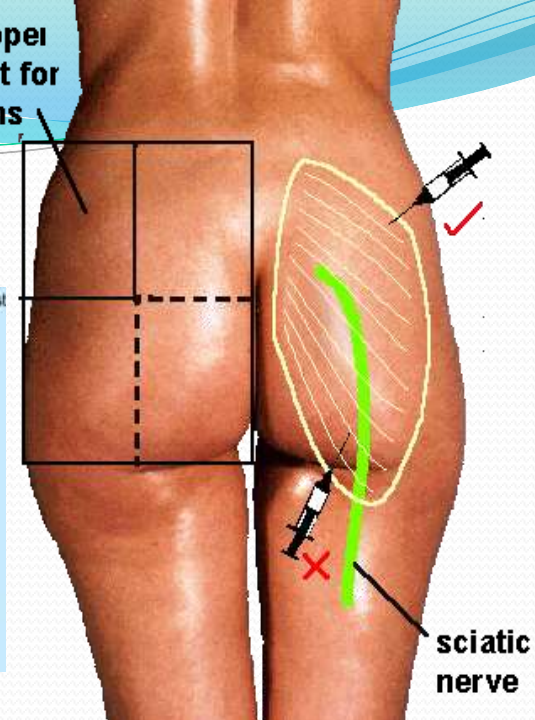


APPLIED ANATOMY

IM injections

- **Safe only in the superolateral quadrant of the buttock. OR**
- **Superior to a line extending from the PSIS to the superior border of the greater trochanter.**
- **Intramuscular injections can also be given safely into the anterolateral part of the thigh.**

outer upper quadrant for injections

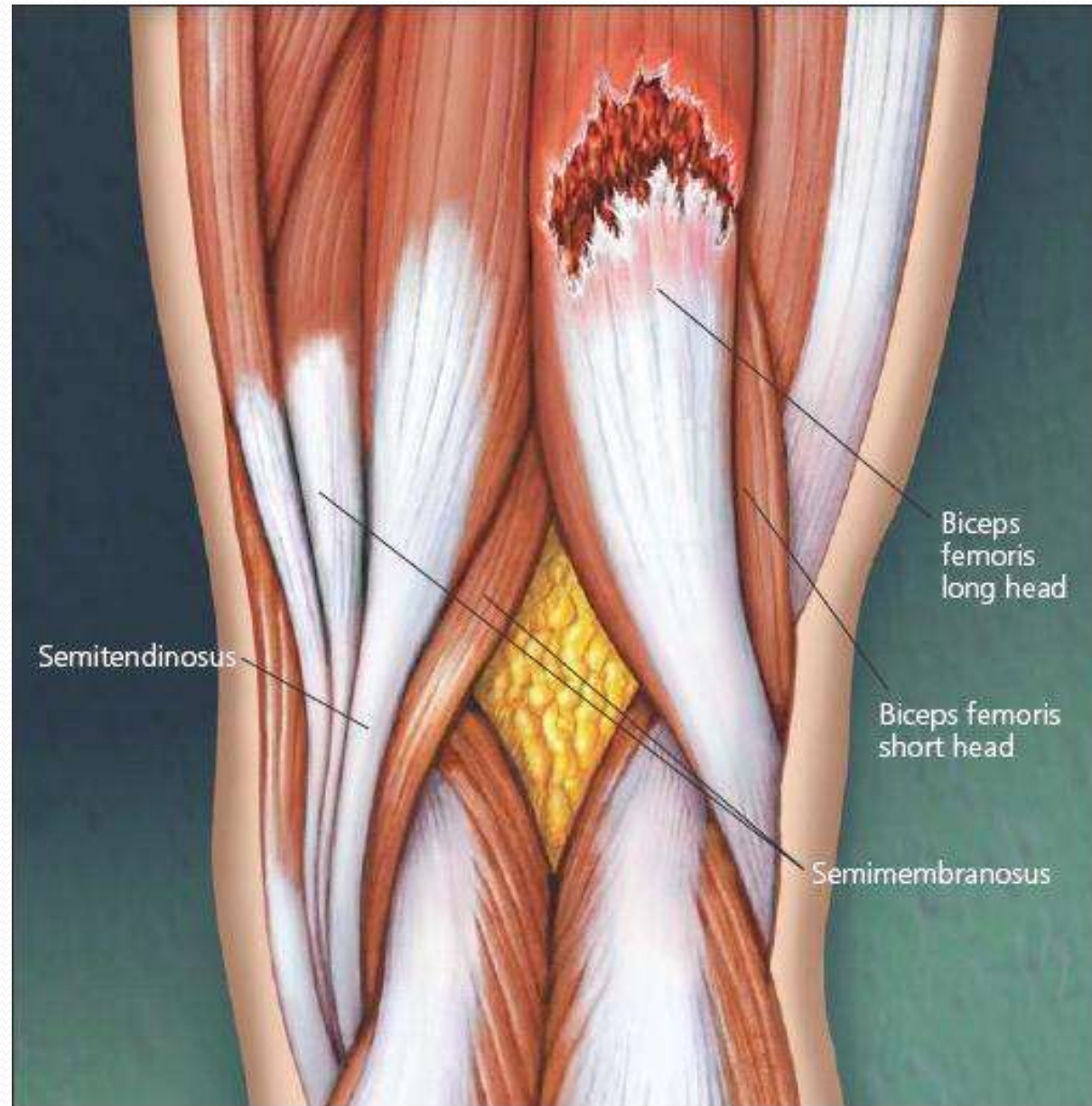


Hamstring injuries

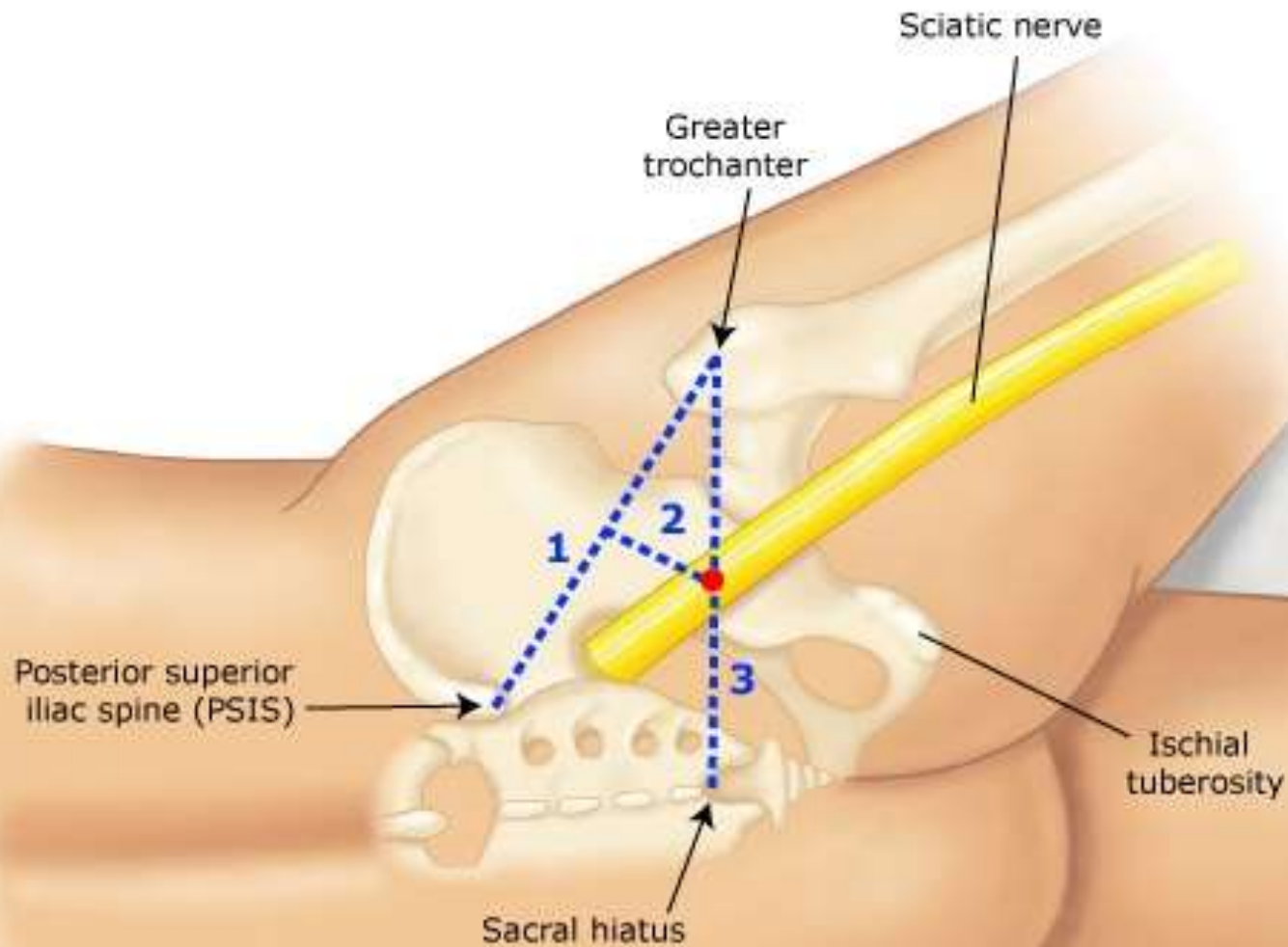
- **Hamstrings strains (pulled and/or torn hamstrings) are common in people who run and/or kick hard (e.g., in running, jumping and quick-start sports such as baseball and soccer).**
- **The violent muscular exertion required to excel in these sports may tear part of the proximal attachments of the hamstrings to the ischial tuberosity.**



Hamstring injuries



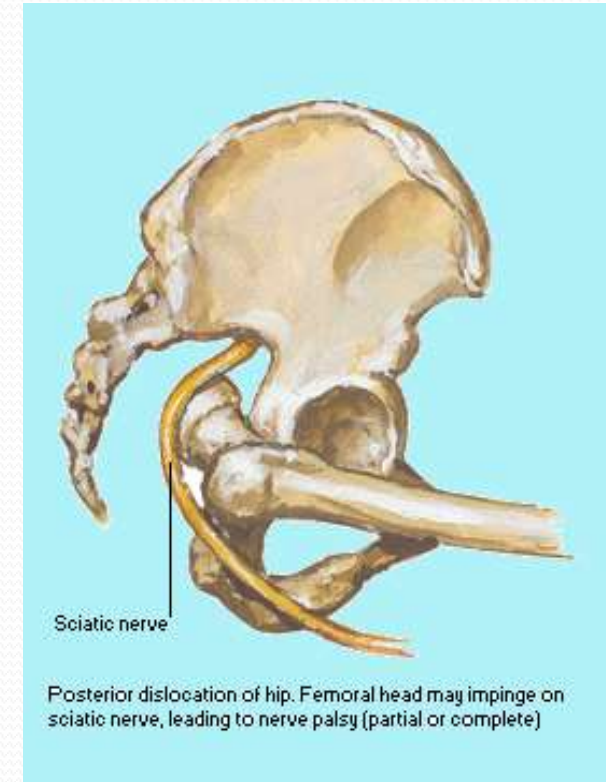
Anesthetic block of the sciatic nerve



SCIATIC NERVE INJURY

Commonly injured in following conditions:

- **IVD prolapse**
- **Dislocation of hip joint**
- **Piriformis syndrome**
- **Intramuscular injection**
- **Penetrating wound and fracture of pelvis**
 - **Most lesions are incomplete**
 - **90% cases involve common peroneal**



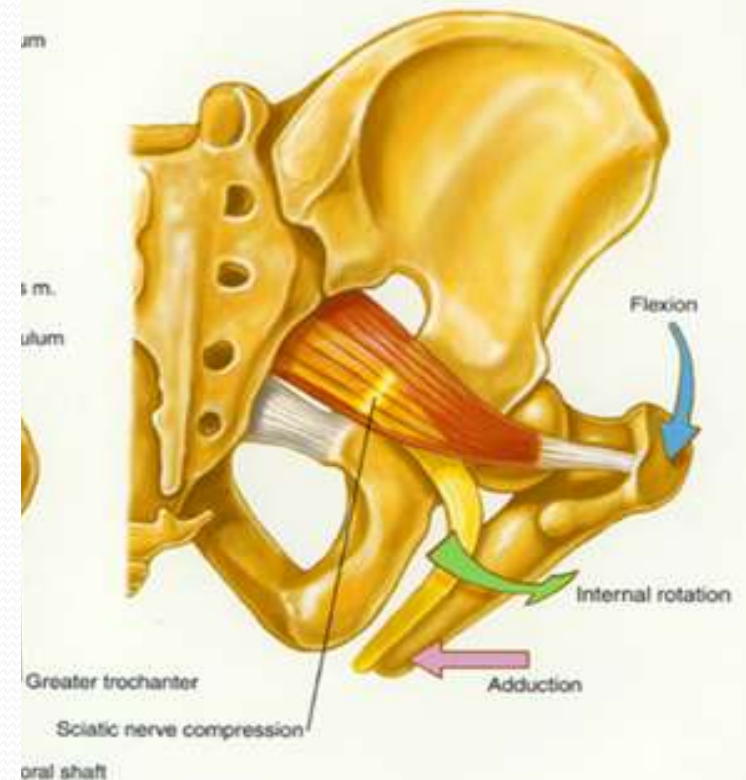
SCIATIC NERVE INJURY

- **Muscles**
 - Hamstring muscles and all the muscles below knee;
- **Motor loss**
 - Severe impairment in knee flexion
 - Loss of all movements at foot
- **Deformity**
 - Foot drop due to weight of foot.
- **Sensory loss**
 - All sensation below knee except the medial aspect of leg and foot up to ball of big toe.
 - Loss of sensation of sole makes the patient vulnerable to trophic ulcers

Injury to sciatic nerve

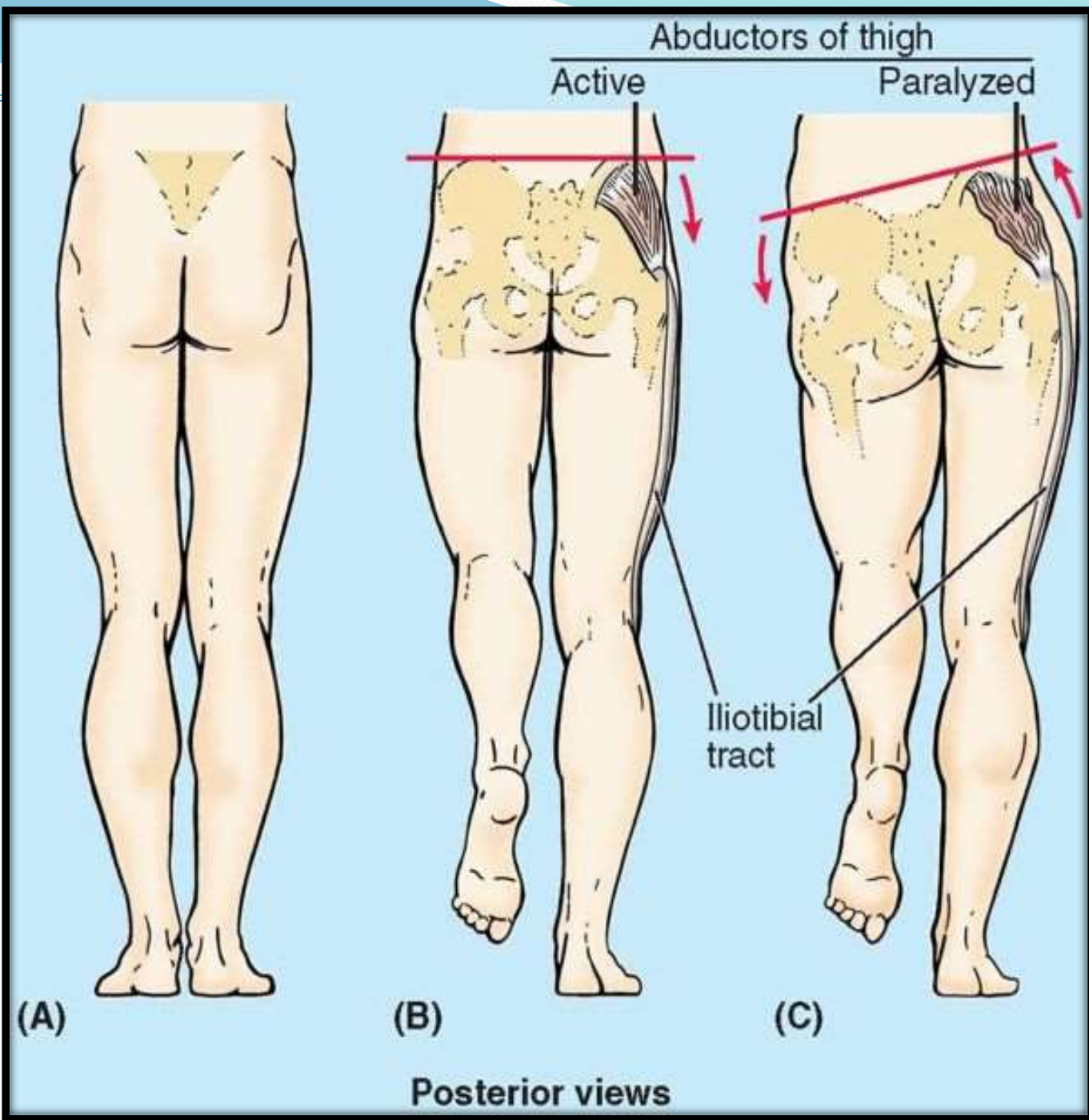
Piriformis syndrome

- A pain in the buttock may result from compression of the sciatic nerve by the piriformis muscle.
- Individuals involved in sports that require excessive use of the gluteal muscles (e.g., ice skaters, cyclists, and rock climbers) are more likely to develop this syndrome.



INJURY TO SUPERIOR GLUTEAL NERVE

- Muscles paralyzed
 - Gluteus medius
 - Gluteus minimus
- Motor loss
 - Loss of abduction of hip
 - **Drooping of pelvis on one side when ipsilateral foot is lifted off the ground.**



INJURY TO INFERIOR GLUTEAL NERVE

- Muscle paralyzed:
 - Gluteus maximus muscle
- Motor loss
 - Impairment of hip extension and lateral rotation
 - Difficulty in raising the body from sitting or stooping position

Inferior Gluteal Nerve Palsy



Difficulty in arising from chair is often an early complaint



Difficulty stepping into bus



Difficulty in climbing stairs is often an early symptom due to weakness of pelvic girdle muscles

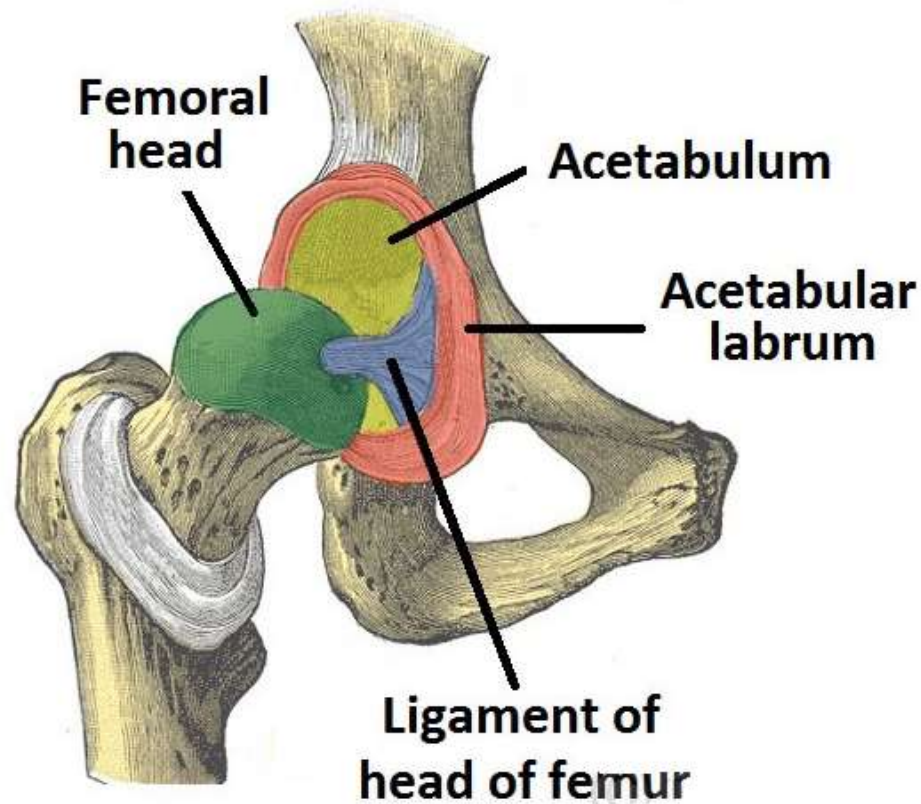


Hip Joint

The Hip Joint

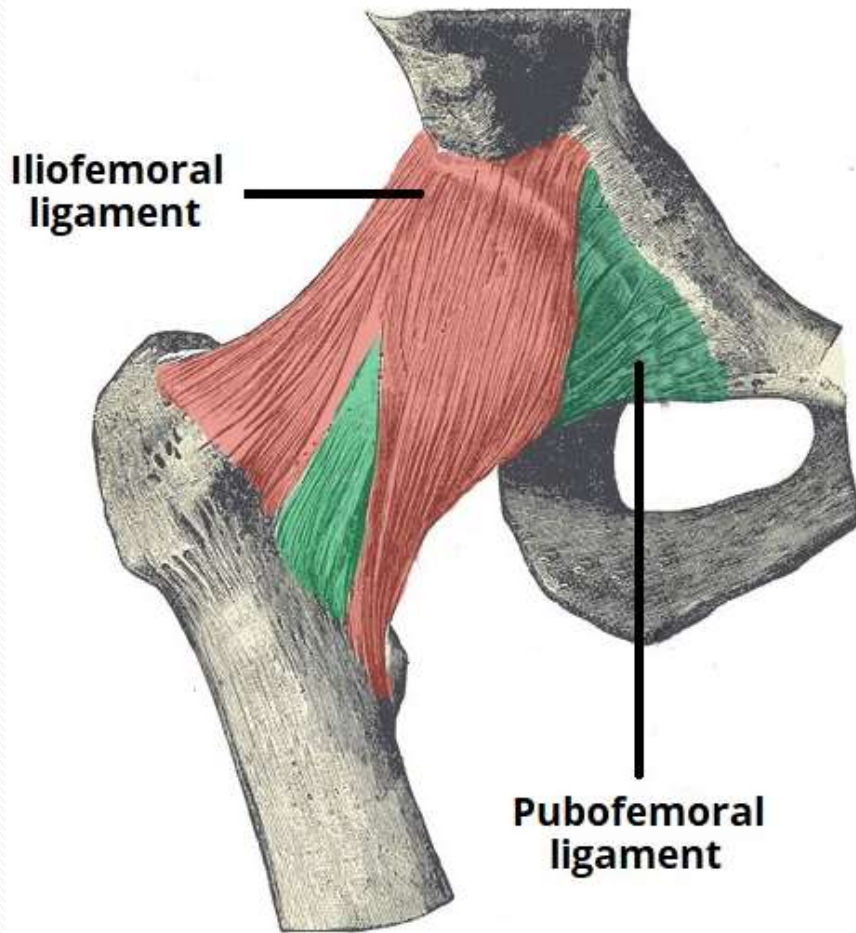
- Ball and socket synovial joint
- Formed by an articulation between acetabulum and the head of the femur.
- Designed for stability and **weight-bearing**

The Hip Joint: Types & articular surfaces

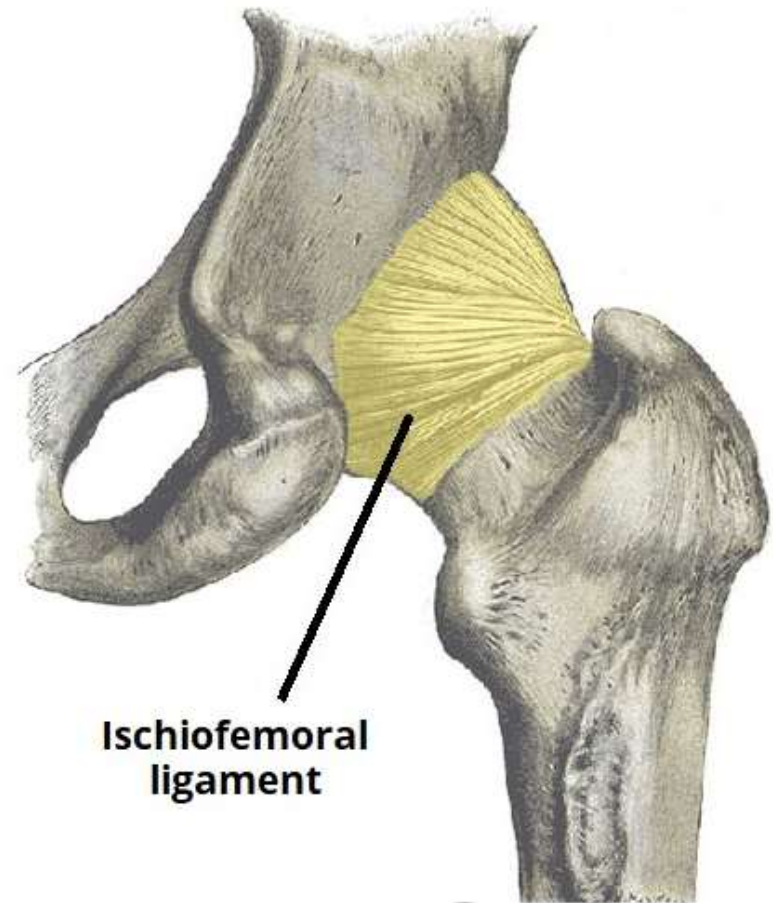


Ligaments & their stabilizing role

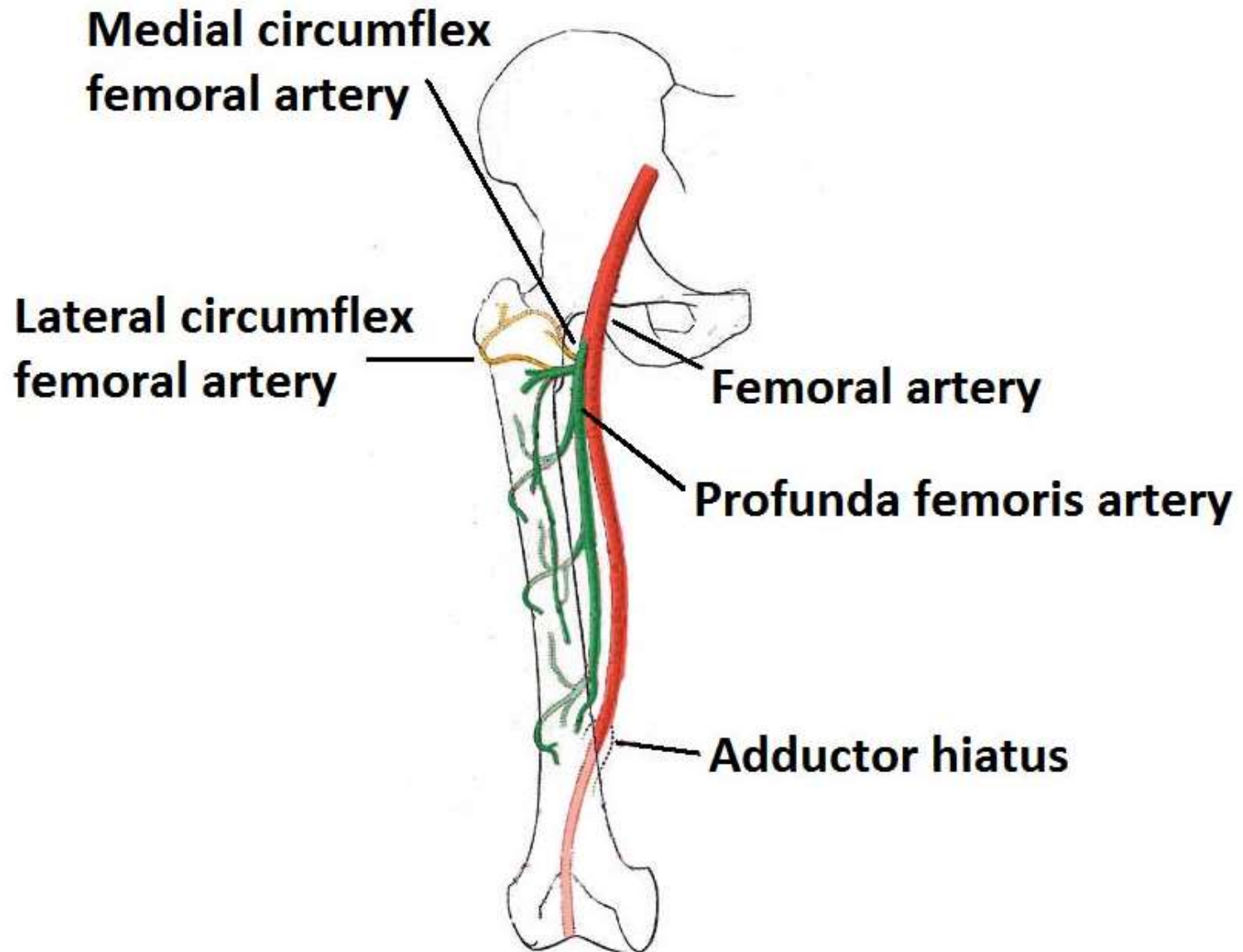
Anterior



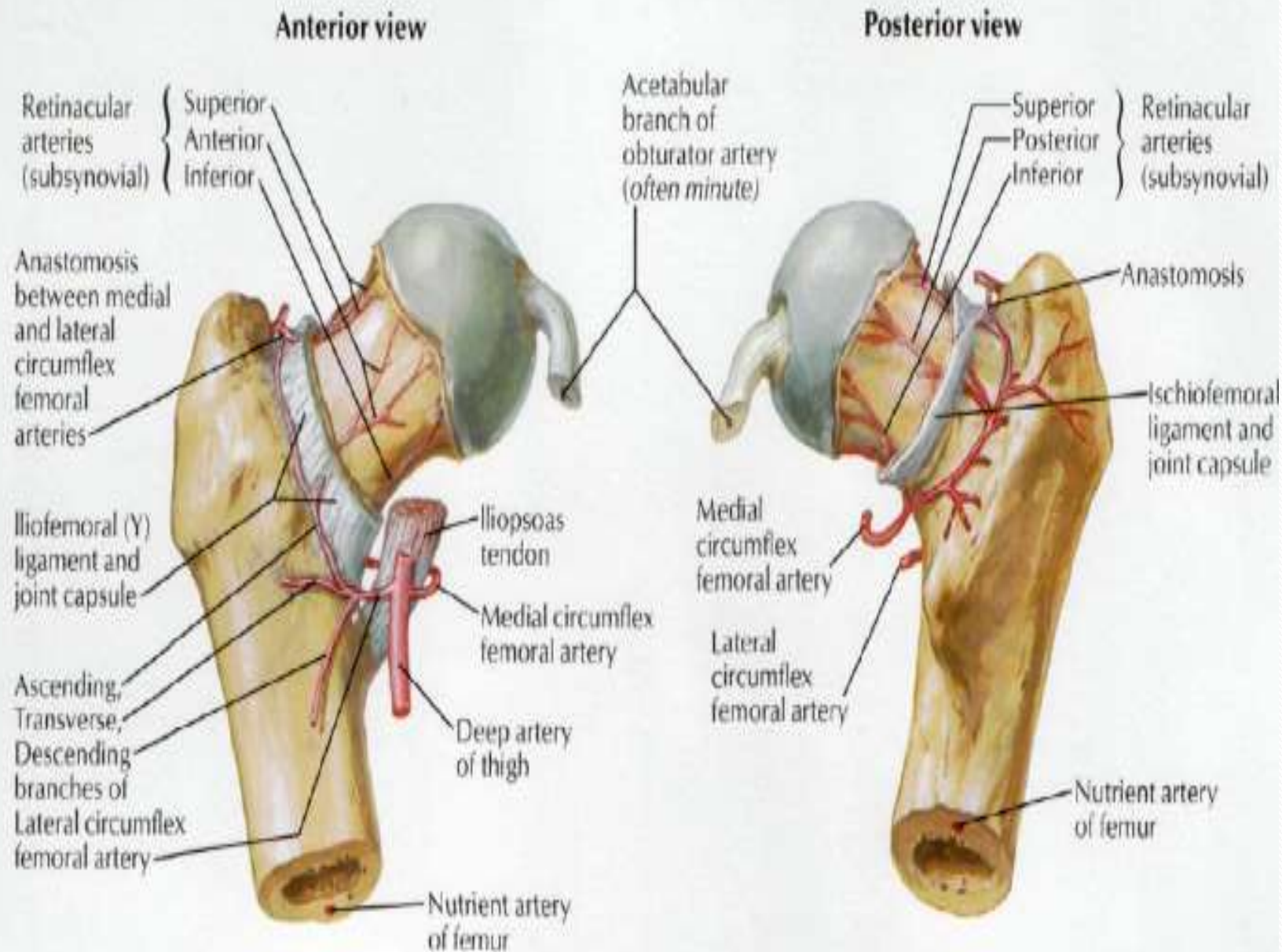
Posterior



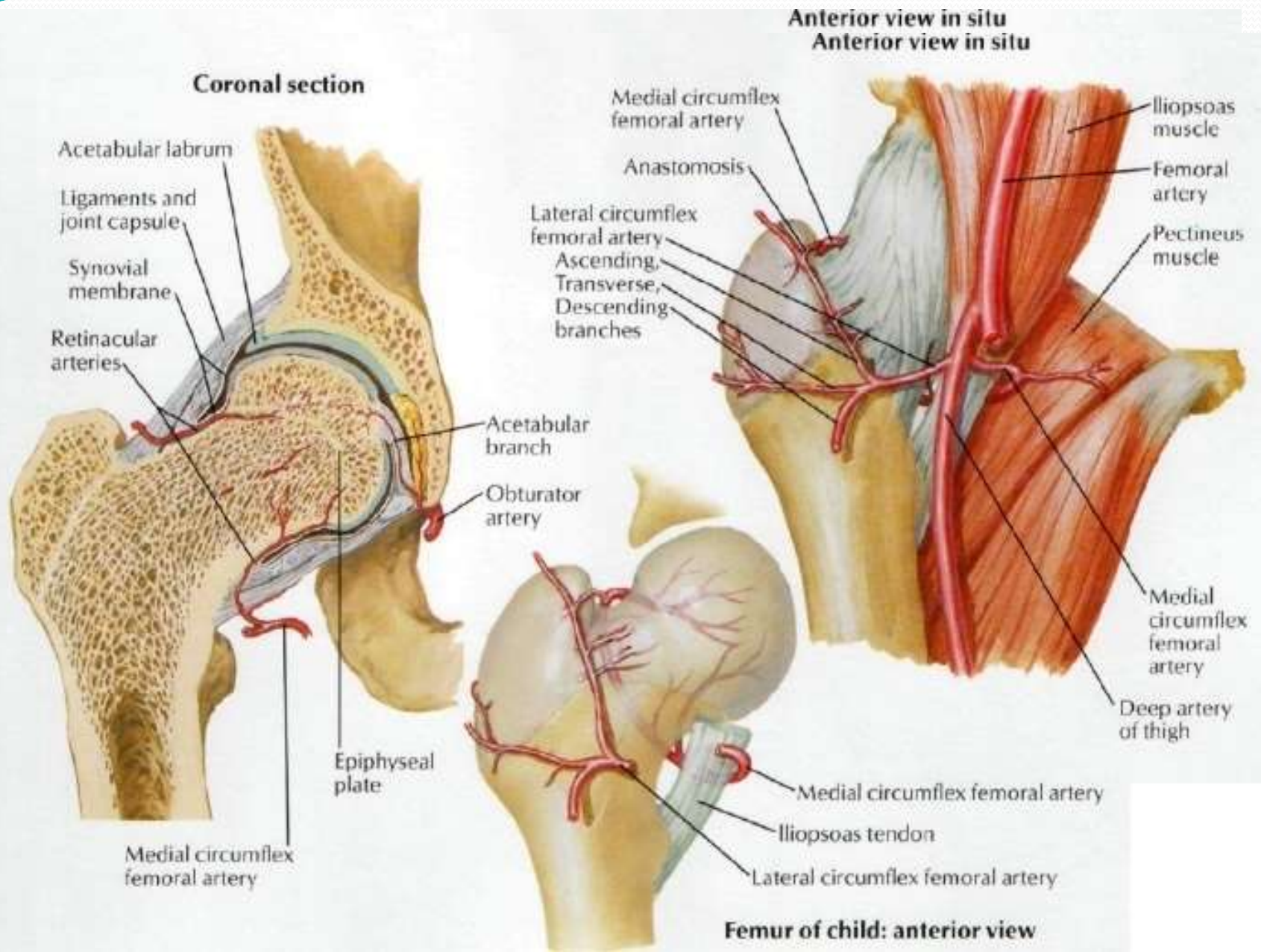
Neurovascular Supply



Arteries of femoral head & neck

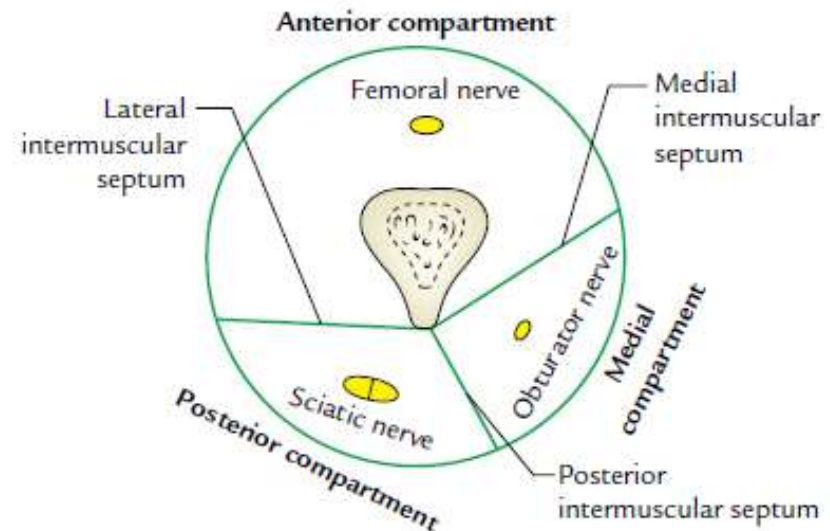
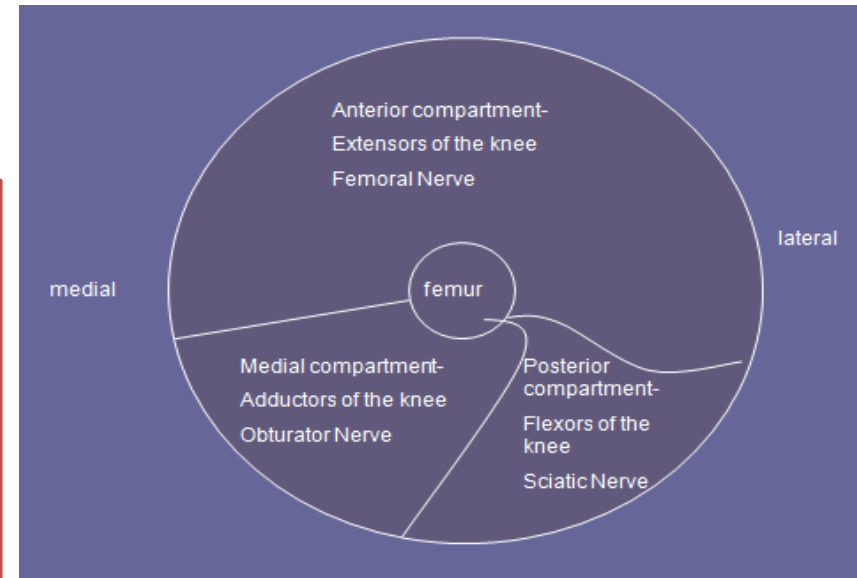


Arteries of femoral head & neck



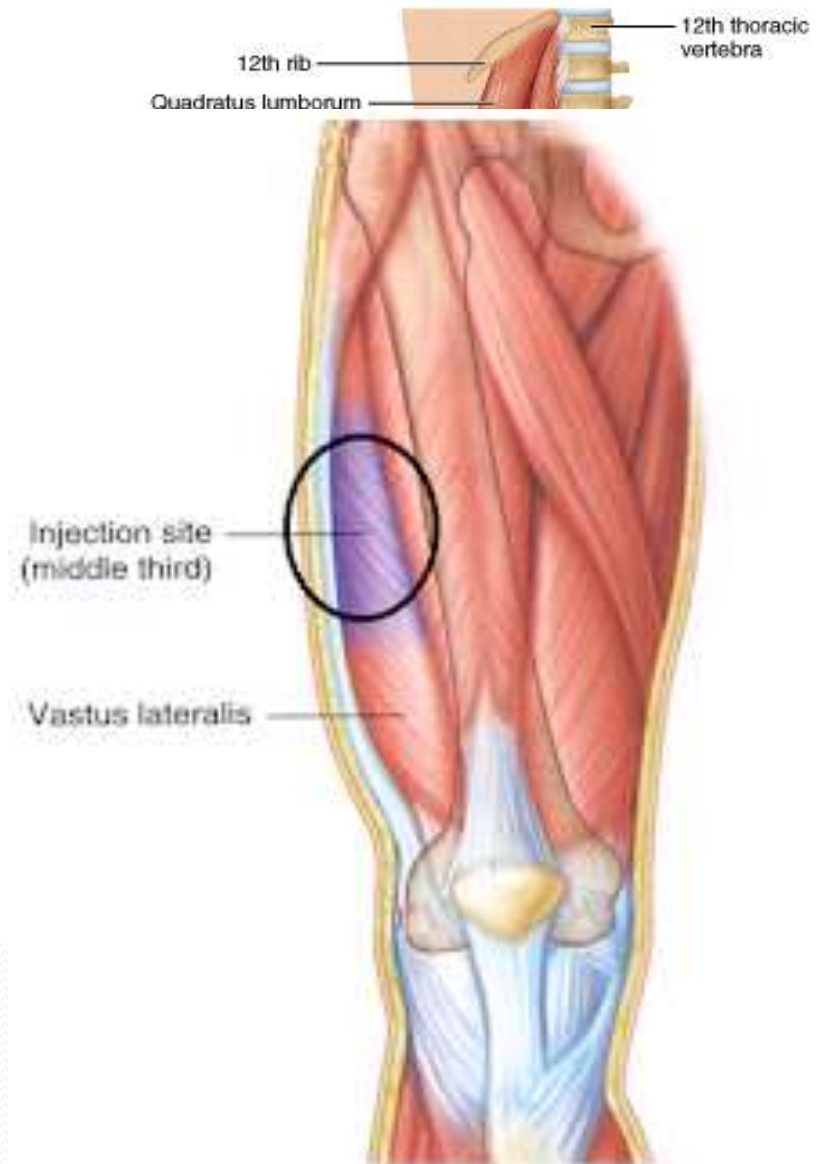
Thigh/Femoral Region

- **Thigh muscles** are organized into 3 compartments by **intermuscular septa**
 - **Anterior or extensor**
 - **Medial or adductor**
 - **Posterior or flexor**

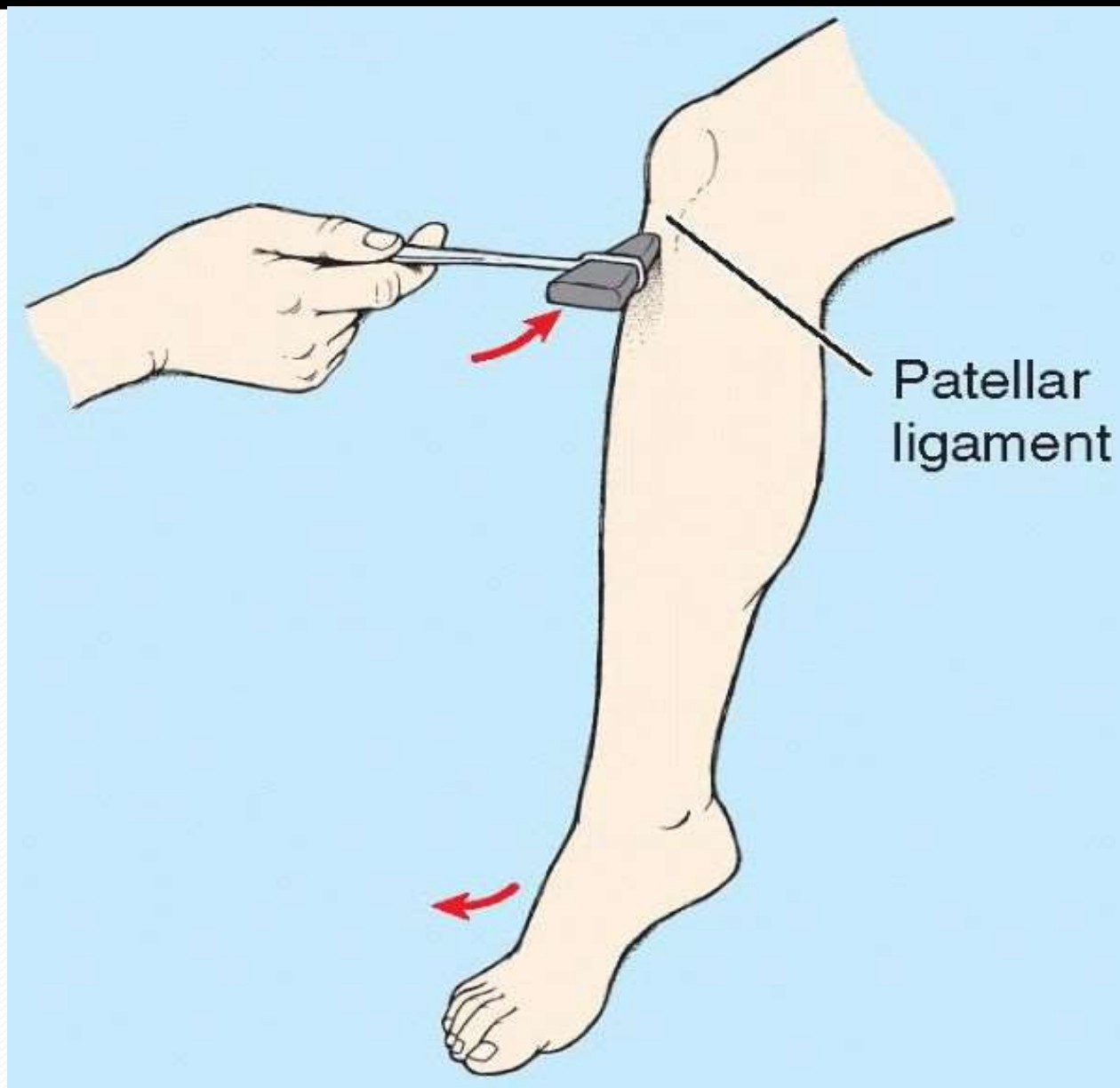


Anterior thigh muscles

- Found in anterior compartment of thigh
- ❑ Flexors of hip and extensors of knee
- **Sartorius**
- **Quadriceps femoris**
- ✓ Rectus femoris
- ✓ Vastus medialis
- ✓ Vastus lateralis
- ✓ Vastus intermedius
- ❖ **All insert into tibial tuberosity by a common distal tendon - quadriceps tendon**

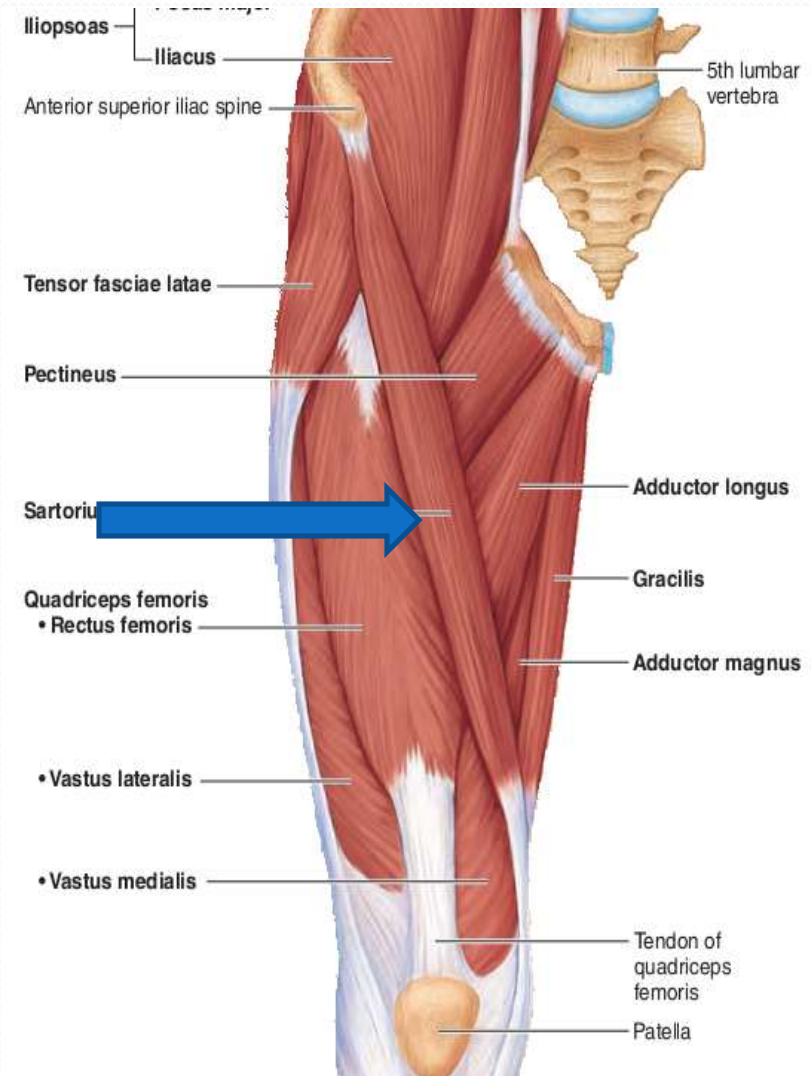


Patellar tendon reflex



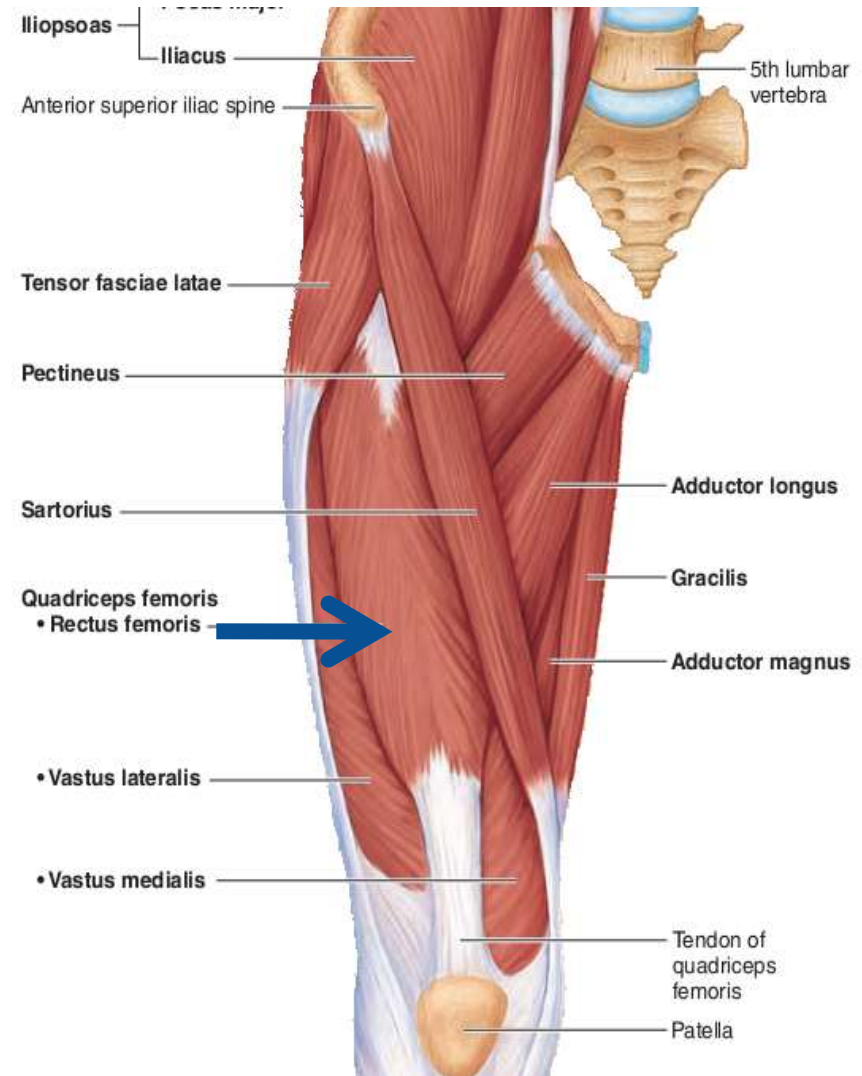
Sartorius

- Long, ribbon-like muscle
- The most superficial muscle in the anterior thigh
- Passes obliquely (lateral to medial)
- **Innervation**
 - **femoral nerve**
- **Action**
 - Flexes, abducts and laterally rotates thigh at hip joint
 - Flexes leg at knee joint
 - It provide **cross-legged sitting** position; "tailor's muscle"
 - **"Honey moon muscle"**



Rectus femoris

- **Origin** - anterior inferior iliac spine (by its straight head) and acetabular margin (by its reflected head)
- **Insertion** - patella and tibial tuberosity
- **Action** - **extends knee and flexes thigh at hip**
- **Innervation** - femoral nerve



Muscles of Hip

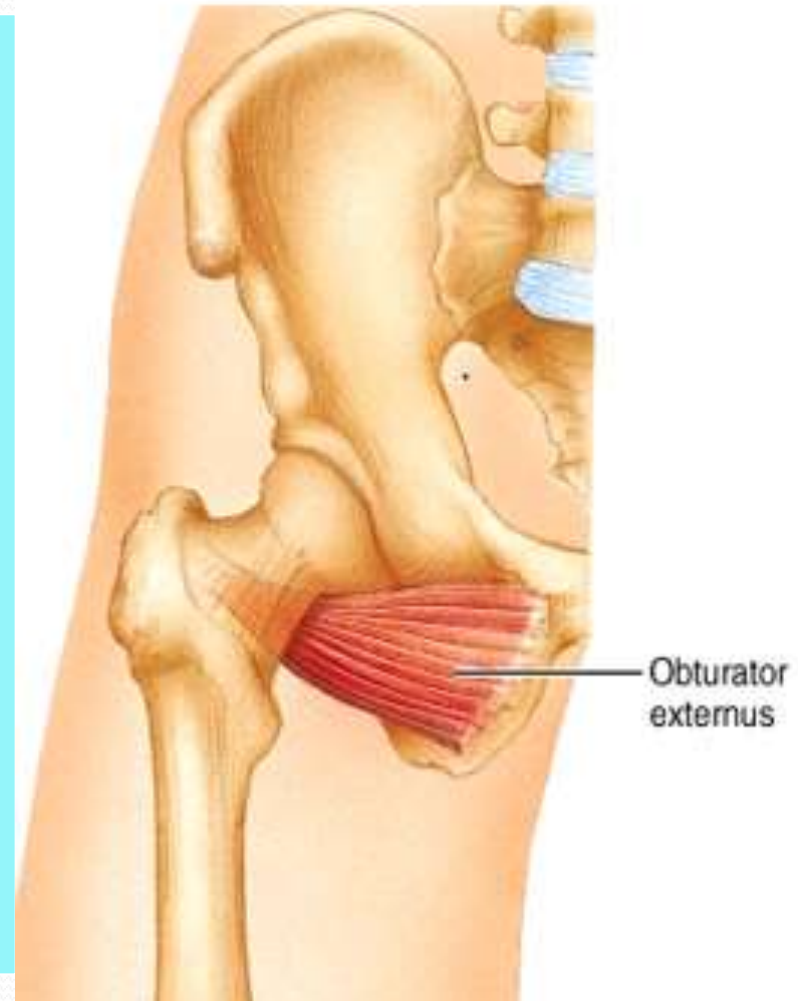
- Medial muscles of the hip
 - **Psoas major** – chief flexor
 - **Psoas minor**
 - **Iliacus**
- **Iliacus**
- Origin - iliac fossa
- Insertion - lesser trochanter of femur
- Action - **hip flexion**
- Innervation - femoral nerve abdominal branch



What is psoas positive ?

Medial thigh muscles

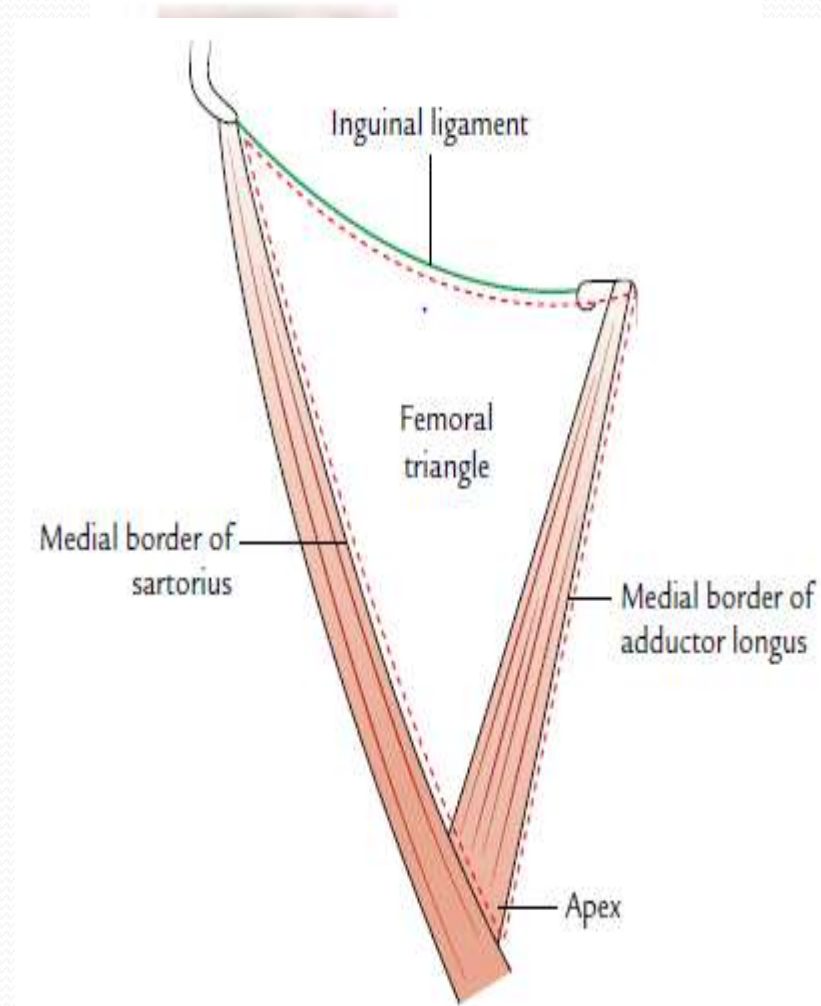
- Collectively called **adductor group**
- Found in medial compartment
- **Adductor muscles are**
 - Adductor magnus
 - Adductor longus
 - **Adductor brevis**
 - **Pectineus**(by two)
 - **Gracilis**
 - **Obturator externus**
 - **All innervated by Obturator nerve**



Femoral triangle

□ Boundaries

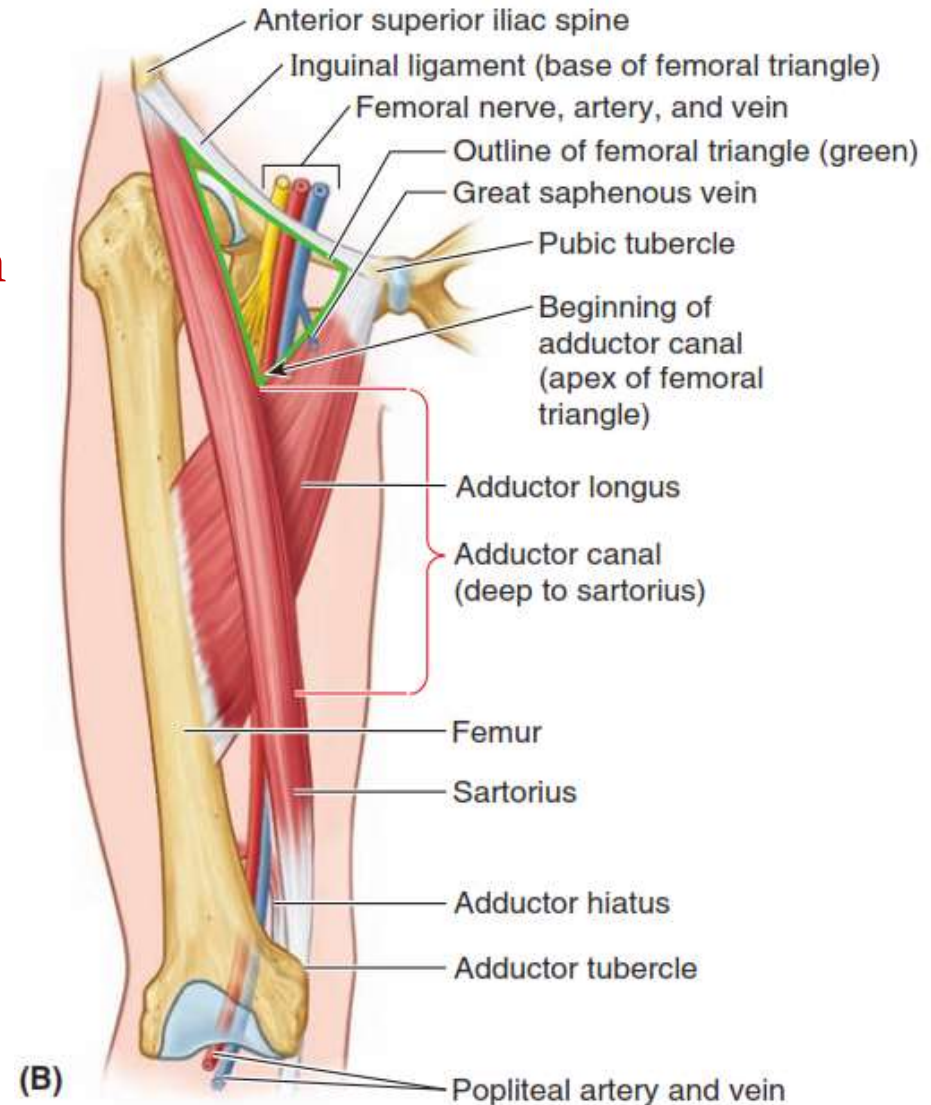
- Medial – **medial boarder of adductor longus**
- Lateral - **medial boarder of Sartorius**
- Superior - **inguinal ligament**
- Floor - pectineus, psoas and iliacus
- Roof - formed by
 - superficial fascia, containing superficial inguinal lymph nodes and great saphenous vein with its tributaries



Femoral triangle

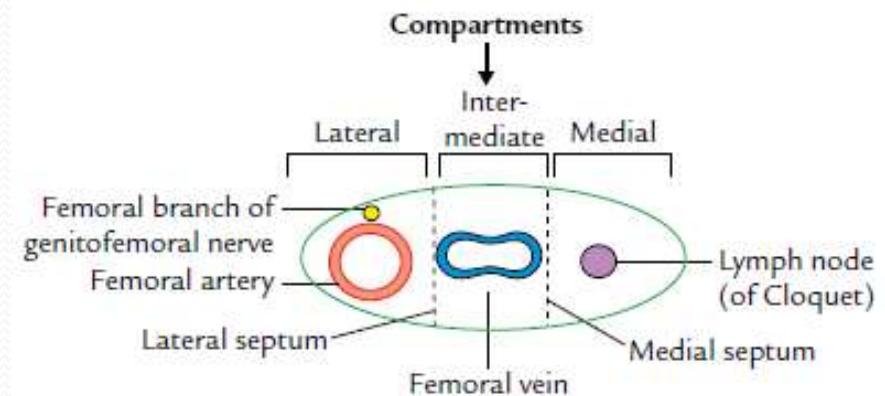
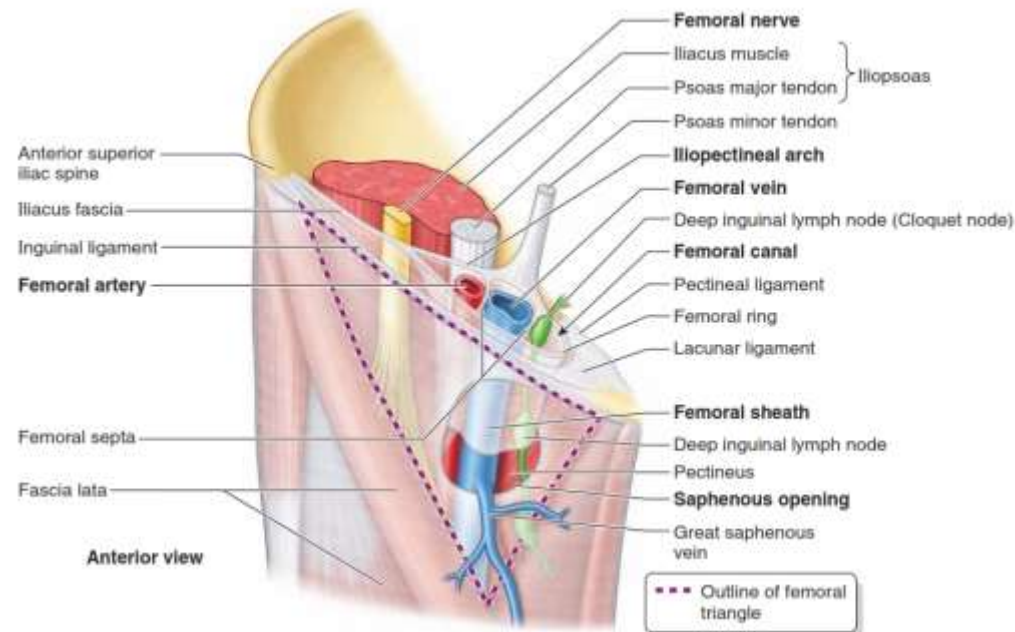
□ Contents

- **Femoral sheath**
- **Femoral artery and its branch**
- **Femoral vein and its tributaries**
- **Femoral nerve and its branches**
- **Deep inguinal lymph nodes**
- **Profunda femoris artery and vein and their circumflex branches**



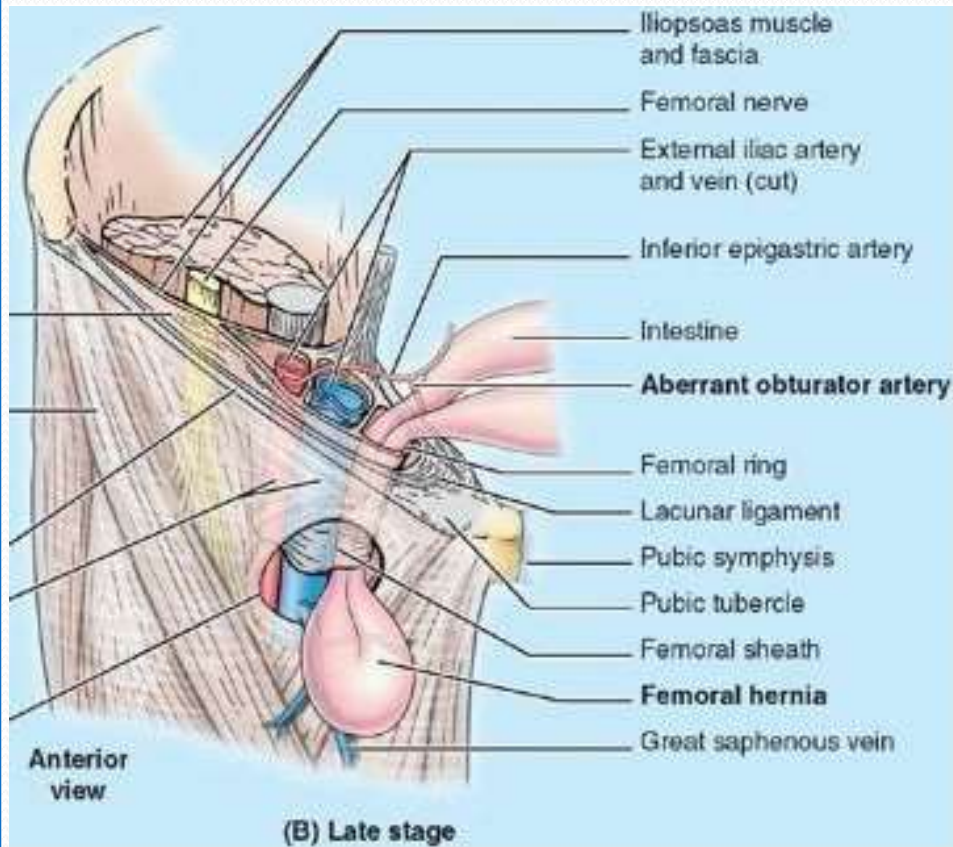
Femoral sheath

- Prolongation of extra peritoneal areolar tissue
- **Has 3 compartments and two septa**
 - Lateral for the **femoral artery** and **genital branch of genitofemoral nerve**
 - Middle for the **femoral vein**
 - Medial for the **femoral canal**
 - contain lymph vessels and deep lymph nodes
 - **It is the site of femoral hernia**
 - **It is most common in female why?**



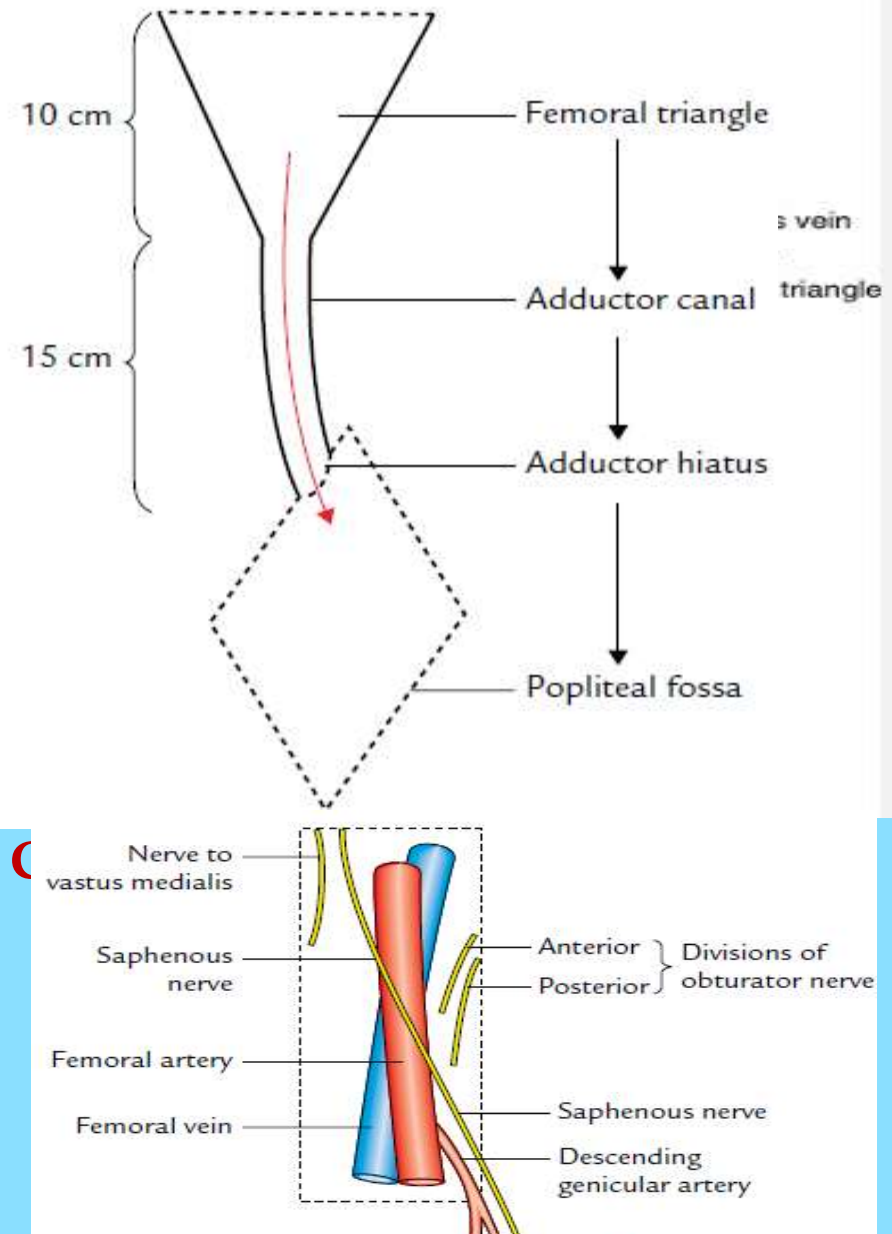
Femoral canal

- Medial part of the femoral sheath
- Small, vertically placed gap
- Larger in female due to the greater width of the female pelvis
 - Femoral hernias are more common in female



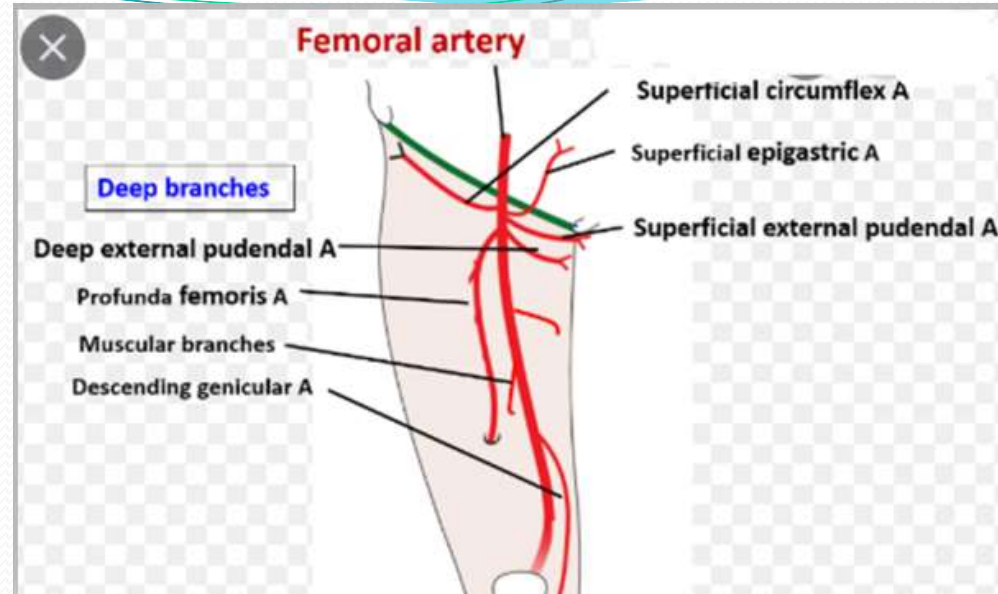
Adductor (sub sartorial) canal

- Leads on from the apex of the femoral triangle
- Ends 10cm above the **adductor tubercle**
- Triangular
- Its walls (boundaries) are:
 - **Posteriorly** - adductor longus and magnus
 - **Anterolaterally** - vastus medialis
 - **Anteromedial** - sartorius



Blood vessels

- **Femoral artery**
- **Origin**
 - Main continuation of **external iliac** distal to inguinal ligament
- **Distribution:** anterior and anteromedial thigh
- **Branches**
 - Superficial branches of femoral artery
 - Superficial external pudendal artery
 - Superficial epigastric artery
 - Superficial circumflex iliac artery

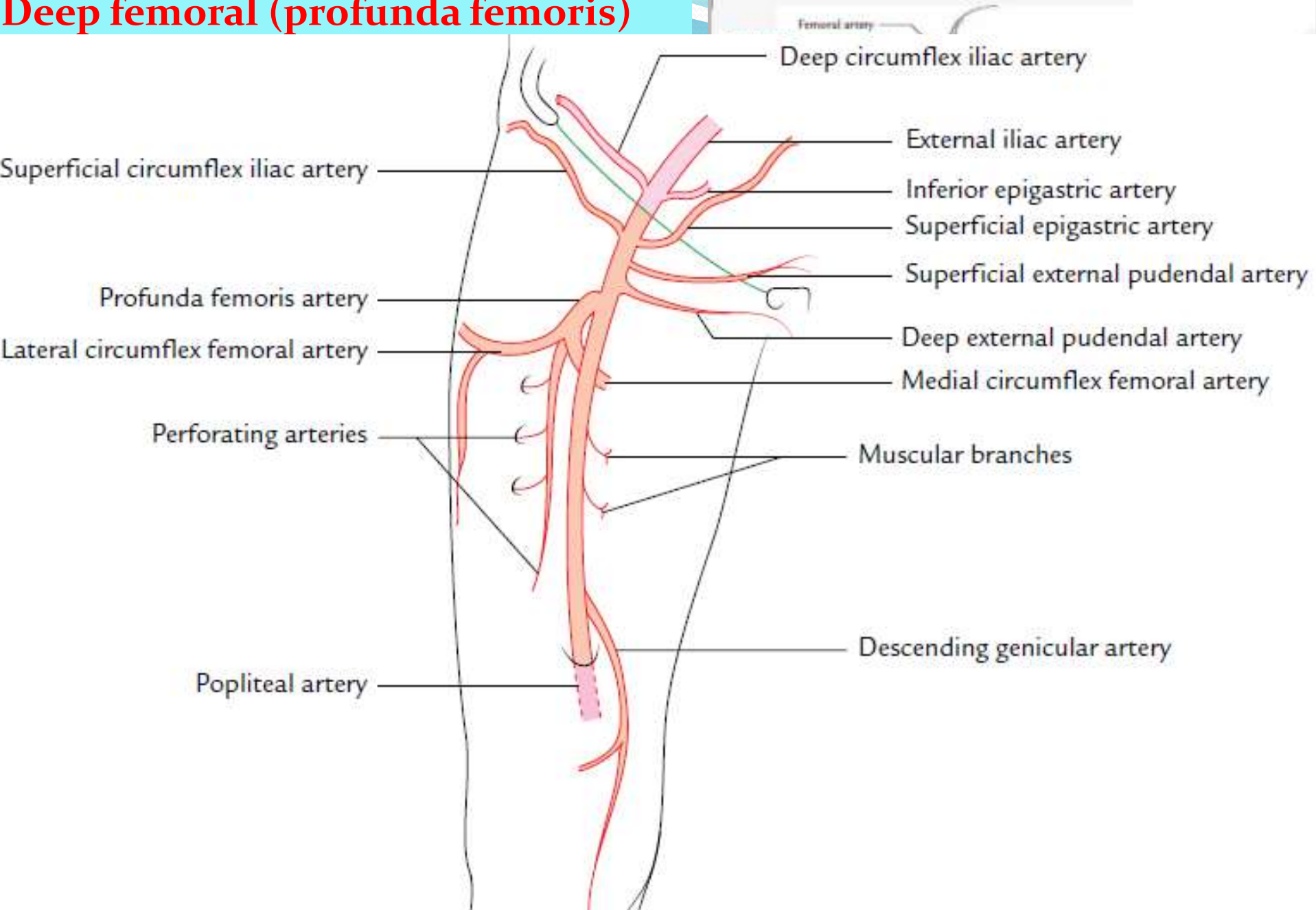


Clinical correlation

Compression, palpation and cannulation of femoral artery:

- The femoral artery can be compressed against the femoral head at the midinguinal point to control the bleeding in the distal part of the limb.
- The pulsations of the femoral artery are felt by the clinicians in the femoral triangle just below the midinguinal point.
- Since the femoral artery is quite superficial in the femoral triangle, it is the preferred artery for cannulation and injecting dye to perform procedures like angiography. It is also the preferred vessel for performing the coronary angiography and angioplasty.

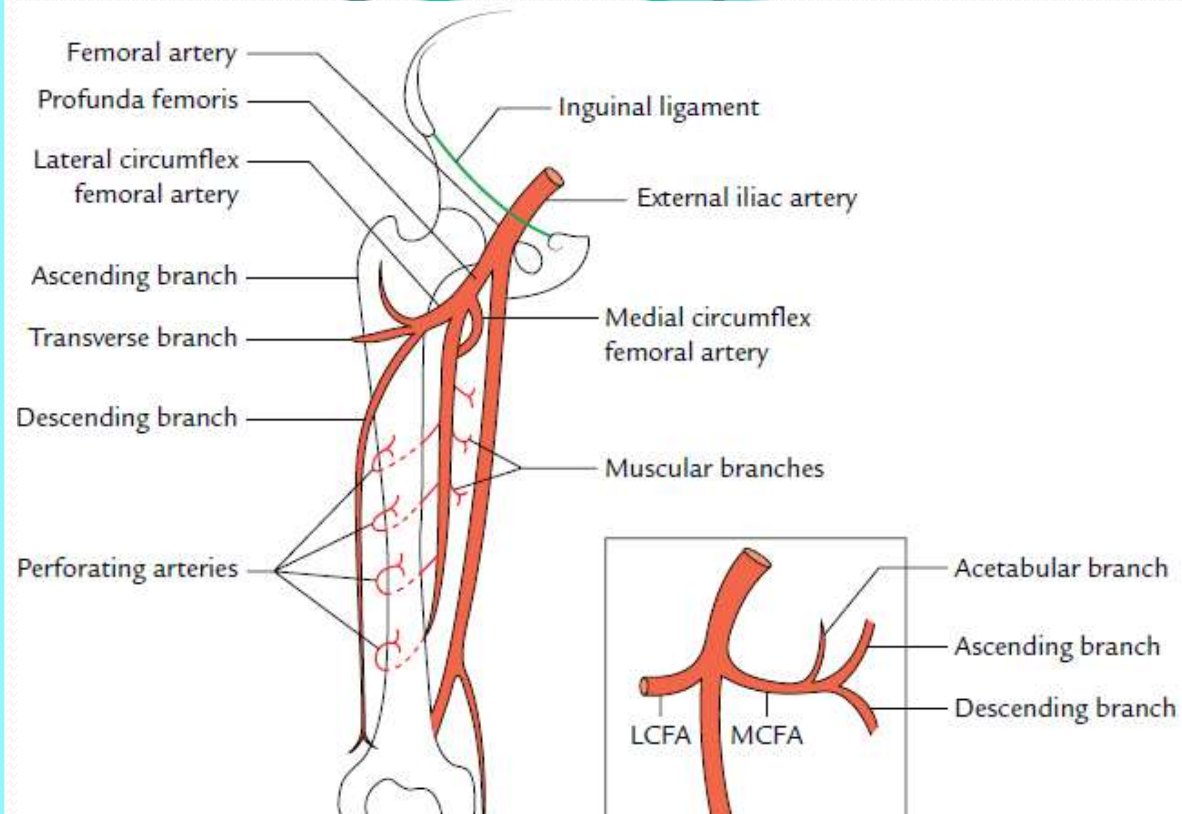
Deep femoral (profunda femoris)



and descending branches

Deep branch of femoral artery

- Four perforating branches and terminates at the fourth perforating

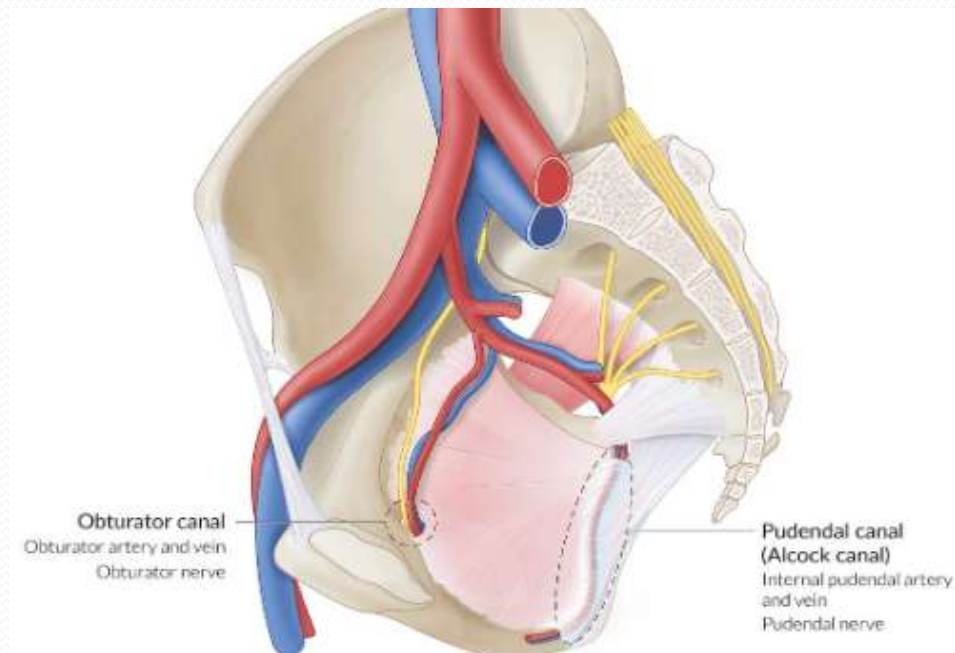
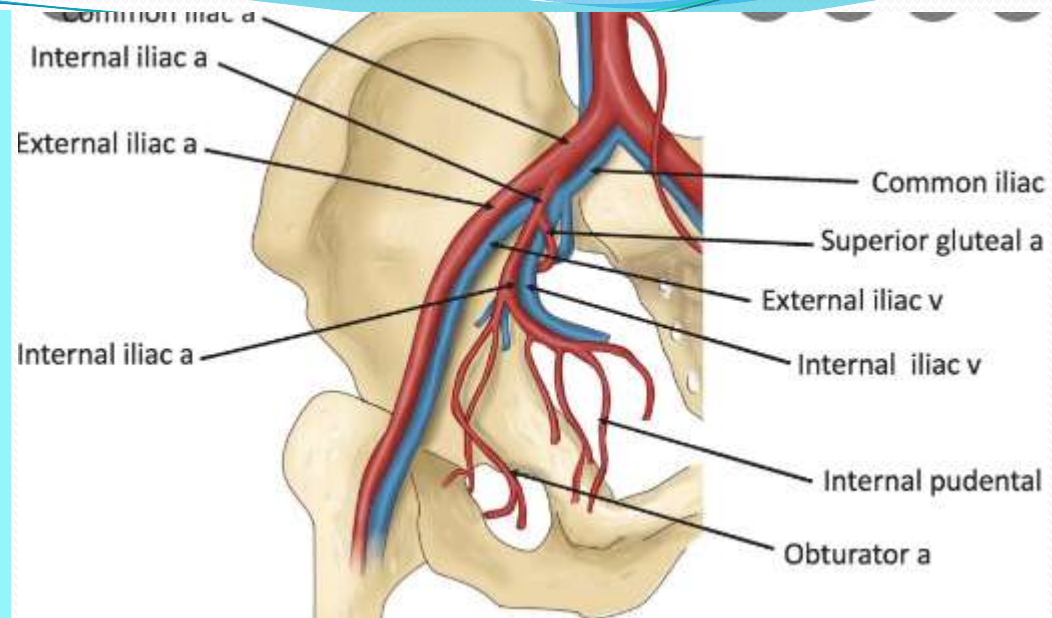


Clinical correlation

Clinical significance of profunda femoris artery: The profunda femoris artery is of great clinical importance because it is deeply located and lies in close proximity to the femoral shaft; hence, it is prone to injury in fracture of femoral shaft. The artery is also liable to injury during surgical procedure of fixing metallic screws in the femur by an orthopedic surgeon.

Obturator artery

- Arises from **internal iliac artery**
- **Passes through obturator foramen**
- Enters medial compartment of the thigh
- **Divides into anterior and posterior branches**
- Supplies obturator externus and adductors of thigh
- Posterior branch gives off acetabular branch to head of femur



Arterial anastomosis on hip joint

❑ Trochanteric anastomosis

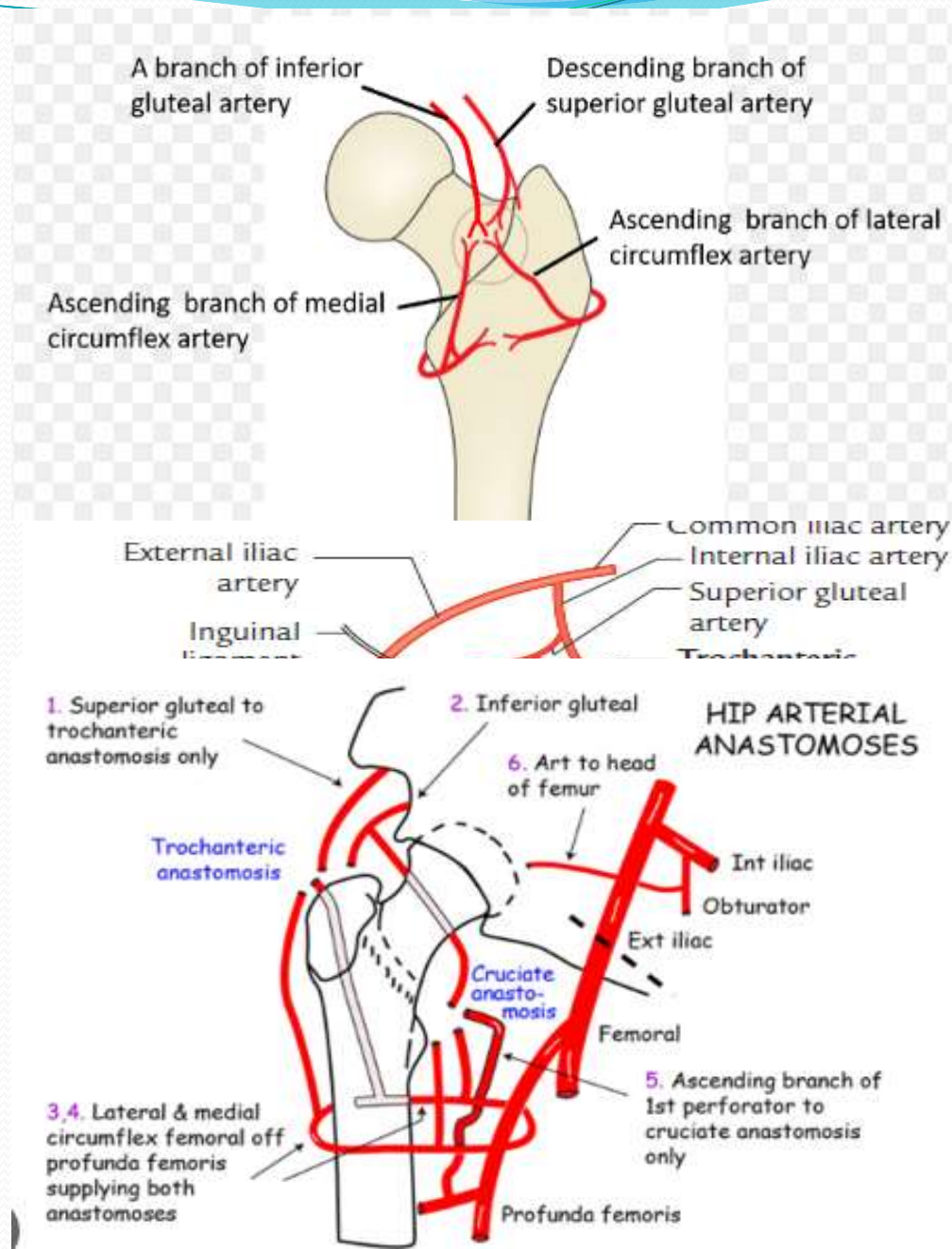
❖ Lie at greater trochanter

- Superior gluteal artery
- Inferior gluteal artery
- Medial circumflex femoral artery
- Lateral circumflex femoral artery

❑ Cruciate anastomosis

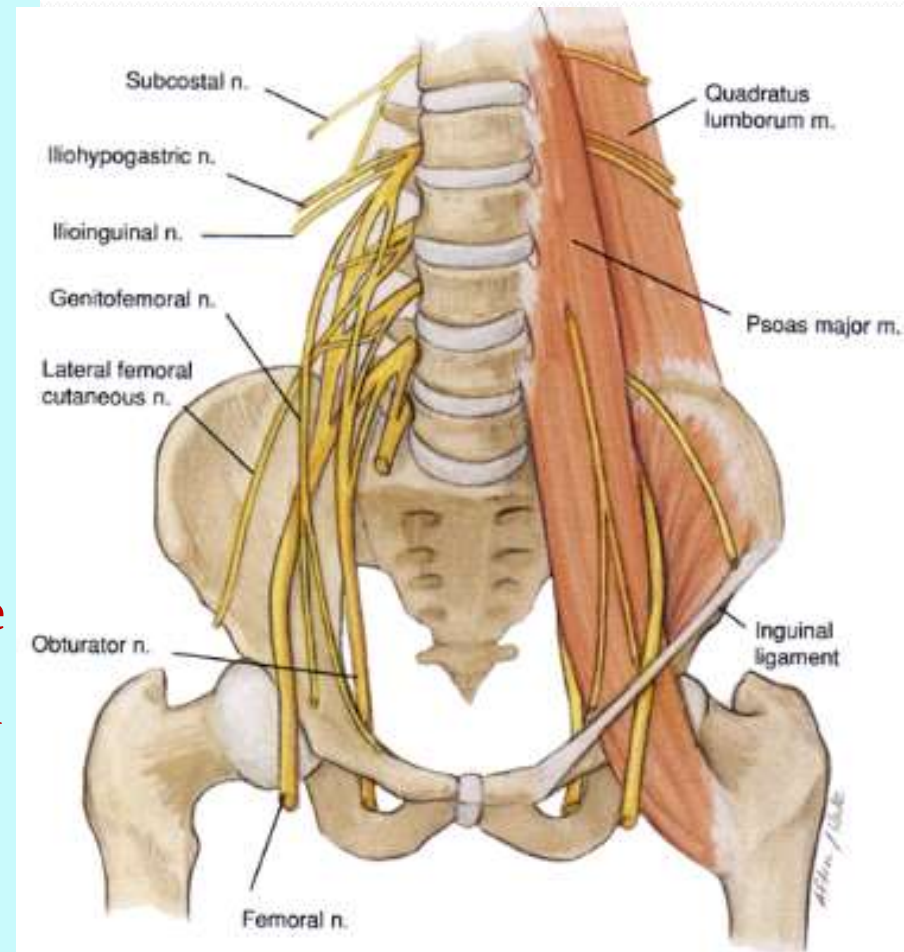
❖ Lie at the lesser trochanter

- Inferior gluteal artery
- Medial circumflex femoral artery
- Lateral circumflex femoral artery
- 1st perforating branch



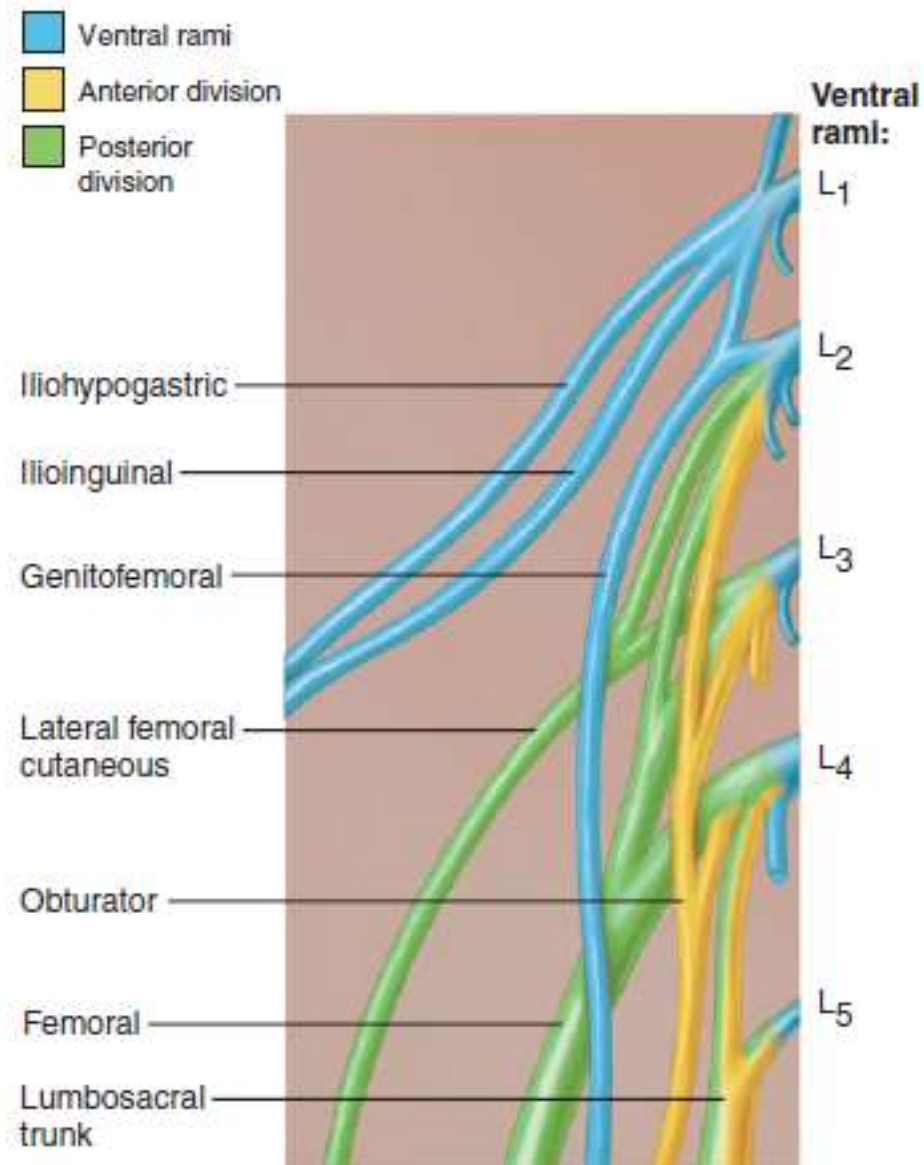
Lumbar plexus

- It is formed from the anterior rami of (L1 - L4)
- The trunks of the plexus traverse psoas major and **emerge from its lateral border**
 - There are two exceptions:
 - **Obturator nerve appears at the medial border of psoas tendon**
 - **Genitofemoral nerve emerges on the anterior aspect of the muscle by piercing the psoas major.**



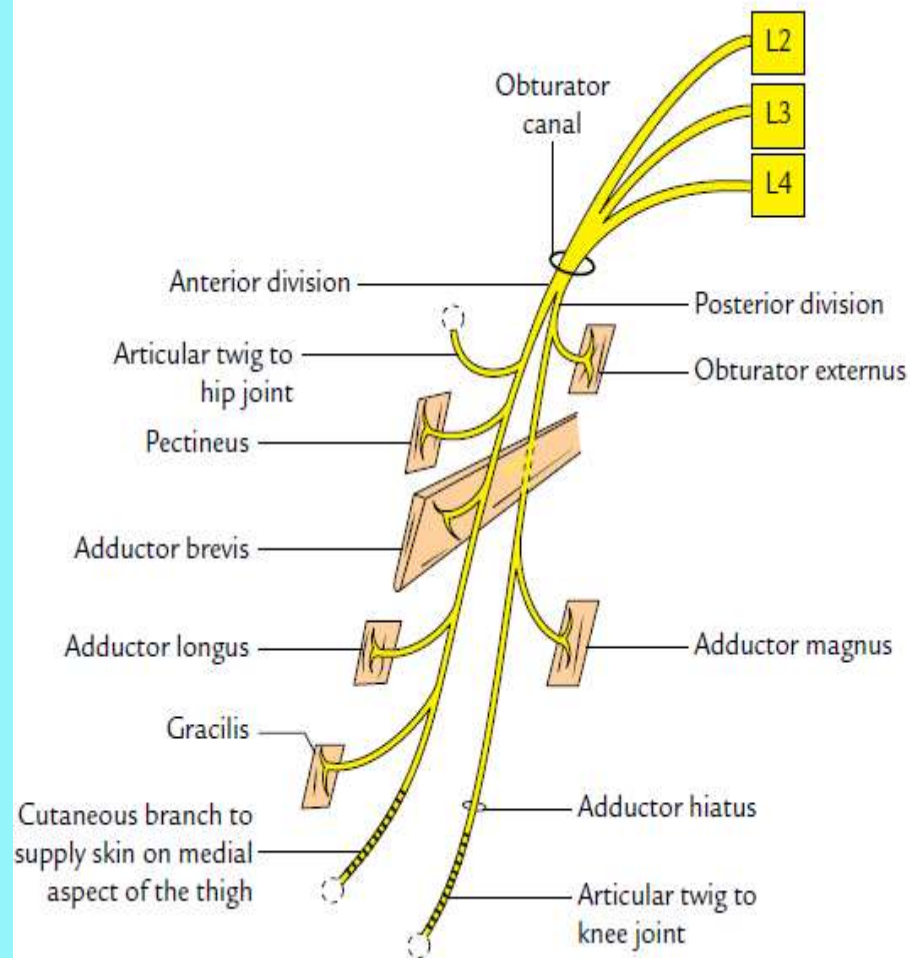
Lumbar plexus

- Branches of the lumbar plexus
 - Obturator nerve(L2,4)
 - Femoral nerve (L2-4)
 - Lumbosacral trunk(L4,5)
 - Ilioinguinal nerve (L1)
 - Iliohypogastric nerve(L1)
 - Genitofemoral nerve(L1,2)
 - Lateral cutaneous nerve of the thigh(L2,3)



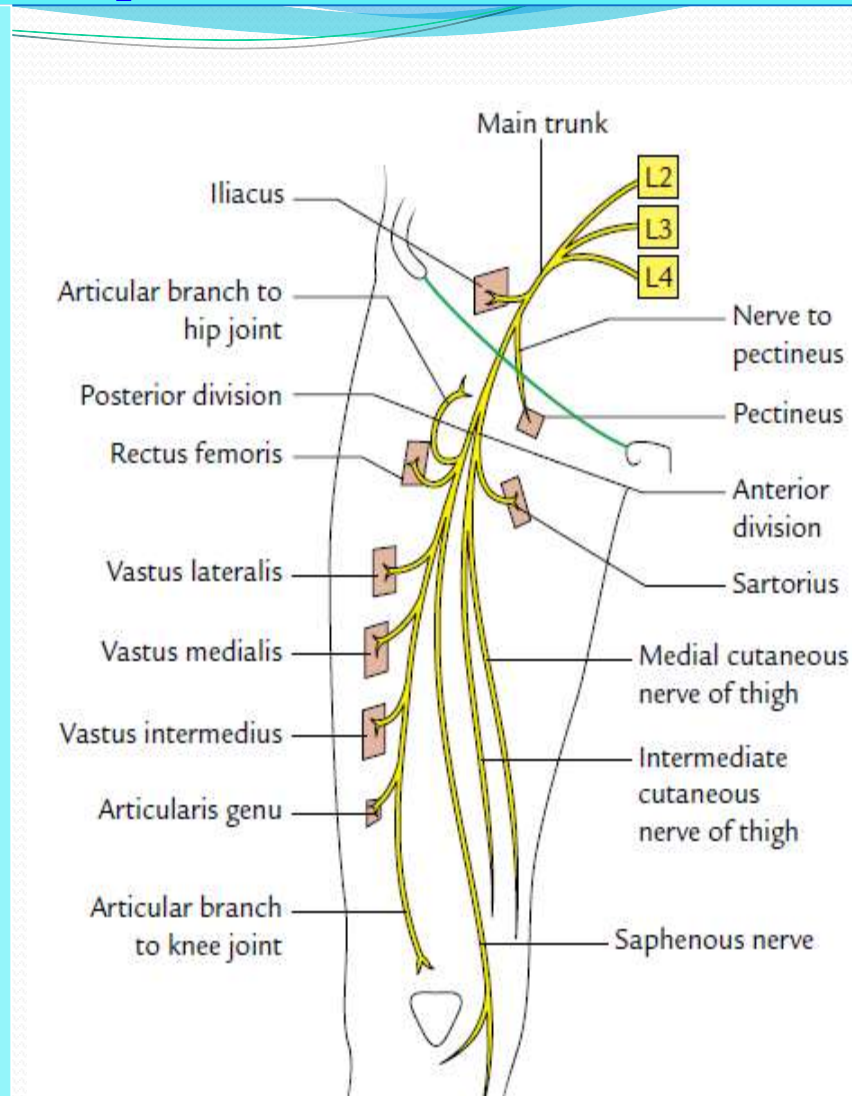
Obturator nerve (L2- L4)

- Emerges from the **medial border of the psoas major**
- Passes into the lesser pelvis Passing through the *Obturator foramen* to the medial thigh
- Divides into **anterior and posterior divisions** by **adductor brevis muscle**
- The nerve to gracilis becomes the **only cutaneous branch**
- Has articular branches for **both the hip and knee joint**
- ***It supply all adductor muscles in thigh***



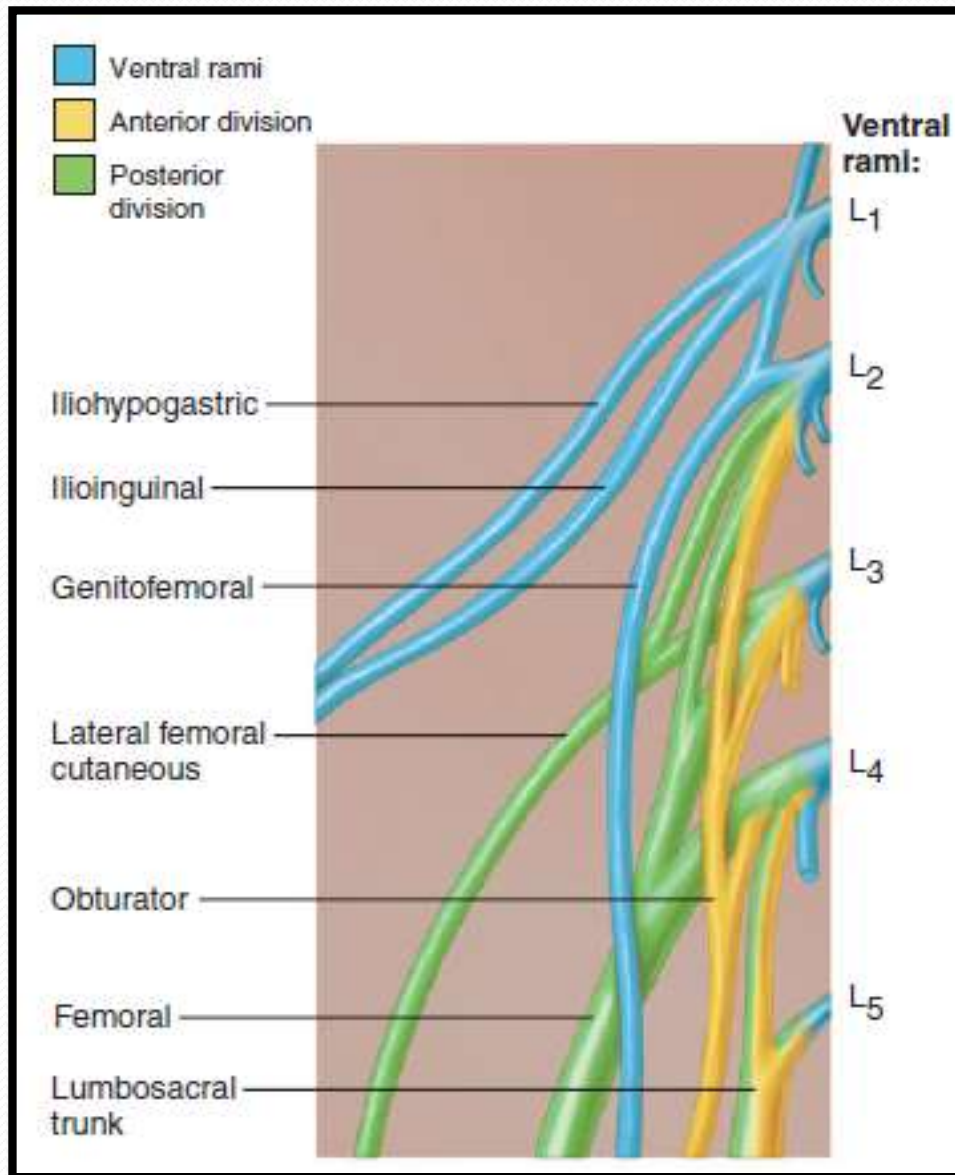
Femoral nerve (L2 - L4)

- Emerges from the **lateral border of the psoas major**
- Lies in front of the iliacus
- Passes deep to the inguinal ligament to the anterior thigh lies **lateral to the femoral artery**
- Breaks up into motor and sensory branches
- Supply the flexors of the hip and extensors of the knee
- **Two branches follow the artery on its lateral side into the adductor canal:**
 - **Nerve to vastus medialis (motor)**
 - **Saphenous nerve (sensory)**



Lumbosacral trunk (L4 & L5)

- Passes over the ala (wing) of the sacrum
- Descends into the pelvis
- Participate in the formation of the sacral plexus with the anterior rami of S1 - S4 nerves



Genitofemoral nerve (L1 & L2)

- **Pierces the psoas major** and runs deep to the psoas fascia
- Divides lateral to the common and external iliac arteries into **femoral and genital branches**
- The lateral cutaneous nerve of the thigh, or lateral femoral cutaneous nerve (L2, L3)
- Runs inferolaterally on the iliacus and enters the thigh deep to the inguinal ligament/iliopubic tract, just medial to the ASIS
- **Supplies skin on the anterolateral surface of the thigh**

